Analysis Fourier and Distributions II

Function spaces and distributions

Multiindex notation

Let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$ be a multiindex. We write

$$|\alpha| := \alpha_1 + \cdots + \alpha_n, \qquad \partial^\alpha := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

The basic test–function spaces

Let $\Omega \subset \mathbb{R}^n$ be open.

$\mathcal{D}(\Omega)$: compactly supported smooth functions. We set

$$\mathcal{D}(\Omega) := C_c^\infty(\Omega),$$

the space of all C^∞ functions with compact support in Ω. A sequence (φ_j) converges to φ in $\mathcal{D}(\Omega)$ if and only if there exists a compact set $K \subset \Omega$ such that $\operatorname{supp} \varphi_j \subset K$ for all j and

$$\sup_{x \in K} |\partial^\alpha \varphi_j(x) - \partial^\alpha \varphi(x)| \longrightarrow 0 \qquad \text{for every multiindex } \alpha.$$

This defines the standard locally convex topology on $\mathcal{D}(\Omega)$.

$\mathcal{E}(\Omega)$: all smooth functions. We set

$$\mathcal{E}(\Omega) := C^\infty(\Omega).$$

A sequence (f_j) converges to f in $\mathcal{E}(\Omega)$ if and only if for every compact set $K \subset \Omega$ and every multiindex α we have

$$\sup_{x \in K} |\partial^\alpha f_j(x) - \partial^\alpha f(x)| \longrightarrow 0.$$

Thus all derivatives converge uniformly on every compact subset of Ω.

$\mathcal{S}(\mathbb{R}^n)$**: Schwartz space.** The Schwartz space $\mathcal{S}(\mathbb{R}^n)$ consists of all $\phi \in C^\infty(\mathbb{R}^n)$ such that for every pair of multiindices α, β the seminorm

$$p_{\alpha,\beta}(\phi) := \sup_{x\in\mathbb{R}^n} (1+|x|)^{|\alpha|} \, |\partial^\beta \phi(x)|$$

is finite. The family $\{p_{\alpha,\beta}\}$ defines the natural Fréchet topology on $\mathcal{S}(\mathbb{R}^n)$. Functions in $\mathcal{S}$ are smooth and decrease, together with all derivatives, faster than any power of $|x|$.

Distributions and their dual spaces

Distributions. The space of (Schwartz) distributions on Ω is the topological dual

$$\mathcal{D}'(\Omega) := \{T : \mathcal{D}(\Omega) \to \mathbb{C} \text{ linear and continuous}\}.$$

Elements of $\mathcal{D}'(\Omega)$ are called distributions. A basic class of examples is provided by locally integrable functions: if $f \in L^1_{\text{loc}}(\Omega)$ we define the *regular distribution* $T_f \in \mathcal{D}'(\Omega)$ by

$$\langle T_f, \varphi\rangle := \int_\Omega f(x)\, \varphi(x)\, dx, \qquad \varphi \in \mathcal{D}(\Omega).$$

Compactly supported distributions. The dual of $\mathcal{E}(\mathbb{R}^n) = C^\infty(\mathbb{R}^n)$ is denoted by

$$\mathcal{E}'(\mathbb{R}^n) := \{\Lambda : \mathcal{E}(\mathbb{R}^n) \to \mathbb{C} \text{ linear and continuous}\}.$$

A fundamental theorem identifies $\mathcal{E}'(\mathbb{R}^n)$ with the space of distributions (in $\mathcal{D}'(\mathbb{R}^n)$) which have compact support. Thus $\mathcal{E}'(\mathbb{R}^n)$ is often referred to as the space of *compactly supported distributions.*

Tempered distributions. The dual of the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ is the space of *tempered distributions*

$$\mathcal{S}'(\mathbb{R}^n) := \{T : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C} \text{ linear and continuous}\}.$$

Tempered distributions are precisely those distributions that grow at most polynomially at infinity and are stable under the Fourier transform.

Inclusions. On the test–function level we have the inclusions

$$\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset \mathcal{E}(\mathbb{R}^n),$$

and hence, by duality, the reversed inclusions on the distribution side:

$$\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n).$$

Order and regularity of distributions

Order of a distribution. A distribution $T \in \mathcal{D}'(\Omega)$ is said to be of order $\leq m$ if for every compact $K \subset \Omega$ there exists a constant $C_K > 0$ such that

$$|\langle T, \varphi\rangle| \leq C_K \max_{|\alpha|\leq m} \sup_{x\in K} |\partial^\alpha \varphi(x)| \qquad \text{for all } \varphi \in \mathcal{D}(\Omega) \text{ with } \operatorname{supp}\varphi \subset K.$$

The smallest such m (if it exists) is called the *order* of T. For example, the Dirac delta δ_x has order 0, while $\partial^\alpha \delta_x$ has order $|\alpha|$ and the principal value distribution p.v.$(1/x)$ has order 1 on $\mathbb{R}$.

Regular and tempered distributions. If $f \in L^1_{\text{loc}}(\Omega)$, the distribution T_f defined by

$$\langle T_f, \varphi\rangle = \int_\Omega f(x)\varphi(x)\,dx$$

is called a *regular distribution*. When f has at most polynomial growth on $\mathbb{R}^n$, the same formula defines a tempered distribution in $\mathcal{S}'(\mathbb{R}^n)$. In this way many classical functions (polynomials, exponentials of purely imaginary arguments, etc.) are naturally embedded into the spaces of distributions and tempered distributions.

Distributions as dual pairings. A distribution on Ω is a continuous linear functional

$$T : \mathcal{D}(\Omega) \to \mathbb{C}, \qquad \varphi \mapsto \langle T, \varphi\rangle.$$

If $f \in L^1_{\text{loc}}(\Omega)$, the associated *regular distribution* T_f is defined by

$$\langle T_f, \varphi\rangle := \int_\Omega f(x)\,\varphi(x)\,dx, \qquad \varphi \in \mathcal{D}(\Omega).$$

The brackets $\langle T, \varphi\rangle$ denote the action of the distribution T on the test function φ; they play the role of evaluation for distributions, in analogy with $\lambda(v)$ or $\langle \lambda, v\rangle$ in linear algebra.

Distributions: Regular, Singular, Tempered, and Their Examples

A distribution on $\mathbb{R}$ is a continuous linear functional on the space $\mathcal{D}(\mathbb{R}) = C_0^\infty(\mathbb{R})$ of compactly supported smooth test functions. For $T \in \mathcal{D}'(\mathbb{R})$ and $\varphi \in \mathcal{D}(\mathbb{R})$ we write $\langle T, \varphi\rangle$ for the action of T on φ.

1. Regular distributions

If $f \in L^1_{\text{loc}}(\mathbb{R})$, the *regular distribution* associated with f is

$$\boxed{\langle T_f, \varphi\rangle = \int_\mathbb{R} f(x)\,\varphi(x)\,dx}.$$

Examples:

$$T_{x^2}(\varphi) = \int x^2\varphi(x)\,dx, \qquad T_{|x|}(\varphi) = \int |x|\varphi(x)\,dx,$$
$$T_{\mathbf{1}_{[-1,1]}}(\varphi) = \int_{-1}^{1} \varphi(x)\,dx, \qquad T_{1/\sqrt{|x|}}(\varphi) = \int_{\mathbb{R}} \frac{\varphi(x)}{\sqrt{|x|}}\,dx.$$

These are all valid distributions even if the corresponding functions are not smooth or even nowhere classically differentiable.

2. Distributional derivatives

For any $T \in \mathcal{D}'(\mathbb{R})$, its derivative ∂T is defined by

$$\boxed{\langle \partial T, \varphi \rangle := -\langle T, \varphi' \rangle.}$$

If $T = T_f$ is regular, then

$$\langle \partial T_f, \varphi \rangle = -\int f(x)\varphi'(x)\,dx.$$

Examples:

$$(|x|)' = \operatorname{sgn}(x) \quad \text{(distributionally)}$$

even though $|x|$ is not differentiable at 0.

$$H'(x) = \delta_0$$

where H is the Heaviside step function. This produces the Dirac delta as the derivative of a jump.

If f has a jump discontinuity at $x = a$, then

$$\partial T_f = T_{f'} + (\text{jump at } a)\,\delta_a.$$

3. Purely singular distributions

These do not come from locally integrable functions.

Dirac delta and its derivatives.

$$\langle \delta_a, \varphi \rangle = \varphi(a), \qquad \langle \delta_a^{(k)}, \varphi \rangle = (-1)^k\,\varphi^{(k)}(a).$$

Principal value distribution.

$$\boxed{\left\langle PV\left(\frac{1}{x}\right), \varphi \right\rangle = \lim_{\varepsilon \to 0} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\, dx.}$$

This is a classical example of a singular distribution of order one.

4. Tempered distributions

The space of tempered distributions $\mathcal{S}'(\mathbb{R})$ is the dual of the Schwartz space $\mathcal{S}(\mathbb{R})$. These are distributions that grow at most polynomially.

Examples:

$$T_{x^k}, \qquad T_{\sqrt{1+x^2}}, \qquad \delta_0,\ \delta_0^{(k)}, \qquad PV\left(\frac{1}{x}\right),$$
$$\widehat{f} \in \mathcal{S}' \quad \text{for any } f \in L^1(\mathbb{R}).$$

All Dirac deltas and principal value kernels are tempered.

5. Why we write T_f instead of just f

The notation T_f emphasizes that we view f as a linear functional on test functions:

$$\langle T_f, \varphi \rangle = \int f\varphi.$$

- f is an ordinary function; T_f is an element of $\mathcal{D}'$.
- Differential operators apply naturally to T_f, even when f is not classically differentiable.
- If $f = g$ almost everywhere, then $T_f = T_g$ as distributions; this is not visible from pointwise notation.

Thus the distributional viewpoint is strictly more flexible than the classical one.

6. Convolution of distributions with test functions

If $\phi \in C_0^\infty(\mathbb{R}^n)$ and $T \in \mathcal{D}'(\mathbb{R}^n)$, the convolution

$$\boxed{(T * \phi)(x) := \langle T, \phi(x - \cdot) \rangle}$$

is always a smooth function.

Example A: Convolution with a regular distribution. Let $T = T_f$ where $f \in L^1_{\text{loc}}$. Then

$$(T_f * \phi)(x) = \int_{\mathbb{R}} f(y)\,\phi(x-y)\,dy,$$

which is the classical convolution.

Example B: Convolution with Dirac delta.

$$\delta_a * \phi = \phi(\cdot - a).$$

Example C: Convolution smooths distributions. Convolving the principal value or δ' with a mollifier produces smooth approximations.

7. Summary table

Distribution	Defined by	Example	Singular?	Tempered?
Regular	$\int f\varphi$	T_{x^2}, $T_{\lvert x\rvert}$	No	Yes
Derivative of regular	$-\int f\varphi'$	$(\lvert x\rvert)' = \operatorname{sgn}(x)$	Sometimes	Often
Dirac	$\varphi(a)$	δ_0	Yes	Yes
Dirac derivatives	$(-1)^k\varphi^{(k)}(a)$	δ', δ''	Very	Yes
Principal value	symmetric limit	$PV(1/x)$	Yes	Yes

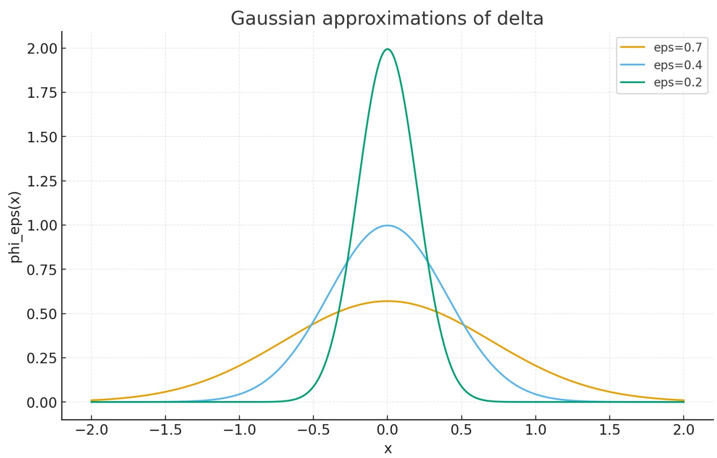


Figure 1: Gaussian approximations of the Dirac delta at 0: $\phi_\varepsilon(x) = \frac{1}{\sqrt{2\pi}\varepsilon} e^{-\frac{x^2}{2\varepsilon^2}}$ for several values of ε. As $\varepsilon \to 0$, the peak becomes narrower and higher, while the area stays equal to 1, illustrating convergence to δ_0 in the sense of distributions.

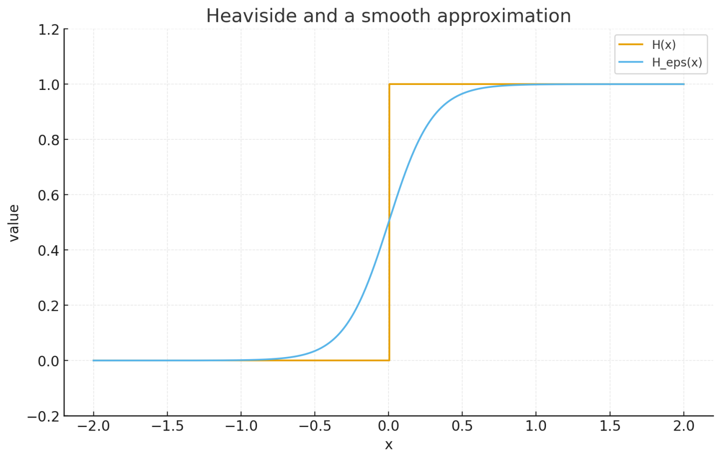


Figure 2: The Heaviside step function $H(x)$ and a smooth approximation $H_\varepsilon(x) = \frac{1}{2}(1 + \tanh(x/\varepsilon))$. The jump at 0 is smoothed out over a small interval of width $\sim \varepsilon$. Distributionally, $H' = \delta_0$, while H'_ε is a smooth "bump" converging to δ_0.

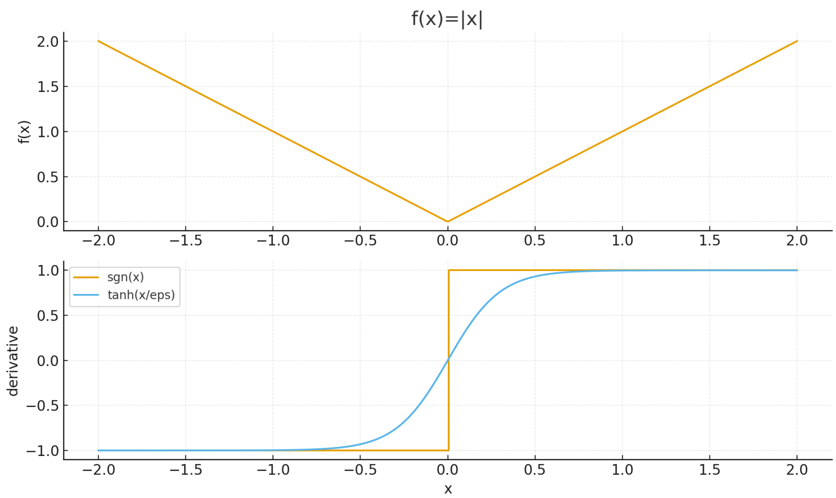


Figure 3: Top: the function $f(x) = |x|$. Bottom: the sign function $\operatorname{sgn}(x)$ (distributional derivative of $|x|$) and a smooth approximation $\tanh(x/\varepsilon)$. This illustrates how distributional derivatives can be visualised via smoothing: the derivative of the mollified $|x|$ converges to $\operatorname{sgn}(x)$ in the sense of distributions.

Principal value distribution as a tempered distribution

The function $f(x) = 1/x$ does not define a distribution, because the integral

$$\int_{-\infty}^{\infty} \frac{\varphi(x)}{x}\,dx$$

diverges for general test functions φ. The singularity at $x = 0$ is not locally integrable, so $1/x \notin L^1_{\mathrm{loc}}$ and hence does not define a regular distribution.

Nevertheless, the *principal value* distribution

$$\left\langle PV\left(\frac{1}{x}\right), \varphi \right\rangle := \lim_{\varepsilon\to 0} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\,dx$$

exists for every $\varphi \in \mathcal{D}(\mathbb{R})$. Since φ is smooth, the quantity $\varphi(x) - \varphi(0) = O(x)$, and the symmetric cancellation near 0 guarantees convergence. The mapping $\varphi \mapsto \langle PV(1/x), \varphi\rangle$ is linear and continuous on $\mathcal{S}(\mathbb{R})$, and hence

$$PV\left(\frac{1}{x}\right) \in \mathcal{S}'(\mathbb{R})$$

is a tempered distribution.

A smooth regularisation is given by

$$f_\varepsilon(x) = \frac{x}{x^2+\varepsilon^2},$$

which satisfies $f_\varepsilon \to PV(1/x)$ in $\mathcal{S}'$.

A. Smooth regularisations of $1/x$

A standard smooth odd regularisation of $1/x$ is given by

$$f_\varepsilon(x) = \frac{x}{x^2+\varepsilon^2}, \qquad \varepsilon > 0.$$

Each f_ε is smooth on $\mathbb{R}$, odd, and satisfies

$$f_\varepsilon(x) \sim \frac{1}{x} \quad \text{for } |x| \gg \varepsilon, \qquad f_\varepsilon(0) = 0.$$

Thus f_ε looks like $1/x$ away from the origin, but the singularity at $x = 0$ has been "rounded off".

In the sense of distributions we have

$$f_\varepsilon \longrightarrow PV\left(\frac{1}{x}\right) \quad \text{in } \mathcal{S}'(\mathbb{R}),$$

i.e.

$$\lim_{\varepsilon\to 0} \int_{\mathbb{R}} f_\varepsilon(x)\,\varphi(x)\,dx = \left\langle PV\left(\frac{1}{x}\right), \varphi \right\rangle = \lim_{\varepsilon\to 0} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\,dx$$

for all $\varphi \in \mathcal{S}(\mathbb{R})$.

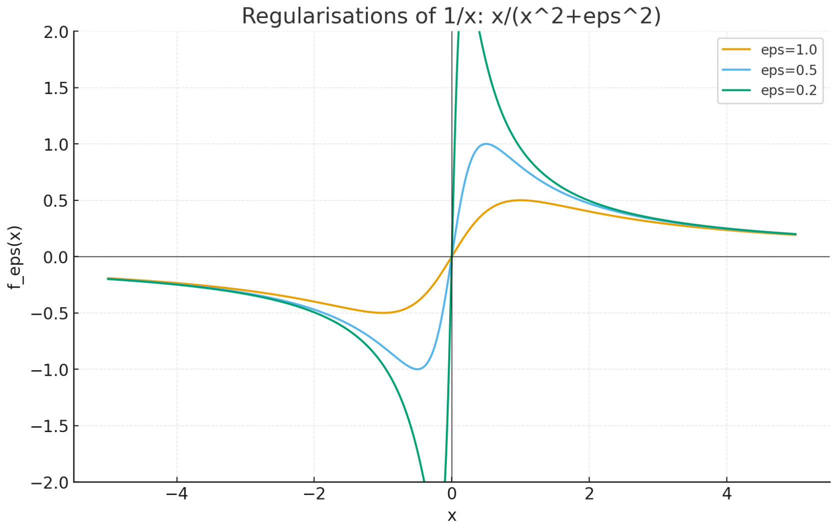


Figure 4: Regularisations $f_\varepsilon(x) = x/(x^2 + \varepsilon^2)$ of $1/x$ for several values of ε.

B. Comparison with $1/x$

The function $1/x$ is not locally integrable at 0, so it does not define a regular distribution. Nevertheless, away from the origin the regularisation $f_\varepsilon(x) = x/(x^2+\varepsilon^2)$ agrees with $1/x$ up to an error of order $O(\varepsilon^2/x^3)$:

$$\frac{x}{x^2+\varepsilon^2} = \frac{1}{x}\,\frac{1}{1+\varepsilon^2/x^2} = \frac{1}{x} - \frac{\varepsilon^2}{x^3} + O\!\left(\frac{\varepsilon^4}{x^5}\right) \qquad (|x| \gg \varepsilon).$$

Thus the regularisation captures the correct singular behaviour at infinity, while remaining smooth at 0. The principal value distribution $PV(1/x)$ can be viewed as the "limit" of these regularisations.

C. Fourier transform and the sign function

A key fact from harmonic analysis is the distributional Fourier transform identity

$$\mathcal{F}\!\left(PV\frac{1}{x}\right)(\xi) = -\,i\pi\ \mathrm{sgn}(\xi),$$

where $\mathrm{sgn}(\xi)$ is the sign function,

$$\mathrm{sgn}(\xi) = \begin{cases} -1, & \xi < 0, \\ 0, & \xi = 0, \\ 1, & \xi > 0. \end{cases}$$

This shows that $PV(1/x)$ is (up to constants) the kernel of the Hilbert transform and connects singular integral operators with tempered distributions.

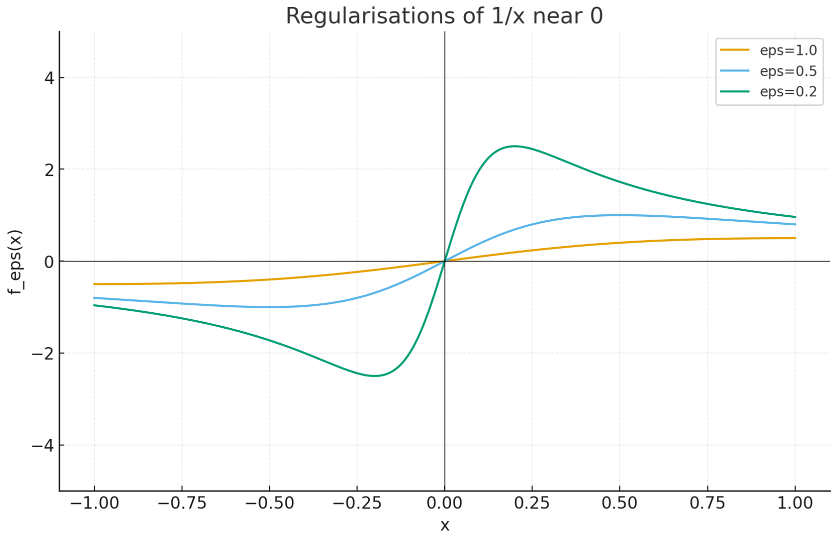


Figure 5: Zoom near 0 of the regularisations $f_\varepsilon(x)$. As $\varepsilon \to 0$, the functions become steeper near 0 and approximate the singular behaviour of $1/x$ in the principal value sense.

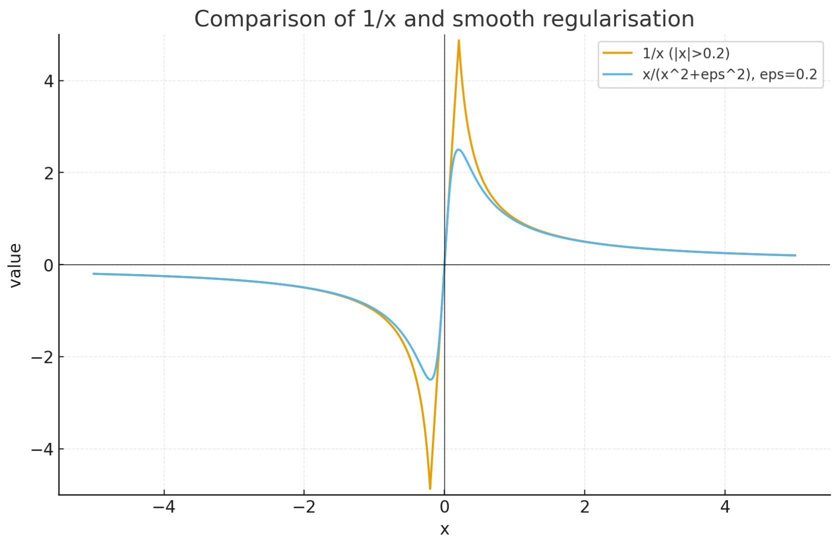


Figure 6: Comparison of $1/x$ (away from a small neighbourhood of 0) with a fixed regularisation $f_\varepsilon(x) = x/(x^2 + \varepsilon^2)$. The curves almost coincide for $|x| \gg \varepsilon$, while the regularisation remains bounded at 0.

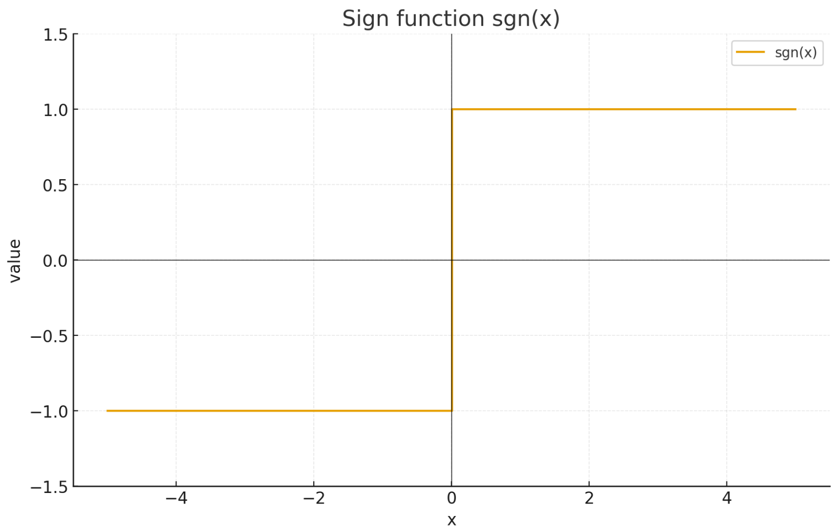


Figure 7: The sign function $\operatorname{sgn}(x)$, which appears as the Fourier transform (up to the factor $-i\pi$) of the principal value distribution $PV(1/x)$.

Exercise 3.1. Approximation in $\mathcal{E}(\Omega)$ and $\mathcal{D}(\Omega)$

(a) Suppose that $\{\phi_j\}_{j=1}^{\infty} \subset \mathcal{E}(\Omega)$ is a sequence such that $\phi_j \to \phi$ in $\mathcal{E}(\Omega)$, and $\chi \in \mathcal{D}(\Omega)$. Show that

$$\chi\phi_j \longrightarrow \chi\phi \quad \text{in } \mathcal{D}(\Omega).$$

(b) Show that if $\psi \in \mathcal{E}(\Omega)$, then there exists a sequence $\{\phi_j\}_{j=1}^{\infty} \subset \mathcal{D}(\Omega)$ such that $\phi_j \to \psi$ in $\mathcal{E}(\Omega)$.

Solution

(a) Let $\chi \in \mathcal{D}(\Omega)$ and let $K := \operatorname{supp}\chi \subset \Omega$, which is compact. Convergence $\phi_j \to \phi$ in $\mathcal{E}(\Omega)$ means that for every multiindex β,

$$\sup_{x\in K} |\partial^\beta \phi_j(x) - \partial^\beta \phi(x)| \longrightarrow 0.$$

For any multiindex α, by Leibniz' rule,

$$\partial^\alpha(\chi\phi_j) = \sum_{\beta\le\alpha} \binom{\alpha}{\beta} (\partial^{\alpha-\beta}\chi)\,(\partial^\beta\phi_j).$$

Each $\partial^{\alpha-\beta}\chi$ is continuous on K and fixed, while $\partial^\beta\phi_j \to \partial^\beta\phi$ uniformly on K. Hence $\partial^\alpha(\chi\phi_j) \to \partial^\alpha(\chi\phi)$ uniformly on K. Moreover, $\operatorname{supp}(\chi\phi_j) \subset K$ for all j. Therefore $\chi\phi_j \to \chi\phi$ in $\mathcal{D}(\Omega)$.

(b) Let $\psi \in \mathcal{E}(\Omega)$. Choose an exhaustion of Ω by compact sets $K_1 \subset K_2 \subset \cdots$ with $\bigcup_j K_j = \Omega$. For each j one can construct $\chi_j \in \mathcal{D}(\Omega)$ such that

$$0 \le \chi_j \le 1, \qquad \chi_j \equiv 1 \text{ on } K_j.$$

(For instance by smoothing suitable cut–off functions, cf. Lemma 1.14.) Define

$$\phi_j := \chi_j\psi \in \mathcal{D}(\Omega).$$

Let $K \subset \Omega$ be any compact set. Then $K \subset K_J$ for some J, and for all $j \ge J$ we have $\chi_j \equiv 1$ on K, hence $\phi_j = \psi$ on K. It follows that for every multiindex α and every compact $K \subset \Omega$,

$$\sup_{x\in K} |\partial^\alpha\phi_j(x) - \partial^\alpha\psi(x)| = 0$$

for all sufficiently large j. Thus $\phi_j \to \psi$ in $\mathcal{E}(\Omega)$.

$\square$

Exercise 3.2. Tempered and general distributions

(a) Prove Lemma 2.11. *(See the textbook; the proof is analogous to that of Lemma 2.9.)*

(b) Show that we have the inclusions

$$\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n).$$

Solution (b)

We know that

$$\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset \mathcal{E}(\mathbb{R}^n).$$

If $T \in \mathcal{E}'(\mathbb{R}^n)$ is continuous on $\mathcal{E}(\mathbb{R}^n)$, then its restriction to the subspace $\mathcal{S}(\mathbb{R}^n)$ is also continuous, so $T \in \mathcal{S}'(\mathbb{R}^n)$. Hence $\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$.

Similarly, if $T \in \mathcal{S}'(\mathbb{R}^n)$ is continuous on $\mathcal{S}(\mathbb{R}^n)$, then the restriction of T to $\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$ is a continuous linear functional on $\mathcal{D}(\mathbb{R}^n)$, i.e. $T \in \mathcal{D}'(\mathbb{R}^n)$. Thus $\mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$. Combining both inclusions we obtain

$$\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n).$$

$\square$

Exercise 3.3. Sequences as distributions

Let $\{a_j\}_{j=1}^{\infty} \subset \mathbb{R}$ be a sequence of real numbers. For $\phi \in C^\infty(\mathbb{R})$ define

$$u[\phi] := \sum_{j=1}^{\infty} a_j \, \phi(j),$$

provided that the sum converges. Give necessary and sufficient conditions on (a_j) such that:

(a) $u \in \mathcal{E}'(\mathbb{R})$,

(b) $u \in \mathcal{S}'(\mathbb{R})$,

(c) $u \in \mathcal{D}'(\mathbb{R})$.

Solution

We may write formally

$$u = \sum_{j=1}^{\infty} a_j \, \delta_j, \qquad \delta_j[\phi] = \phi(j).$$

(a) Membership in $\mathcal{E}'(\mathbb{R})$. The distribution u is compactly supported iff the closure of the set $\{j \in \mathbb{N} : a_j \neq 0\}$ is compact in $\mathbb{R}$, which happens exactly when this set is finite. Hence

$$u \in \mathcal{E}'(\mathbb{R}) \quad \Longleftrightarrow \quad a_j = 0 \text{ for all but finitely many } j.$$

In this case u is a finite linear combination of Dirac deltas.

(b) Membership in $\mathcal{S}'(\mathbb{R})$. Recall that $\mathcal{S}(\mathbb{R})$ has seminorms

$$p_{m,k}(\phi) = \sup_{x\in\mathbb{R}} (1+|x|)^m |\phi^{(k)}(x)|.$$

For each m and $\phi \in \mathcal{S}(\mathbb{R})$ we have

$$|\phi(j)| \leq \frac{p_{m,0}(\phi)}{(1+|j|)^m}, \qquad j \in \mathbb{Z}.$$

If there exist constants $C > 0$ and $N \in \mathbb{N}$ such that

$$|a_j| \leq C(1+|j|)^N \qquad \text{for all } j,$$

then choosing $m > N + 2$ we obtain

$$\sum_{j=1}^{\infty} |a_j \phi(j)| \leq C\, p_{m,0}(\phi) \sum_{j=1}^{\infty} (1+|j|)^{N-m} < \infty,$$

since $\sum_j (1+|j|)^{N-m}$ converges for $m > N+1$. Thus u is a well-defined continuous linear functional on $\mathcal{S}(\mathbb{R})$, so $u \in \mathcal{S}'(\mathbb{R})$.

Conversely, if $u \in \mathcal{S}'(\mathbb{R})$ then the map $\phi \mapsto \sum_j a_j \phi(j)$ is continuous with respect to the Schwartz seminorms. Testing this continuity on functions that are concentrated near each integer j shows that the sequence (a_j) cannot grow faster than some polynomial; hence there exist C, N such that $|a_j| \leq C(1+|j|)^N$ for all j.

Therefore

$$u \in \mathcal{S}'(\mathbb{R}) \quad \Longleftrightarrow \quad |a_j| \leq C(1+|j|)^N \text{ for some } C, N.$$

(c) Membership in $\mathcal{D}'(\mathbb{R})$. For $\phi \in \mathcal{D}(\mathbb{R})$ the support of ϕ is compact, hence there are only finitely many integers j with $\phi(j) \neq 0$. Thus the sum

$$u[\phi] = \sum_{j=1}^{\infty} a_j \phi(j)$$

is always finite and therefore convergent, without any restriction on (a_j). Each δ_j is a distribution, and for test functions the sum defining u is locally finite, so u is a continuous linear functional on $\mathcal{D}(\mathbb{R})$.

Consequently, *every* sequence (a_j) defines an element of $\mathcal{D}'(\mathbb{R})$; no additional conditions are needed.

□

Exercise 3.4. Plane waves as tempered distributions

For $\xi \in \mathbb{R}^n$ define $e_\xi(x) = e^{i\xi\cdot x}$ and let $T_{e_\xi} \in \mathcal{S}'(\mathbb{R}^n)$ be the regular distribution associated with e_ξ:

$$\langle T_{e_\xi}, \varphi\rangle = \int_{\mathbb{R}^n} e^{i\xi\cdot x}\varphi(x)\,dx, \qquad \varphi \in \mathcal{S}(\mathbb{R}^n).$$

Show that $T_{e_\xi} \in \mathcal{S}'(\mathbb{R}^n)$ and that $T_{e_\xi} \to 0$ as $|\xi| \to \infty$ in the topology of $\mathcal{S}'$.

Solution

Since e_ξ is smooth and bounded, it is of polynomial growth. Thus for each fixed ξ the map

$$\varphi \mapsto \int_{\mathbb{R}^n} e^{i\xi\cdot x}\varphi(x)\,dx$$

defines a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$, so $T_{e_\xi} \in \mathcal{S}'(\mathbb{R}^n)$.

Using the Fourier transform

$$\widehat{\varphi}(\xi) = \int_{\mathbb{R}^n} e^{-ix\cdot\xi}\varphi(x)\,dx,$$

we note that

$$\langle T_{e_\xi}, \varphi\rangle = \int_{\mathbb{R}^n} e^{i\xi\cdot x}\varphi(x)\,dx = \widehat{\varphi}(-\xi).$$

Since $\varphi \in \mathcal{S}(\mathbb{R}^n)$, its Fourier transform $\widehat{\varphi}$ is also in $\mathcal{S}(\mathbb{R}^n)$, in particular $\widehat{\varphi}(\eta) \to 0$ as $|\eta| \to \infty$. Hence

$$\langle T_{e_\xi}, \varphi\rangle = \widehat{\varphi}(-\xi) \to 0 \quad \text{as } |\xi| \to \infty,$$

for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$. This is precisely the statement that $T_{e_\xi} \to 0$ in $\mathcal{S}'$.

Exercise 3.5. Fourier transforms of some L^1 functions

We use the convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-ix\xi} f(x)\,dx.$$

(a) Let $f(x) = \dfrac{\sin x}{1+x^2}$. Compute $\widehat{f}$.

Solution (a)

Let $g(x) = \dfrac{1}{1+x^2}$, so that

$$f(x) = \frac{\sin x}{1+x^2} = \frac{1}{2i}\Big(e^{ix}g(x) - e^{-ix}g(x)\Big).$$

Using the frequency–shift property $\mathcal{F}(e^{\pm ix}g)(\xi) = \widehat{g}(\xi \mp 1)$ and the known pair

$$g(x) = \frac{1}{1+x^2} \quad \Longrightarrow \quad \widehat{g}(\xi) = \pi e^{-|\xi|},$$

we obtain

$$\widehat{f}(\xi) = \frac{1}{2i}\Big(\widehat{g}(\xi-1) - \widehat{g}(\xi+1)\Big) = \frac{\pi}{2i}\Big(e^{-|\xi-1|} - e^{-|\xi+1|}\Big).$$

(b) Let $f(x) = \dfrac{1}{\varepsilon^2 + x^2}$ with $\varepsilon > 0$. Compute $\widehat{f}$.

Solution (b)

We write

$$f(x) = \frac{1}{\varepsilon^2 + x^2} = \frac{1}{\varepsilon^2} g\Big(\frac{x}{\varepsilon}\Big), \qquad g(x) = \frac{1}{1+x^2}.$$

The scaling property yields, for $h_a(x) = g(x/a)$,

$$\widehat{h_a}(\xi) = a\,\widehat{g}(a\xi).$$

With $a = \varepsilon$ we have

$$\widehat{g(x/\varepsilon)}(\xi) = \varepsilon\,\widehat{g}(\varepsilon\xi) = \varepsilon\pi e^{-\varepsilon|\xi|}.$$

Thus

$$\widehat{f}(\xi) = \frac{1}{\varepsilon^2}\widehat{g(x/\varepsilon)}(\xi) = \frac{\pi}{\varepsilon} e^{-\varepsilon|\xi|}.$$

(c) Let

$$f(x) = \sqrt{\frac{\sigma}{t}}\, e^{-\sigma\frac{(x-y)^2}{t}}, \qquad \sigma > 0,\ t > 0,\ y \in \mathbb{R}.$$

Compute $\widehat{f}$.

Solution (c)

Set $a = \sigma/t > 0$, so

$$f(x) = \sqrt{a}\, e^{-a(x-y)^2}.$$

We use the Gaussian transform

$$\int_{\mathbb{R}} e^{-a(x-y)^2} e^{-ix\xi}\, dx = \sqrt{\frac{\pi}{a}}\, e^{-\frac{\xi^2}{4a}} e^{-iy\xi}.$$

Therefore

$$\widehat{f}(\xi) = \sqrt{a}\,\sqrt{\frac{\pi}{a}}\, e^{-\frac{\xi^2}{4a}} e^{-iy\xi} = \sqrt{\pi}\, e^{-\frac{\xi^2}{4a}} e^{-iy\xi}.$$

Substituting $a = \sigma/t$ gives

$$\widehat{f}(\xi) = \sqrt{\pi}\, \exp\Big(-\frac{t\,\xi^2}{4\sigma}\Big)\, e^{-iy\xi}.$$

(d*) Let $f(x) = 1/\cosh x$. Show that

$$\widehat{f}(\xi) = \int_{\mathbb{R}} \frac{e^{-ix\xi}}{\cosh x}\, dx = \frac{\pi}{\cosh\left(\frac{\pi\xi}{2}\right)}.$$

Solution (sketch, d*)

The function $f(x) = \operatorname{sech} x$ is meromorphic with simple poles at $x = i\pi(n+\frac{1}{2})$, $n \in \mathbb{Z}$. By integrating $e^{-ix\xi}/\cosh x$ over a suitable rectangular contour and summing the residues, one obtains

$$\int_{\mathbb{R}} \frac{e^{-ix\xi}}{\cosh x}\, dx = \frac{\pi}{\cosh(\pi\xi/2)}.$$

Thus

$$\widehat{f}(\xi) = \frac{\pi}{\cosh(\pi\xi/2)}.$$

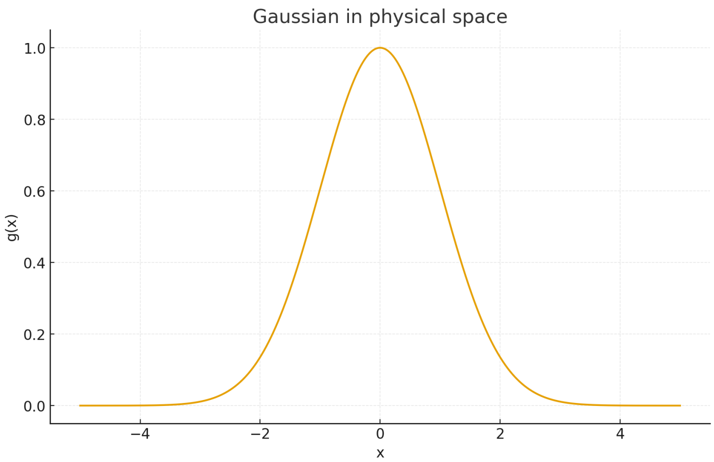


Figure 8: Gaussian $g(x) = e^{-x^2/2}$ in physical space.

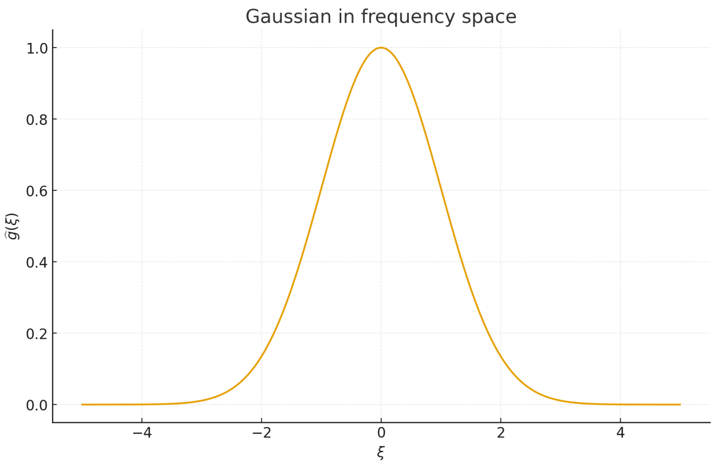


Figure 9: Fourier transform $\widehat{g}(\xi) \propto e^{-\xi^2/2}$ in frequency space. The shape is again Gaussian, illustrating the self-reproducing property of Gaussians under the Fourier transform.

Plane waves as inverse Fourier transforms of point masses. In the sense of tempered distributions we have

$$\mathcal{F}\Big(e^{i\xi_0\cdot x}\Big) = (2\pi)^n\delta_{\xi_0}, \qquad e^{i\xi_0\cdot x} = \mathcal{F}^{-1}\Big((2\pi)^n\delta_{\xi_0}\Big).$$

Thus the regular distribution $T_{e_{\xi_0}}$ associated with the plane wave $e^{i\xi_0\cdot x}$ is the inverse Fourier transform of the point mass at ξ_0. For $\varphi \in \mathcal{S}(\mathbb{R}^n)$ this gives

$$\langle T_{e_\xi}, \varphi\rangle = \int_{\mathbb{R}^n} e^{i\xi\cdot x}\varphi(x)\,dx = \widehat{\varphi}(-\xi),$$

so that $T_{e_\xi} \to 0$ as $|\xi| \to \infty$ because $\widehat{\varphi} \in \mathcal{S}(\mathbb{R}^n)$ and hence $\widehat{\varphi}(-\xi) \to 0$ as $|\xi| \to \infty$.

The Gaussian example illustrates this behaviour: for a Gaussian $g(x) = e^{-x^2/2}$, the Fourier transform $\widehat{g}$ is again Gaussian. As the Gaussian in frequency concentrates and tends to a Dirac distribution δ_{ξ_0}, the inverse Fourier transform tends to a plane wave $e^{i\xi_0\cdot x}$, which is completely delocalised in physical space.

Exercise 3.6 (Approximation by compactly supported C^1 functions)

Suppose $f \in C^1(\mathbb{R}^n)$ and that $f, D_jf \in L^1(\mathbb{R}^n)$. Fix $\varepsilon > 0$. Show that there exists $f_\varepsilon \in C_0^1(\mathbb{R}^n)$ such that

$$\|f - f_\varepsilon\|_{L^1(\mathbb{R}^n)} + \|D_jf - D_jf_\varepsilon\|_{L^1(\mathbb{R}^n)} < \frac{\varepsilon}{2}.$$

Solution. *Step 1: Cut off the tails.* Because $f, D_jf \in L^1(\mathbb{R}^n)$, we can choose $R > 0$ so large that

$$\int_{|x|\geq R} \Big(|f(x)| + |D_jf(x)|\Big)\,dx < \frac{\varepsilon}{8}. \tag{1}$$

Let $\chi_R \in C_0^\infty(\mathbb{R}^n)$ be a smooth cut-off with

$$\chi_R(x) = 1 \text{ for } |x| < R, \qquad \chi_R(x) = 0 \text{ for } |x| \geq 2R, \qquad |D\chi_R(x)| \leq C$$

for some constant $C > 0$ independent of R. Set

$$g_R(x) := \chi_R(x)f(x).$$

Then $g_R \in C_0^1(\mathbb{R}^n)$ and $\operatorname{supp} g_R \subset B_{2R}(0)$.

On the set $\{|x| < R\}$ we have $g_R = f$, whereas on $\{|x| \geq R\}$, $|f(x) - g_R(x)| = |1 - \chi_R(x)|\,|f(x)| \leq |f(x)|$. Hence

$$\|f - g_R\|_{L^1} \leq \int_{|x|\geq R} |f(x)|\,dx < \frac{\varepsilon}{8}. \tag{2}$$

For the derivative we use $D_j(\chi_R f) = \chi_R D_j f + (D_j\chi_R) f$ and compute

$$D_j f - D_j g_R = (1-\chi_R) D_j f - (D_j \chi_R)\, f.$$

Thus

$$\begin{aligned} \|D_j f - D_j g_R\|_{L^1} &\le \int_{|x|\ge R} |D_j f(x)|\,dx + \int_{R\le|x|\le 2R} |D_j\chi_R(x)|\,|f(x)|\,dx \\ &\le \int_{|x|\ge R} |D_j f(x)|\,dx + C\int_{|x|\ge R} |f(x)|\,dx. \end{aligned} \tag{3}$$

By enlarging R if necessary we may assume that

$$\int_{|x|\ge R} |D_j f(x)|\,dx < \frac{\varepsilon}{8}, \qquad C\int_{|x|\ge R} |f(x)|\,dx < \frac{\varepsilon}{8}.$$

Together with (**??**) and (**??**) this gives

$$\|f - g_R\|_{L^1} + \|D_j f - D_j g_R\|_{L^1} < \frac{\varepsilon}{4}.$$

Step 2: Smoothing by convolution. Let $\rho \in C_0^\infty(\mathbb{R}^n)$ be a standard mollifier with $\rho \ge 0$, $\int \rho = 1$ and set $\rho_\delta(x) = \delta^{-n}\rho(x/\delta)$. Define

$$f_\varepsilon := g_R * \rho_\delta.$$

Since g_R has compact support and ρ_δ has compact support, $f_\varepsilon \in C_0^\infty(\mathbb{R}^n) \subset C_0^1(\mathbb{R}^n)$.

Standard properties of mollifiers yield

$$\|g_R - g_R * \rho_\delta\|_{L^1} \to 0, \qquad \|D_j g_R - (D_j g_R) * \rho_\delta\|_{L^1} \to 0$$

as $\delta \to 0$. Hence we can choose $\delta > 0$ so small that

$$\|g_R - f_\varepsilon\|_{L^1} + \|D_j g_R - D_j f_\varepsilon\|_{L^1} < \frac{\varepsilon}{4}.$$

Step 3: Conclusion. By the triangle inequality,

$$\begin{aligned} \|f - f_\varepsilon\|_{L^1} &\le \|f - g_R\|_{L^1} + \|g_R - f_\varepsilon\|_{L^1}, \\ \|D_j f - D_j f_\varepsilon\|_{L^1} &\le \|D_j f - D_j g_R\|_{L^1} + \|D_j g_R - D_j f_\varepsilon\|_{L^1}. \end{aligned}$$

Each term on the right can be made $< \varepsilon/4$, so

$$\|f - f_\varepsilon\|_{L^1} + \|D_j f - D_j f_\varepsilon\|_{L^1} < \frac{\varepsilon}{2},$$

as required.

Exercise 3.7 (Fourier transform of a compactly supported L^1 function)

Suppose $f \in L^1(\mathbb{R}^n)$ with $\operatorname{supp} f \subset B_R(0)$ for some $R > 0$.

(a) Show that $\widehat{f} \in C^\infty(\mathbb{R}^n)$ and that for any multi–index α,

$$\sup_{\xi \in \mathbb{R}^n} \left| D^\alpha \widehat{f}(\xi) \right| \le R^{|\alpha|} \, \|f\|_{L^1(\mathbb{R}^n)}.$$

(b) Show that $\widehat{f}$ is real analytic, with an infinite radius of convergence, i.e.

$$\widehat{f}(\xi) = \sum_\alpha \frac{D^\alpha \widehat{f}(0)}{\alpha!} \, \xi^\alpha \qquad \text{for all } \xi \in \mathbb{R}^n.$$

(c) Show that if $\widehat{f}(\xi)$ vanishes on an open set, then it must vanish everywhere. *(Hint: use part (i) of Lemma 3.2.)*

You may assume the following version of Taylor's theorem: if $g \in C^{k+1}(B_r(0))$, then for $x \in B_r(0)$,

$$g(x) = \sum_{|\alpha| \le k} D^\alpha g(0) \, \frac{x^\alpha}{\alpha!} + \sum_{|\beta| = k+1} R_\beta(x) \, x^\beta,$$

where the remainder satisfies

$$|R_\beta(x)| \le \frac{1}{\beta!} \max_{|\alpha| = |\beta|} \max_{y \in B_r(0)} |D^\alpha g(y)|.$$

Suppose $f \in L^1(\mathbb{R}^n)$, with $\operatorname{supp} f \subset B_R(0)$ for some $R > 0$, and define

$$\widehat{f}(\xi) := \int_{\mathbb{R}^n} e^{-2\pi i x \cdot \xi} f(x) \, dx.$$

(a) Smoothness and derivative bounds. We first show that $\widehat{f} \in C^\infty(\mathbb{R}^n)$ and that, for any multi-index α,

$$\sup_{\xi \in \mathbb{R}^n} \left| D^\alpha \widehat{f}(\xi) \right| \le R^{|\alpha|} \, \|f\|_{L^1(\mathbb{R}^n)}.$$

Proof. Because f has compact support, for each multi-index α we may differentiate under the integral sign:

$$D^\alpha_\xi \widehat{f}(\xi) = \int_{\mathbb{R}^n} D^\alpha_\xi \Big(e^{-2\pi i x \cdot \xi} \Big) f(x) \, dx = \int_{\mathbb{R}^n} (-2\pi i x)^\alpha e^{-2\pi i x \cdot \xi} f(x) \, dx.$$

Hence

$$|D^\alpha \widehat{f}(\xi)| \le \int_{\mathbb{R}^n} |x^\alpha| \, |f(x)| \, dx.$$

On the support of f we have $|x_k| \le R$ for each coordinate, so $|x^\alpha| \le R^{|\alpha|}$. Therefore

$$|D^\alpha \widehat{f}(\xi)| \le R^{|\alpha|} \int_{\mathbb{R}^n} |f(x)| \, dx = R^{|\alpha|} \, \|f\|_{L^1(\mathbb{R}^n)},$$

and this upper bound is independent of ξ. Thus

$$\sup_{\xi\in\mathbb{R}^n} |D^\alpha \widehat{f}(\xi)| \le R^{|\alpha|}\, \|f\|_{L^1}.$$

Since each derivative is obtained as an L^1–limit under the integral, $D^\alpha \widehat{f}$ is continuous, and hence $\widehat{f} \in C^\infty(\mathbb{R}^n)$. □

(b) Real analyticity and infinite radius of convergence. We show that $\widehat{f}$ is real analytic with infinite radius of convergence, i.e.

$$\widehat{f}(\xi) = \sum_\alpha \frac{D^\alpha \widehat{f}(0)}{\alpha!}\, \xi^\alpha \qquad \text{for all } \xi \in \mathbb{R}^n.$$

Proof. Apply the multivariable Taylor theorem to

$$g(\xi) := \widehat{f}(\xi)$$

at the point $0 \in \mathbb{R}^n$. For $x \in B_r(0)$ we have

$$\widehat{f}(x) = \sum_{|\alpha|\le k} D^\alpha \widehat{f}(0)\, \frac{x^\alpha}{\alpha!} + \sum_{|\beta|=k+1} R_\beta(x)\, x^\beta,$$

where each remainder term satisfies

$$|R_\beta(x)| \le \frac{1}{\beta!} \max_{|\gamma|=|\beta|} \max_{y\in B_r(0)} |D^\gamma \widehat{f}(y)|.$$

Using the estimate from part (a) we obtain

$$|R_\beta(x)| \le \frac{1}{\beta!} R^{|\beta|}\, \|f\|_{L^1}.$$

Hence, for $|x| < r$,

$$\begin{aligned} \left| \sum_{|\beta|=k+1} R_\beta(x)\, x^\beta \right| &\le \sum_{|\beta|=k+1} \frac{R^{|\beta|}\|f\|_{L^1}}{\beta!}\, |x|^{|\beta|} \\ &\le \|f\|_{L^1} \sum_{m=k+1}^{\infty} \frac{(CR|x|)^m}{m!}, \end{aligned}$$

where $C > 0$ depends only on the dimension (via the number of multi-indices of length m). The last sum is the tail of an exponential series and therefore tends to 0 as $k \to \infty$, for every $x \in \mathbb{R}^n$ (no restriction on $|x|$).

Letting $k \to \infty$ we conclude that

$$\widehat{f}(\xi) = \sum_\alpha \frac{D^\alpha \widehat{f}(0)}{\alpha!}\, \xi^\alpha \qquad \text{for all } \xi \in \mathbb{R}^n,$$

so $\widehat{f}$ is real analytic with infinite radius of convergence. □

(c) Vanishing on an open set implies $\widehat{f} \equiv 0$. Suppose that $\widehat{f}(\xi) = 0$ for all ξ in some nonempty open set $\Omega \subset \mathbb{R}^n$. Choose $\xi_0 \in \Omega$ and let $B_r(\xi_0) \subset \Omega$ be a ball contained in Ω. On $B_r(\xi_0)$ we have $\widehat{f} \equiv 0$, and hence, for every multi-index α,

$$D^\alpha \widehat{f}(\xi_0) = 0.$$

By part (b), the Taylor expansion of $\widehat{f}$ at ξ_0 has infinite radius of convergence:

$$\widehat{f}(\xi) = \sum_\alpha \frac{D^\alpha \widehat{f}(\xi_0)}{\alpha!} (\xi - \xi_0)^\alpha \qquad \text{for all } \xi \in \mathbb{R}^n.$$

Since all derivatives at ξ_0 vanish, every coefficient in this series is zero, and consequently $\widehat{f}(\xi) = 0$ for all $\xi \in \mathbb{R}^n$.

Thus, if $\widehat{f}$ vanishes on any nonempty open set, it must vanish identically. □

Examples illustrating Exercise 3.7

Example 1 (Indicator of an interval). In one dimension, let

$$f(x) = \mathbf{1}_{[-1,1]}(x).$$

Then $f \in L^1(\mathbb{R})$ and $\operatorname{supp} f \subset [-1,1] = B_1(0)$, so $R = 1$ and $\|f\|_{L^1} = 2$. With the Fourier transform

$$\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-2\pi i x\xi} f(x)\, dx,$$

we compute

$$\widehat{f}(\xi) = \int_{-1}^{1} e^{-2\pi i x\xi}\, dx = \frac{\sin(2\pi\xi)}{\pi\xi},$$

where $\widehat{f}(0)$ is defined by continuity, giving $\widehat{f}(0) = 2$.

For any $k \in \mathbb{N}$ we have

$$\widehat{f}^{(k)}(\xi) = \int_{-1}^{1} (-2\pi i x)^k e^{-2\pi i x\xi}\, dx,$$

so that

$$\left|\widehat{f}^{(k)}(\xi)\right| \le \int_{-1}^{1} |2\pi x|^k\, dx \le (2\pi)^k \int_{-1}^{1} |x|^k\, dx \le (2\pi)^k \cdot 2.$$

This is of the form

$$\sup_{\xi\in\mathbb{R}} \left|\widehat{f}^{(k)}(\xi)\right| \le C^k \|f\|_{L^1},$$

and illustrates the bound from part (a) of Exercise 3.7 with $R = 1$.

Around $\xi = 0$ we expand

$$e^{-2\pi i x\xi} = \sum_{m=0}^{\infty} \frac{(-2\pi i x\xi)^m}{m!},$$

and since $|x| \le 1$ on the support of f, we may integrate term-by-term:

$$\widehat{f}(\xi) = \sum_{m=0}^{\infty} \frac{(-2\pi i\xi)^m}{m!} \int_{-1}^{1} x^m \, dx.$$

The inner integral is

$$\int_{-1}^{1} x^m \, dx = \begin{cases} 0, & m \text{ odd}, \\ \dfrac{2}{m+1}, & m \text{ even}, \end{cases}$$

so

$$\widehat{f}(\xi) = \sum_{k=0}^{\infty} \frac{(-2\pi i\xi)^{2k}}{(2k)!} \, \frac{2}{2k+1}.$$

This is a power series in ξ which converges for all $\xi \in \mathbb{R}$, and precisely recovers $\widehat{f}(\xi) = \sin(2\pi\xi)/(\pi\xi)$. It is a concrete instance of the statement in part (b) that $\widehat{f}$ is real analytic with infinite radius of convergence.

Note also that $\widehat{f}(\xi)$ has only isolated zeros (at $\xi = k/2$, $k \in \mathbb{Z} \setminus \{0\}$), so it does *not* vanish on any nonempty open interval. This is consistent with part (c): if $\widehat{f}$ vanished on an open set, analyticity would force all derivatives at a point of that set to be zero and hence $\widehat{f} \equiv 0$.

Example 2 (Smooth bump function). Let

$$f(x) = \begin{cases} e^{-\frac{1}{1-x^2}}, & |x| < 1, \\ 0, & |x| \ge 1, \end{cases} \qquad f \in C_c^{\infty}(\mathbb{R}),\ \operatorname{supp}(f) \subset [-1, 1].$$

Its Fourier transform is

$$\widehat{f}(\xi) = \int_{-1}^{1} e^{-2\pi i x\xi} \, e^{-\frac{1}{1-x^2}} \, dx,$$

which does not admit a simple closed form, but it satisfies all conclusions of Exercise 3.7:

(a) **Differentiating under the integral.** For every integer $k \ge 0$,

$$\widehat{f}^{(k)}(\xi) = \int_{-1}^{1} (-2\pi i x)^k e^{-2\pi i x\xi} \, e^{-\frac{1}{1-x^2}} \, dx.$$

Since $|x| \le 1$,

$$\left|\widehat{f}^{(k)}(\xi)\right| \le \int_{-1}^{1} |2\pi x|^k e^{-\frac{1}{1-x^2}} \, dx \le C^k \, \|f\|_{L^1}, \qquad \sup_{\xi \in \mathbb{R}} \left|\widehat{f}^{(k)}(\xi)\right| < \infty.$$

This matches exactly the estimate

$$\sup_{\xi \in \mathbb{R}^n} |D^{\alpha} \widehat{f}(\xi)| \le R^{|\alpha|} \, \|f\|_{L^1}, \qquad R = 1.$$

(b) **Real analyticity.** Expanding the exponential,

$$e^{-2\pi i x\xi} = \sum_{m=0}^{\infty} \frac{(-2\pi i x\xi)^m}{m!},$$

and since $|x| \le 1$ on $\mathrm{supp}(f)$, the series converges absolutely for all ξ. Thus the integral may be computed term-by-term:

$$\widehat{f}(\xi) = \sum_{m=0}^{\infty} \frac{(-2\pi i\xi)^m}{m!} \int_{-1}^{1} x^m \, e^{-\frac{1}{1-x^2}} \, dx.$$

This is a power series in ξ with infinite radius of convergence, exactly as stated in Exercise 3.7(b).

(c) **Vanishing on an open set implies $\widehat{f} \equiv 0$.** If $\widehat{f}$ vanished on an open interval, real analyticity would force all derivatives at a point of that interval to vanish, making all coefficients in the Taylor series zero. Hence $\widehat{f} \equiv 0$. But

$$\widehat{f}(0) = \int_{-1}^{1} f(x) \, dx > 0,$$

so this cannot happen for the bump function. This fully matches Exercise 3.7(c).

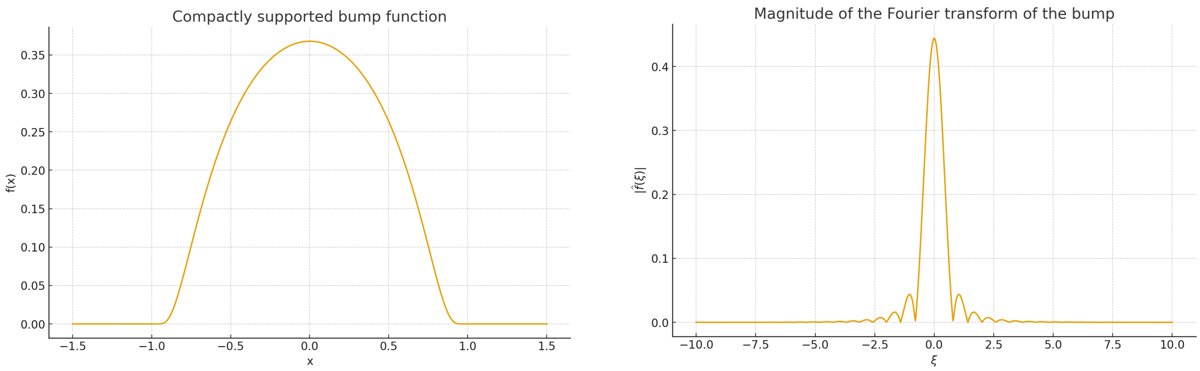


Figure 10: Left: compactly supported bump function $f(x) = e^{-1/(1-x^2)}$ for $|x| < 1$ and 0 otherwise. Right: numerically computed magnitude of its Fourier transform $|\widehat{f}(\xi)|$, illustrating smooth, rapidly decaying behaviour.

Exercise 4.1

Consider the ODE

$$-\phi''(x) + \phi(x) = f(x), \qquad f : \mathbb{R} \to \mathbb{C}. \tag{4}$$

(a) Assume $f \in \mathcal{S}(\mathbb{R})$. Taking the Fourier transform of both sides gives

$$\widehat{-\phi'' + \phi}(\xi) = -\widehat{\phi''}(\xi) + \widehat{\phi}(\xi).$$

Using the standard identities

$$\widehat{g'}(\xi) = (2\pi i\xi)\,\widehat{g}(\xi), \qquad \widehat{g''}(\xi) = -(2\pi\xi)^2\widehat{g}(\xi) = -4\pi^2\xi^2\widehat{g}(\xi),$$

we obtain

$$-\widehat{\phi''}(\xi) + \widehat{\phi}(\xi) = 4\pi^2\xi^2\widehat{\phi}(\xi) + \widehat{\phi}(\xi) = (4\pi^2\xi^2+1)\widehat{\phi}(\xi).$$

Thus the transformed equation is

$$(4\pi^2\xi^2+1)\,\widehat{\phi}(\xi) = \widehat{f}(\xi),$$

and hence

$$\boxed{\widehat{\phi}(\xi) = \frac{\widehat{f}(\xi)}{1+4\pi^2\xi^2}.}$$

Since the multiplier $(1+4\pi^2\xi^2)^{-1}$ is smooth and rapidly decaying, multiplication by it preserves the Schwartz class. Thus $\phi \in \mathcal{S}(\mathbb{R})$.

The homogeneous equation $-\psi''+\psi = 0$ has solutions $\psi(x) = Ce^x + De^{-x}$, none of which lie in $\mathcal{S}$ unless $C = D = 0$. Hence the solution in $\mathcal{S}$ is unique.

(b) Define the Green's function

$$G(x) = \begin{cases} \frac{1}{2}e^{x}, & x<0, \\ \frac{1}{2}e^{-x}, & x \geq 0. \end{cases}$$

A direct calculation yields

$$\widehat{G}(\xi) = \frac{1}{1+4\pi^2\xi^2}.$$

Thus from part (a),

$$\widehat{\phi}(\xi) = \widehat{f}(\xi)\,\widehat{G}(\xi) = \widehat{f*G}\,(\xi).$$

By injectivity of the Fourier transform,

$$\boxed{\phi(x) = (f*G)(x) = \int_{\mathbb{R}} f(y)\,G(x-y)\,dy.}$$

Exercise 4.1

Consider the ODE

$$-\phi''(x) + \phi(x) = f(x), \qquad f:\mathbb{R}\to\mathbb{C}. \tag{5}$$

(a) Consider the ODE

$$-\phi''(x) + \phi(x) = f(x), \qquad f:\mathbb{R}\to\mathbb{C}. \tag{6}$$

(a) Assume $f \in \mathcal{S}(\mathbb{R})$. Taking the Fourier transform of both sides gives

$$\widehat{-\phi'' + \phi}(\xi) = -\widehat{\phi''}(\xi) + \widehat{\phi}(\xi).$$

Using the standard identities

$$\widehat{g'}(\xi) = (2\pi i\xi)\,\widehat{g}(\xi), \qquad \widehat{g''}(\xi) = -(2\pi\xi)^2\widehat{g}(\xi) = -4\pi^2\xi^2\widehat{g}(\xi),$$

we obtain

$$-\widehat{\phi''}(\xi) + \widehat{\phi}(\xi) = 4\pi^2\xi^2\widehat{\phi}(\xi) + \widehat{\phi}(\xi) = (4\pi^2\xi^2 + 1)\widehat{\phi}(\xi).$$

Thus the transformed equation is

$$(4\pi^2\xi^2 + 1)\,\widehat{\phi}(\xi) = \widehat{f}(\xi),$$

and hence

$$\boxed{\widehat{\phi}(\xi) = \frac{\widehat{f}(\xi)}{1 + 4\pi^2\xi^2}.}$$

Since the multiplier $(1 + 4\pi^2\xi^2)^{-1}$ is smooth and rapidly decaying, multiplication by it preserves the Schwartz class. Thus $\phi \in \mathcal{S}(\mathbb{R})$.

The homogeneous equation $-\psi'' + \psi = 0$ has solutions $\psi(x) = Ce^x + De^{-x}$, none of which lie in $\mathcal{S}$ unless $C = D = 0$. Hence the solution in $\mathcal{S}$ is unique.

It comes from the Fourier transform of derivatives. Start with

$$-\phi''(x) + \phi(x) = f(x).$$

Taking the Fourier transform in x and using linearity,

$$\widehat{-\phi'' + \phi}(\xi) = -\widehat{\phi''}(\xi) + \widehat{\phi}(\xi).$$

We use the standard identity (with the convention $\widehat{g}(\xi) = \int_{\mathbb{R}} e^{-2\pi i x\xi} g(x)\,dx$):

$$\widehat{g'}(\xi) = (2\pi i\xi)\,\widehat{g}(\xi).$$

Applying it twice gives

$$\widehat{g''}(\xi) = (2\pi i\xi)^2\widehat{g}(\xi) = -4\pi^2\xi^2\widehat{g}(\xi).$$

Taking $g = \phi$, we obtain

$$\widehat{\phi''}(\xi) = -4\pi^2\xi^2\widehat{\phi}(\xi).$$

Substituting this into the transformed equation,

$$\widehat{-\phi'' + \phi}(\xi) = -\widehat{\phi''}(\xi) + \widehat{\phi}(\xi) = -\Big(-4\pi^2\xi^2\widehat{\phi}(\xi)\Big) + \widehat{\phi}(\xi) = (4\pi^2\xi^2 + 1)\widehat{\phi}(\xi).$$

(b) Define the Green's function

$$G(x) = \begin{cases} \frac{1}{2}e^{x}, & x < 0, \\ \frac{1}{2}e^{-x}, & x \geq 0. \end{cases}$$

A direct calculation yields

$$\widehat{G}(\xi) = \frac{1}{1 + 4\pi^2\xi^2}.$$

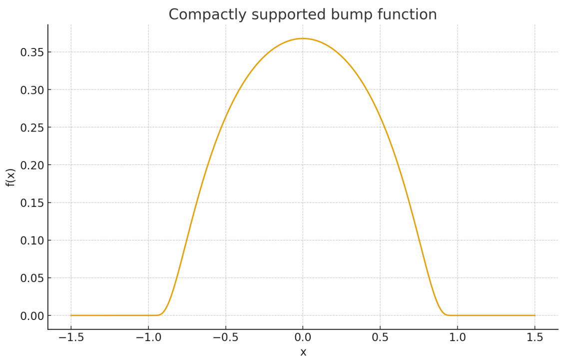
Thus

$$\widehat{\phi}(\xi) = \widehat{f}(\xi)\,\widehat{G}(\xi) = \widehat{f * G}(\xi),$$

and by injectivity of the Fourier transform,

$$\boxed{\phi(x) = (f * G)(x) = \int_{\mathbb{R}} f(y)\,G(x-y)\,dy.}$$

Illustrations



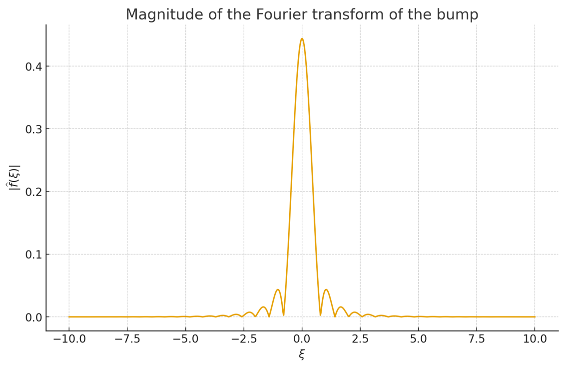


Figure 11: Left: $f(x) = e^{-1/(1-x^2)}$ on $|x| < 1$. Right: numerical magnitude of $\widehat{f}(\xi)$.

1. Compactly supported bump function and its Fourier transform.

2. Green's function for $-\phi'' + \phi = f$.

3. Example convolution $f * G$.

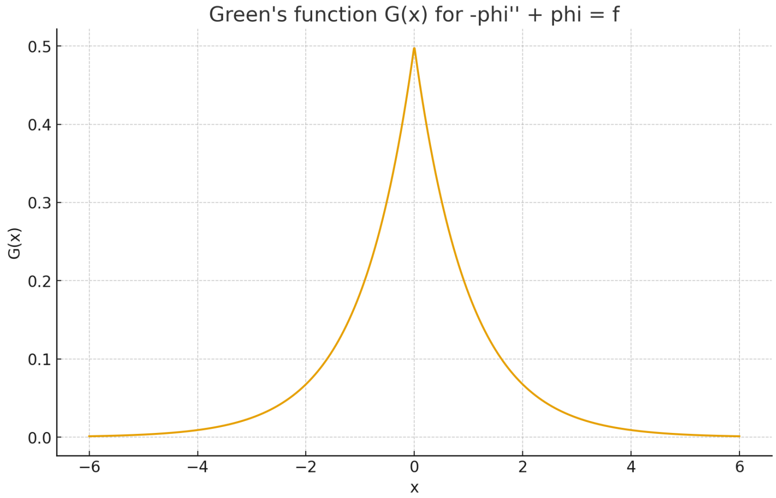


Figure 12: Green's function $G(x) = \frac{1}{2}e^{-|x|}$.

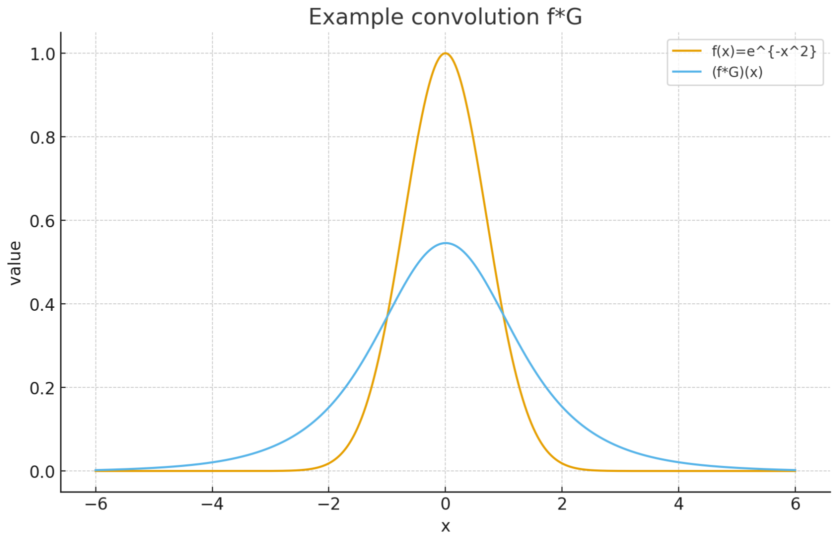


Figure 13: Convolution of G with $f(x) = e^{-x^2}$.

Exercise 4.2 (Radial Fourier Transform in $\mathbb{R}^3$)

Suppose $f \in L^1(\mathbb{R}^3)$ is a radial function, i.e. $f(Rx) = f(x)$ whenever $R \in SO(3)$ is a rotation.

(a) Show that $\widehat{f}$ is radial.

(b) Suppose that $\xi = (0, 0, \zeta)$. By writing the Fourier integral in spherical coordinates, show that
$$\widehat{f}(\xi) = \int_{r=0}^{\infty} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} f(r)\, e^{-i\zeta r \cos\theta}\, r^2 \sin\theta \, d\phi \, d\theta \, dr.$$

(c) Making the substitution $s = \cos\theta$, and using the fact that $\widehat{f}$ is radial, deduce that
$$\widehat{f}(\xi) = 4\pi \int_0^{\infty} f(r)\, \frac{\sin(r|\xi|)}{r|\xi|}\, r^2 \, dr$$
for any $\xi \in \mathbb{R}^3$.

Suppose $f \in L^1(\mathbb{R}^3)$ is a radial function, i.e. $f(Rx) = f(x)$ for all rotations $R \in SO(3)$.

(a) To show that $\widehat{f}$ is radial, take any $R \in SO(3)$:
$$\widehat{f}(R\xi) = \int_{\mathbb{R}^3} f(x) e^{-2\pi i x \cdot (R\xi)} \, dx.$$
With the substitution $x = Ry$ (Jacobian $= 1$),
$$\widehat{f}(R\xi) = \int_{\mathbb{R}^3} f(Ry) e^{-2\pi i (Ry) \cdot (R\xi)} \, dy.$$
Since f is radial, $f(Ry) = f(y)$, and rotations preserve dot products: $(Ry) \cdot (R\xi) = y \cdot \xi$. Hence
$$\widehat{f}(R\xi) = \int_{\mathbb{R}^3} f(y) e^{-2\pi i y \cdot \xi} \, dy = \widehat{f}(\xi),$$
showing that $\widehat{f}$ is radial.

(b) Since $\widehat{f}$ is radial, we may assume $\xi = (0, 0, \zeta)$ with $\zeta = |\xi|$. In spherical coordinates $x = (r, \theta, \phi)$, one has
$$x \cdot \xi = r\zeta \cos\theta, \qquad dx = r^2 \sin\theta \, dr \, d\theta \, d\phi, \qquad f(x) = f(r).$$
Thus
$$\widehat{f}(\xi) = \int_0^{\infty} \int_0^{\pi} \int_0^{2\pi} f(r) e^{-i\zeta r \cos\theta}\, r^2 \sin\theta \; d\phi \, d\theta \, dr.$$

(c) Substituting $s = \cos\theta$ gives

$$\int_0^\pi e^{-i\zeta r\cos\theta}\sin\theta\,d\theta = \int_{-1}^1 e^{-i\zeta rs}\,ds = 2\frac{\sin(\zeta r)}{\zeta r}.$$

Since $\int_0^{2\pi} d\phi = 2\pi$, we obtain

$$\widehat{f}(\xi) = 4\pi\int_0^\infty f(r)\,\frac{\sin(r|\xi|)}{r|\xi|}\,r^2\,dr.$$

Thus $\widehat{f}$ is radial and given by

$$\boxed{\widehat{f}(\xi) = 4\pi\int_0^\infty f(r)\,\frac{\sin(r|\xi|)}{r|\xi|}\,r^2\,dr.}$$

Theory: Radial Fourier transform in $\mathbb{R}^3$

A function $f : \mathbb{R}^3 \to \mathbb{C}$ is called *radial* if there is $F : [0,\infty) \to \mathbb{C}$ such that

$$f(x) = F(|x|) \qquad \text{for all } x \in \mathbb{R}^3.$$

Equivalently, f is invariant under all rotations:

$$f(Rx) = f(x) \qquad \text{for all } R \in SO(3).$$

Fourier transform and rotations. The Fourier transform in $\mathbb{R}^3$ is

$$\widehat{f}(\xi) = \int_{\mathbb{R}^3} f(x)\,e^{-2\pi i x\cdot\xi}\,dx.$$

For any rotation $R \in SO(3)$, one has

$$\widehat{f\circ R}(\xi) = \widehat{f}(R^{-1}\xi).$$

Indeed,

$$\widehat{f\circ R}(\xi) = \int_{\mathbb{R}^3} f(Rx)e^{-2\pi i x\cdot\xi}\,dx = \int_{\mathbb{R}^3} f(y)e^{-2\pi i (Ry)\cdot\xi}\,dy = \int_{\mathbb{R}^3} f(y)e^{-2\pi i y\cdot(R^{-1}\xi)}\,dy = \widehat{f}(R^{-1}\xi),$$

where we used the change of variables $x = Ry$ and the fact that rotations preserve dot products. If f is radial, then $f \circ R = f$ for all R, hence

$$\widehat{f}(R\xi) = \widehat{f}(\xi) \qquad \forall R \in SO(3),$$

and thus $\widehat{f}$ is radial as well.

Spherical coordinates. Since $\widehat{f}$ is radial, its value at ξ depends only on $|\xi|$. Given $\xi \in \mathbb{R}^3$ with $|\xi| > 0$, we choose a rotation R such that

$$R\xi = (0, 0, |\xi|).$$

Using radiality of $\widehat{f}$ we may compute $\widehat{f}(R\xi)$ instead of $\widehat{f}(\xi)$; in spherical coordinates $x = (r, \theta, \phi)$ we have

$$x \cdot R\xi = r|\xi| \cos\theta, \qquad dx = r^2 \sin\theta \, dr \, d\theta \, d\phi, \qquad f(x) = f(r).$$

Therefore,

$$\widehat{f}(\xi) = \int_0^\infty \int_0^\pi \int_0^{2\pi} f(r) \, e^{-ir|\xi| \cos\theta} \, r^2 \sin\theta \, d\phi \, d\theta \, dr. \tag{7}$$

Angular integration and the 3D radial kernel. The ϕ–integral is trivial:

$$\int_0^{2\pi} d\phi = 2\pi.$$

For the θ–integral we use the substitution $s = \cos\theta$, $ds = -\sin\theta \, d\theta$, which gives

$$\int_0^\pi e^{-ir|\xi| \cos\theta} \sin\theta \, d\theta = \int_{-1}^1 e^{-ir|\xi| s} \, ds = 2 \, \frac{\sin(r|\xi|)}{r|\xi|}.$$

Inserting these into (**??**) yields

$$\widehat{f}(\xi) = 4\pi \int_0^\infty f(r) \, \frac{\sin(r|\xi|)}{r|\xi|} \, r^2 \, dr,$$

which is the standard formula for the Fourier transform of a radial L^1-function in $\mathbb{R}^3$.

Remark (connection with Bessel functions). In general dimension n, the Fourier transform of a radial function involves Bessel functions and can be written as

$$\widehat{f}(\xi) = (2\pi)^{n/2} |\xi|^{-(n/2-1)} \int_0^\infty f(r) \, J_{n/2-1}(2\pi r|\xi|) \, r^{n/2} \, dr.$$

For $n = 3$ this reduces to the above formula, since the corresponding Bessel function combination is

$$\frac{\sin(r|\xi|)}{r|\xi|},$$

the spherical Bessel function of order 0.

Examples of radial Fourier transforms in $\mathbb{R}^3$

Example 1 (Indicator of a ball). Let

$$f(x) = \mathbf{1}_{\{|x| \le a\}}(x), \qquad a > 0.$$

Then f is radial with $f(r) = 1$ for $0 \le r \le a$ and 0 otherwise, and the radial Fourier transform formula gives

$$\widehat{f}(\xi) = 4\pi \int_0^a \frac{\sin(r|\xi|)}{r|\xi|} r^2\,dr = \frac{4\pi}{|\xi|} \int_0^a r\sin(r|\xi|)\,dr.$$

Using

$$\int r\sin(\rho r)\,dr = -\frac{r\cos(\rho r)}{\rho} + \frac{\sin(\rho r)}{\rho^2} + C,$$

we obtain, with $\rho = |\xi|$,

$$\int_0^a r\sin(\rho r)\,dr = -\frac{a\cos(\rho a)}{\rho} + \frac{\sin(\rho a)}{\rho^2}.$$

Hence

$$\widehat{f}(\xi) = 4\pi\left(\frac{\sin(a|\xi|)}{|\xi|^3} - \frac{a\cos(a|\xi|)}{|\xi|^2}\right), \qquad \xi \neq 0.$$

The value at $\xi = 0$ is given by continuity and equals the volume of the ball, $\widehat{f}(0) = \frac{4\pi a^3}{3}$.

Example 2 (Gaussian). Consider the radial Gaussian

$$f(x) = e^{-|x|^2}, \qquad x \in \mathbb{R}^3.$$

Using the one-dimensional identity

$$\int_{\mathbb{R}} e^{-x^2} e^{-ix\xi}\,dx = \sqrt{\pi}\,e^{-\xi^2/4},$$

and the factorisation $e^{-|x|^2} = e^{-x_1^2}e^{-x_2^2}e^{-x_3^2}$, we obtain

$$\widehat{f}(\xi) = \left(\sqrt{\pi}e^{-\xi_1^2/4}\right)\left(\sqrt{\pi}e^{-\xi_2^2/4}\right)\left(\sqrt{\pi}e^{-\xi_3^2/4}\right) = \pi^{3/2}e^{-|\xi|^2/4}.$$

Thus

$$\boxed{\widehat{e^{-|x|^2}}(\xi) = \pi^{3/2}e^{-|\xi|^2/4},}$$

which is manifestly radial. This is consistent with the radial integral

$$\widehat{f}(\xi) = 4\pi \int_0^\infty e^{-r^2}\frac{\sin(r|\xi|)}{r|\xi|} r^2\,dr.$$

Example 3 (Yukawa kernel). Let

$$f(x) = \frac{e^{-|x|}}{|x|}, \qquad x \in \mathbb{R}^3,$$

so that $f(r) = e^{-r}/r$ is radial and integrable. Then

$$\widehat{f}(\xi) = 4\pi \int_0^\infty \frac{e^{-r}}{r}\frac{\sin(r|\xi|)}{r|\xi|} r^2\,dr = 4\pi\frac{1}{|\xi|}\int_0^\infty e^{-r}\sin(r|\xi|)\,dr.$$

Using the Laplace transform identity

$$\int_0^\infty e^{-r} \sin(\rho r)\, dr = \frac{\rho}{1+\rho^2},$$

with $\rho = |\xi|$, we find

$$\widehat{f}(\xi) = \frac{4\pi}{1+|\xi|^2}.$$

This radial function is (up to constants) the Fourier symbol of the resolvent $(-\Delta + 1)^{-1}$.

Figures: Yukawa Kernel and its Fourier Transform

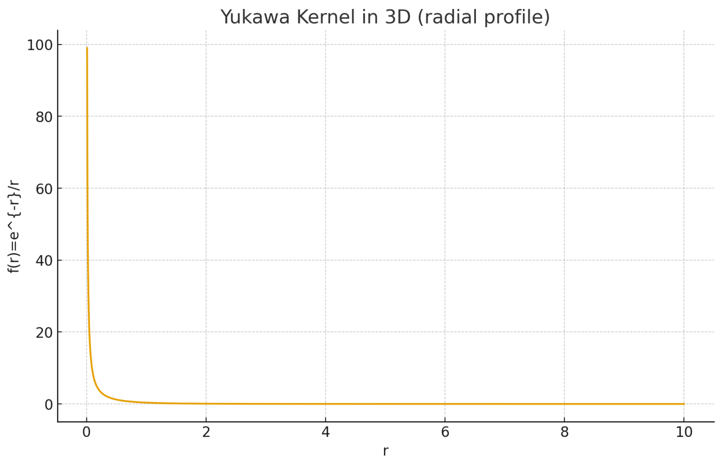


Figure 14: Radial Yukawa kernel in $\mathbb{R}^3$: $f(r) = e^{-r}/r$. The function decays exponentially and has a singularity of order $1/r$ at $r = 0$.

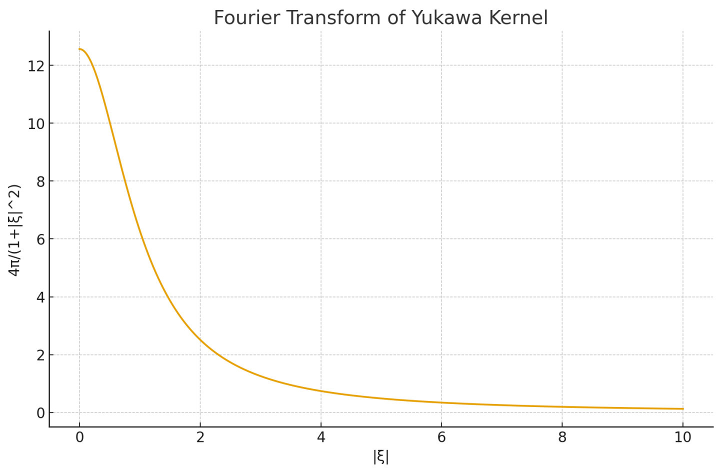


Figure 15: Fourier transform of the Yukawa kernel: $\widehat{f}(\xi) = \dfrac{4\pi}{1+|\xi|^2}$. This is the symbol of the resolvent $(-\Delta + 1)^{-1}$.

Comparison: Coulomb Kernel vs. Yukawa Kernel

The Coulomb kernel in $\mathbb{R}^3$ is

$$K_{\mathrm{C}}(x) = \frac{1}{|x|},$$

and the Yukawa kernel is

$$K_{\mathrm{Y}}(x) = \frac{e^{-|x|}}{|x|}.$$

Both are radial functions of $r = |x|$. The Yukawa kernel introduces exponential screening of the long-range $1/r$ interaction.

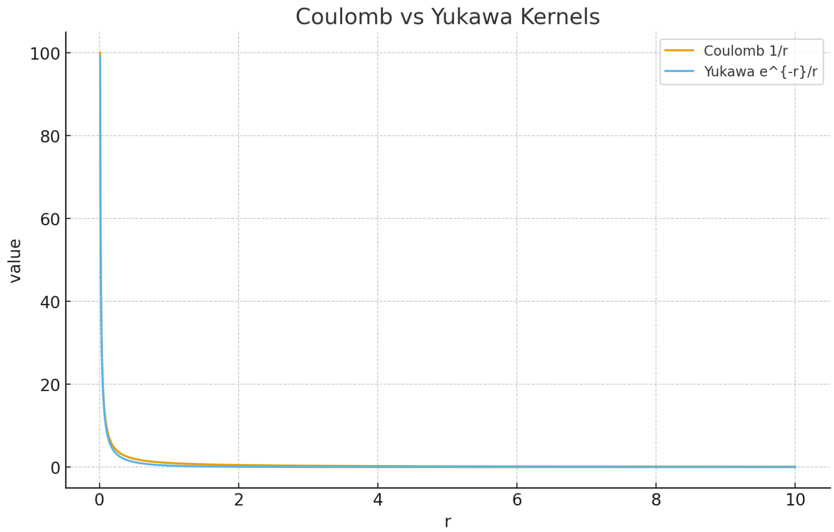


Figure 16: Linear-scale comparison of the Coulomb kernel $1/r$ and the Yukawa kernel e^{-r}/r. The Yukawa potential coincides with Coulomb near $r = 0$ but decays exponentially for large r.

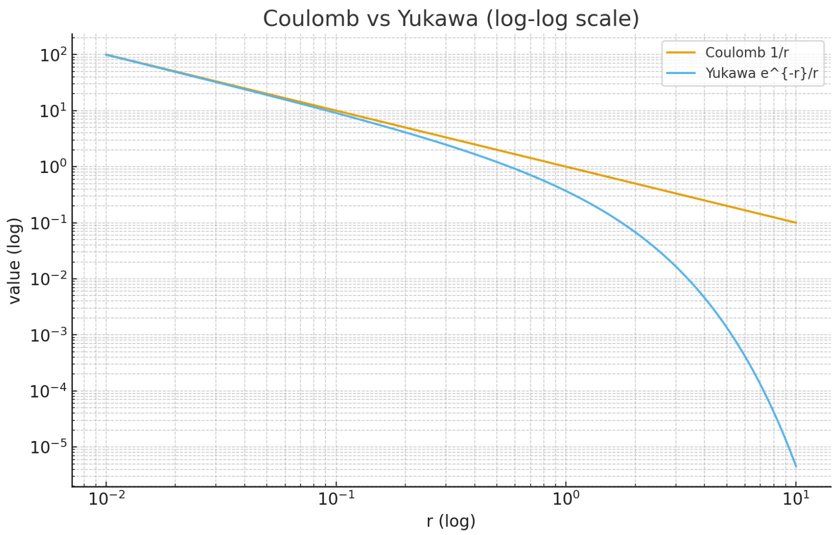


Figure 17: Log–log comparison. The Coulomb kernel exhibits pure algebraic decay r^{-1}, while the Yukawa kernel decays as e^{-r}/r, much faster at large distances.

Exercise 4.3 (Fourier–Plancherel Transform)

Suppose $f, g \in L^2(\mathbb{R}^n)$, and denote the Fourier–Plancherel transform by $\mathcal{F}$. You may assume any results already established for the classical Fourier transform.

(a) Show that

$$(f, g) = \frac{1}{(2\pi)^n}\Big(\mathcal{F}[g], \mathcal{F}[g]\Big).$$

(b) Recall that $\check{f}(y) = f(-y)$. Show that

$$\mathcal{F}[\mathcal{F}[f]] = (2\pi)^n\, \check{f}.$$

Hence, or otherwise, deduce that $\mathcal{F} : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is a bijection, and that $\mathcal{F}^{-1} : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is a bounded linear map.

(c) Show that

$$\mathcal{F}[f](\xi) = \lim_{R\to\infty} \int_{B_R(0)} f(x)e^{-ix\cdot\xi}\, dx,$$

with convergence in $L^2(\mathbb{R}^n)$.

(d) Suppose that $f \in C^1(\mathbb{R}^n)$ and $f, D_j f \in L^2(\mathbb{R}^n)$. Show that $\xi_j\, \mathcal{F}[f](\xi) \in L^2(\mathbb{R}^n)$ and

$$\mathcal{F}[D_j f](\xi) = i\xi_j \mathcal{F}[f](\xi).$$

(e) For $x \in \mathbb{R}$ let $f(x) = \dfrac{\sin x}{x}$.

i. Show that $f \in L^2(\mathbb{R})$.

ii. Show that

$$\mathcal{F}[f](\xi) = \begin{cases} \pi, & -1 < \xi < 1, \\ 0, & |\xi| \geq 1. \end{cases}$$

(f) i. Show that for all $x \in \mathbb{R}^n$,

$$|f * g(x)| \leq \|f\|_{L^2}\|g\|_{L^2}.$$

ii. Show that $f * g \in C^0(\mathbb{R}^n)$ and that

$$f * g = \mathcal{F}^{-1}\Big[\widehat{f}\,\widehat{g}\Big],$$

where

$$\mathcal{F}^{-1}[h](x) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} h(\xi)\, e^{ix\cdot\xi}\, d\xi.$$

Hint for parts (a), (b), (d), (f): approximate by Schwartz functions.

Exercise 4.3 (Fourier–Plancherel Transform)

Suppose $f, g \in L^2(\mathbb{R}^n)$, and denote the Fourier–Plancherel transform by $\mathcal{F}$. You may assume any results already established for the classical Fourier transform.

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$$\mathcal{F}[\mathcal{F}[f]] = (2\pi)^n\, \check{f}.$$

Hence, or otherwise, deduce that $\mathcal{F} : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is a bijection, and that $\mathcal{F}^{-1} : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is a bounded linear map.

(c) Show that
$$\mathcal{F}[f](\xi) = \lim_{R\to\infty} \int_{B_R(0)} f(x) e^{-ix\cdot\xi}\, dx,$$
with convergence in $L^2(\mathbb{R}^n)$.

(d) Suppose that $f \in C^1(\mathbb{R}^n)$ and $f, D_j f \in L^2(\mathbb{R}^n)$. Show that $\xi_j\, \mathcal{F}[f](\xi) \in L^2(\mathbb{R}^n)$ and
$$\mathcal{F}[D_j f](\xi) = i\xi_j \mathcal{F}[f](\xi).$$

(e) For $x \in \mathbb{R}$ let $f(x) = \dfrac{\sin x}{x}$.

i. Show that $f \in L^2(\mathbb{R})$.

ii. Show that
$$\mathcal{F}[f](\xi) = \begin{cases} \pi, & -1 < \xi < 1, \\ 0, & |\xi| \geq 1. \end{cases}$$

(f) i. Show that for all $x \in \mathbb{R}^n$,
$$|f * g(x)| \leq \|f\|_{L^2}\|g\|_{L^2}.$$

ii. Show that $f * g \in C^0(\mathbb{R}^n)$ and that
$$f * g = \mathcal{F}^{-1}\Big[\widehat{f}\,\widehat{g}\Big],$$
where
$$\mathcal{F}^{-1}[h](x) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} h(\xi)\, e^{ix\cdot\xi}\, d\xi.$$

Hint for parts (a), (b), (d), (f): approximate by Schwartz functions.

Theory: Relation between $\mathcal{S}(\mathbb{R}^n)$ and $L^2(\mathbb{R}^n)$

We recall that the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ consists of all smooth functions $f \in C^\infty(\mathbb{R}^n)$ such that for every pair of multi-indices α, β,

$$\sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta f(x)| < \infty.$$

Thus f and all its derivatives decay faster than any polynomial. This leads to four fundamental relations between $\mathcal{S}$, L^2, and the Fourier–Plancherel transform.

1. Inclusion $\mathcal{S} \subset L^2$

From the defining estimate, for each $N \in \mathbb{N}$ there exists $C_N > 0$ such that

$$|f(x)| \leq C_N (1 + |x|)^{-N}.$$

Choosing $N > \frac{n}{2}$ yields

$$\int_{\mathbb{R}^n} |f(x)|^2 \, dx \leq C_N^2 \int_{\mathbb{R}^n} (1 + |x|)^{-2N} dx < \infty.$$

Hence

$$\mathcal{S}(\mathbb{R}^n) \subset L^2(\mathbb{R}^n),$$

and the inclusion map is continuous with respect to the natural topologies.

2. Density of $\mathcal{S}$ in L^2

A key property is that $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$: for every $f \in L^2$ there exists $(f_k) \subset \mathcal{S}$ such that $\|f_k - f\|_{L^2} \to 0$.

One standard construction proceeds as follows:

1. Truncate f to a large ball: $f^{(R)}(x) := f(x)\mathbf{1}_{\{|x| \leq R\}}$. Then $f^{(R)} \to f$ in L^2 as $R \to \infty$.
2. Mollify: choose a smooth compactly supported mollifier η_ε, and set $f^{(R,\varepsilon)} = f^{(R)} * \eta_\varepsilon$. Then $f^{(R,\varepsilon)} \in C_c^\infty(\mathbb{R}^n)$ and $f^{(R,\varepsilon)} \to f^{(R)}$ in L^2 as $\varepsilon \to 0$.

Since $C_c^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$, this proves that

$$\overline{\mathcal{S}(\mathbb{R}^n)}^{L^2} = L^2(\mathbb{R}^n).$$

3. Extension of the Fourier transform to L^2 (Fourier–Plancherel theorem)

On the Schwartz space the Fourier transform is given by

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} \, dx,$$

and satisfies:

- $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ is a bijection.
- *Parseval identity:*

$$\int_{\mathbb{R}^n} f(x)\overline{g(x)}\,dx = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} \widehat{f}(\xi)\,\overline{\widehat{g}(\xi)}\,d\xi.$$

- Therefore,

$$\|\widehat{f}\|_{L^2} = (2\pi)^{n/2}\|f\|_{L^2}.$$

This shows that $\mathcal{F} : \mathcal{S} \to L^2$ is an isometry up to $(2\pi)^{n/2}$. Since $\mathcal{S}$ is dense in L^2, there exists a unique extension

$$\mathcal{F} : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$$

that is bounded and satisfies the same norm identity. The extension is called the *Fourier–Plancherel transform.*

On $\mathcal{S}$ we have the identity

$$\mathcal{F}[\mathcal{F}[f]](\xi) = (2\pi)^n f(-\xi),$$

and by continuity this holds for all $f \in L^2$. In particular $\mathcal{F}$ is invertible and its inverse is the inverse Fourier transform.

4. Why approximation by Schwartz functions works

For many identities one proves the formula first for $f \in \mathcal{S}(\mathbb{R}^n)$, where the Fourier transform is defined by an absolutely convergent integral and all operations are legal. Then, using:

- the density $\mathcal{S} \subset L^2$,
- the fact that $\mathcal{F} : L^2 \to L^2$ is bounded,

one may pass to the limit and obtain the corresponding identity for general $f \in L^2$.

More precisely, if $f_k \in \mathcal{S}$ and $f_k \to f$ in L^2, and if an identity

$$\mathcal{G}(f_k) = \mathcal{H}(f_k)$$

holds for all k, and the operators $\mathcal{G}$ and $\mathcal{H}$ are continuous on L^2, then the same identity holds for f. This mechanism underlies the proofs of Parseval's identity, the differentiation rule $\mathcal{F}[D_j f] = i\xi_j \widehat{f}$, the convolution theorem, and other statements used in Fourier–Plancherel theory.

Theory: Tempered Distributions and the Fourier Transform on $\mathcal{S}'(\mathbb{R}^n)$

The Schwartz space $\mathcal{S}(\mathbb{R}^n)$ consists of rapidly decaying smooth functions. Its topological dual,

$$\mathcal{S}'(\mathbb{R}^n) := \{\text{continuous linear functionals } T : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}\},$$

is called the space of *tempered distributions*. Tempered distributions include all L^p functions ($1 \le p \le \infty$), all locally integrable functions with at most polynomial growth, the Dirac delta and its derivatives, principal value kernels, and many oscillatory objects.

1. Basic Examples

- **Dirac delta:** $\langle \delta, \varphi \rangle = \varphi(0)$.
- **Derivative of delta:** $\langle \delta', \varphi \rangle = -\varphi'(0)$.
- **Polynomial growth functions:** if $|f(x)| \le C(1+|x|)^N$ and f is locally integrable, then
$$\langle f, \varphi \rangle := \int_{\mathbb{R}^n} f(x)\varphi(x)\, dx$$
defines a tempered distribution.
- **Principal value kernels:** p.v.$(1/x)$, the Hilbert transform kernel, etc., belong to $\mathcal{S}'$.

2. Fourier Transform on $\mathcal{S}'$

Since $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ is a bijection, we define the Fourier transform of $T \in \mathcal{S}'$ by duality:

$$\boxed{\langle \mathcal{F}T, \varphi \rangle := \langle T, \mathcal{F}\varphi \rangle, \qquad \varphi \in \mathcal{S}.}$$

This makes $\mathcal{F} : \mathcal{S}' \to \mathcal{S}'$ a continuous bijection. The inversion formula continues to hold:

$$\mathcal{F}^{-1}T = \check{\mathcal{F}T}\,/(2\pi)^n,$$

where $\check{T}(\varphi) := T(\check{\varphi})$ and $\check{\varphi}(x) = \varphi(-x)$.

3. Fourier Transform Identities in $\mathcal{S}'$

The usual differentiation and multiplication rules extend to tempered distributions:

$$\mathcal{F}[D^\alpha T](\xi) = (i\xi)^\alpha \widehat{T}(\xi), \qquad \mathcal{F}[x^\alpha T](\xi) = i^{|\alpha|} D_\xi^\alpha \widehat{T}(\xi).$$

Convolution with a Schwartz function is always well-defined and produces another tempered distribution.

4. Examples of Fourier Transforms in $\mathcal{S}'$

$$\begin{aligned}
\widehat{\delta} &= 1, \\
\widehat{\delta'} &= i\xi, \\
\widehat{1} &= (2\pi)^n \delta, \\
\widehat{\mathrm{p.v.}\tfrac{1}{x}} &= -i\pi \,\mathrm{sgn}(\xi), \\
\widehat{\tfrac{1}{|x|}}(\xi) &= \frac{4\pi}{|\xi|^2} \qquad (\text{in } \mathbb{R}^3).
\end{aligned}$$

Such identities cannot be understood within L^1 or L^2 alone, but follow naturally in the framework of tempered distributions.

Tempered distributions thus provide the natural and maximal domain on which the Fourier transform is an automorphism, preserving the full symmetry between differentiation, multiplication, convolution, and decay.

Exercise 4.3 (Fourier–Plancherel Transform) – Solutions

We use the convention

$$\mathcal{F}[f](\xi) = \widehat{f}(\xi) := \int_{\mathbb{R}^n} f(x)e^{-ix\cdot\xi}\,dx, \qquad \mathcal{F}^{-1}[g](x) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} g(\xi)e^{ix\cdot\xi}\,d\xi.$$

(a) Plancherel identity. For $f, g \in \mathcal{S}(\mathbb{R}^n)$, using inversion we write

$$f(x) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} \widehat{f}(\xi)e^{ix\cdot\xi}\,d\xi,$$

so

$$(f,g) = \int_{\mathbb{R}^n} f(x)\,\overline{g(x)}\,dx = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} \widehat{f}(\xi)\left(\int_{\mathbb{R}^n} e^{ix\cdot\xi}\overline{g(x)}\,dx\right)d\xi.$$

The inner integral equals $\overline{\widehat{g}(\xi)}$, hence

$$(f,g) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} \widehat{f}(\xi)\,\overline{\widehat{g}(\xi)}\,d\xi = \frac{1}{(2\pi)^n}(\widehat{f},\widehat{g}).$$

By density of $\mathcal{S}$ in L^2 and continuity of $\mathcal{F}$ on L^2, the formula extends to arbitrary $f, g \in L^2(\mathbb{R}^n)$.

(b) Double transform and bijectivity. For $f \in \mathcal{S}$,

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-ix\cdot\xi}\,dx,$$

and

$$\mathcal{F}[\widehat{f}](y) = \int_{\mathbb{R}^n} \widehat{f}(\xi)e^{-iy\cdot\xi}\,d\xi = \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} f(x)e^{-ix\cdot\xi}e^{-iy\cdot\xi}\,dx\,d\xi.$$

Swapping the integrals and using that $\int_{\mathbb{R}^n} e^{-i(x+y)\cdot\xi}\,d\xi = (2\pi)^n\delta(x+y)$ gives

$$\mathcal{F}[\widehat{f}](y) = (2\pi)^n f(-y) = (2\pi)^n \check{f}(y).$$

By continuity this identity holds for all $f \in L^2$. In particular, $\mathcal{F}$ is injective and its range is all of L^2; thus $\mathcal{F} : L^2 \to L^2$ is a bijection. The inverse is bounded by the Plancherel identity.

(c) Truncated integrals in L^2. For $R > 0$ set

$$\widehat{f}_R(\xi) := \int_{B_R(0)} f(x)e^{-ix\cdot\xi}\,dx = \mathcal{F}[\chi_{B_R} f](\xi).$$

Then

$$\widehat{f}_R - \widehat{f} = \mathcal{F}[(\chi_{B_R} - 1)f],$$

and by Plancherel,

$$\|\widehat{f}_R - \widehat{f}\|_{L^2} = (2\pi)^{n/2}\,\|(\chi_{B_R} - 1)f\|_{L^2}.$$

Since $\chi_{B_R} \to 1$ pointwise and $|\chi_{B_R}| \le 1$, dominated convergence gives $\|(\chi_{B_R}-1)f\|_{L^2} \to 0$ as $R \to \infty$, hence $\widehat{f}_R \to \widehat{f}$ in $L^2(\mathbb{R}^n)$.

(d) Differentiation. First assume $f \in \mathcal{S}(\mathbb{R}^n)$. Then

$$\mathcal{F}[D_j f](\xi) = \int_{\mathbb{R}^n} (\partial_{x_j} f(x))e^{-ix\cdot\xi}\,dx = -\int_{\mathbb{R}^n} f(x)\,\partial_{x_j}(e^{-ix\cdot\xi})\,dx = i\xi_j\widehat{f}(\xi),$$

since boundary terms vanish due to rapid decay. For $f \in C^1(\mathbb{R}^n)$ with $f, D_j f \in L^2$, choose a sequence $f_k \in \mathcal{S}$ such that $f_k \to f$ and $D_j f_k \to D_j f$ in L^2. The identity for f_k, together with the boundedness of $\mathcal{F}$ on L^2, allows passage to the limit and yields

$$\mathcal{F}[D_j f](\xi) = i\xi_j\widehat{f}(\xi)$$

and hence $\xi_j\widehat{f} \in L^2$.

(e) The function $f(x) = \sin x/x$.

(i) $f \in L^2(\mathbb{R})$. Near $x = 0$ one has $\sin x \sim x$, so $\sin x/x$ is bounded and continuous. For large $|x|$,

$$\left|\frac{\sin x}{x}\right| \le \frac{1}{|x|},$$

hence

$$\left|\frac{\sin x}{x}\right|^2 \le \frac{1}{x^2},$$

and $\int_{|x|>1} x^{-2}\,dx < \infty$. Therefore $f \in L^2(\mathbb{R})$.

(ii) Fourier transform. Let $g(\xi) = \mathbf{1}_{(-1,1)}(\xi)$. Then

$$\mathcal{F}^{-1}[g](x) = \frac{1}{2\pi}\int_{-1}^{1} e^{ix\xi}\, d\xi = \frac{1}{2\pi}\left(\frac{e^{ix} - e^{-ix}}{ix}\right) = \frac{1}{\pi}\frac{\sin x}{x}.$$

Thus

$$\frac{\sin x}{x} = \pi\, \mathcal{F}^{-1}[\mathbf{1}_{(-1,1)}](x).$$

Applying $\mathcal{F}$ and using $\mathcal{F} \circ \mathcal{F}^{-1} = \mathrm{Id}$ on L^2, we obtain

$$\mathcal{F}\left[\frac{\sin x}{x}\right](\xi) = \pi\, \mathbf{1}_{(-1,1)}(\xi) = \begin{cases} \pi, & -1 < \xi < 1, \\ 0, & |\xi| \geq 1. \end{cases}$$

(f) Convolution.

(i) Pointwise bound. For $x \in \mathbb{R}^n$,

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x-y)g(y)\, dy.$$

By Cauchy–Schwarz,

$$|f * g(x)| \leq \left(\int_{\mathbb{R}^n} |f(x-y)|^2 dy\right)^{1/2} \left(\int_{\mathbb{R}^n} |g(y)|^2 dy\right)^{1/2} = \|f\|_{L^2}\|g\|_{L^2}.$$

(ii) Continuity and Fourier representation. For $h \in \mathbb{R}^n$,

$$(f * g)(x+h) - (f * g)(x) = \int_{\mathbb{R}^n} [f(x+h-y) - f(x-y)]g(y)\, dy.$$

Again by Cauchy–Schwarz,

$$|(f * g)(x+h) - (f * g)(x)| \leq \|f(\cdot + h) - f\|_{L^2}\, \|g\|_{L^2},$$

and since translations are continuous in L^2, the right-hand side tends to 0 as $h \to 0$, so $f * g$ is continuous.

If $f, g \in \mathcal{S}$, the convolution theorem gives

$$\mathcal{F}[f * g](\xi) = \widehat{f}(\xi)\, \widehat{g}(\xi),$$

and thus

$$f * g = \mathcal{F}^{-1}[\widehat{f}\, \widehat{g}].$$

For $f, g \in L^2$, one approximates by Schwartz functions and uses the boundedness of $\mathcal{F}$ and density of $\mathcal{S}$ in L^2 to extend this identity.

Exercise 4.4 (Yukawa Green's function in $\mathbb{R}^3$)

Work in $\mathbb{R}^3$. For $k > 0$, define

$$G(x) = \frac{e^{-k|x|}}{4\pi|x|}, \qquad x \in \mathbb{R}^3 \setminus \{0\}.$$

(a) Show that $G \in L^1(\mathbb{R}^3)$.

(b) Show that

$$\widehat{G}(\xi) = \frac{1}{|\xi|^2 + k^2}, \qquad \xi \in \mathbb{R}^3.$$

Hint: Use Exercise 4.2, part (c).

Solution to Exercise 4.4

We work in $\mathbb{R}^3$ and define, for $k > 0$,

$$G(x) = \frac{e^{-k|x|}}{4\pi|x|}, \qquad x \in \mathbb{R}^3 \setminus \{0\}.$$

(a) $G \in L^1(\mathbb{R}^3)$. Since G is radial, write $r = |x|$ and

$$|G(x)| = \frac{e^{-kr}}{4\pi r}.$$

In spherical coordinates,

$$\int_{\mathbb{R}^3} |G(x)|\, dx = \int_0^\infty \int_0^\pi \int_0^{2\pi} \frac{e^{-kr}}{4\pi r}\, r^2 \sin\theta\, d\phi\, d\theta\, dr.$$

The angular integrals give

$$\int_0^\pi \int_0^{2\pi} \sin\theta\, d\phi\, d\theta = 4\pi,$$

hence

$$\int_{\mathbb{R}^3} |G(x)|\, dx = \int_0^\infty \frac{e^{-kr}}{4\pi r}\, r^2 \cdot 4\pi\, dr = \int_0^\infty e^{-kr} r\, dr.$$

The last integral converges (in fact $\int_0^\infty e^{-kr} r\, dr = 1/k^2$), so $G \in L^1(\mathbb{R}^3)$.

(b) Fourier transform of G. By Exercise 4.2(c), if $f \in L^1(\mathbb{R}^3)$ is radial, $f(x) = f(r)$, then for $\xi \in \mathbb{R}^3$,

$$\widehat{f}(\xi) = 4\pi \int_0^\infty f(r)\, \frac{\sin(r|\xi|)}{r|\xi|}\, r^2\, dr.$$

Here $f(r) = G(r) = \dfrac{e^{-kr}}{4\pi r}$, so with $\rho = |\xi|$ we get

$$\widehat{G}(\xi) = 4\pi \int_0^\infty \frac{e^{-kr}}{4\pi r} \frac{\sin(\rho r)}{\rho r}\, r^2\, dr = \int_0^\infty e^{-kr}\, \frac{\sin(\rho r)}{\rho r}\, r\, dr = \frac{1}{\rho} \int_0^\infty e^{-kr} \sin(\rho r)\, dr.$$

Using the standard integral (obtained, for example, via the imaginary part of a Laplace transform),

$$\int_0^\infty e^{-kr}\sin(\rho r)\,dr = \frac{\rho}{k^2+\rho^2}, \qquad k>0,\ \rho\geq 0,$$

we obtain

$$\widehat{G}(\xi) = \frac{1}{\rho}\cdot\frac{\rho}{k^2+\rho^2} = \frac{1}{k^2+\rho^2} = \frac{1}{|\xi|^2+k^2}.$$

Thus

$$\boxed{\widehat{G}(\xi) = \frac{1}{|\xi|^2+k^2}, \quad \xi\in\mathbb{R}^3.}$$

Theory: G as the Green's function of $(-\Delta + k^2)$

The function

$$G(x) = \frac{e^{-k|x|}}{4\pi|x|}, \qquad k>0,$$

is called the Yukawa potential in $\mathbb{R}^3$. We show that G is the fundamental solution (Green's function) for the operator

$$L := -\Delta + k^2,$$

in the sense that

$$(-\Delta + k^2)G = \delta_0$$

in $\mathcal{S}'(\mathbb{R}^3)$.

From Exercise 4.4(b) we already know that

$$\widehat{G}(\xi) = \frac{1}{|\xi|^2+k^2}, \qquad \xi\in\mathbb{R}^3.$$

On the other hand, for $u\in\mathcal{S}'(\mathbb{R}^3)$ we have

$$\widehat{-\Delta u}(\xi) = |\xi|^2\widehat{u}(\xi), \qquad \widehat{k^2 u}(\xi) = k^2\widehat{u}(\xi).$$

Therefore

$$(\widehat{-\Delta + k^2})G(\xi) = (|\xi|^2+k^2)\,\widehat{G}(\xi) = (|\xi|^2+k^2)\,\frac{1}{|\xi|^2+k^2} = 1.$$

Since the Fourier transform of the Dirac delta is $\widehat{\delta_0}(\xi) = 1$, we conclude that

$$(\widehat{-\Delta + k^2})G = \widehat{\delta_0} \quad\Longrightarrow\quad (-\Delta + k^2)G = \delta_0$$

in the sense of tempered distributions.

For comparison, the Coulomb (or Newton) potential

$$\Phi(x) = \frac{1}{4\pi|x|}$$

satisfies

$$-\Delta\Phi = \delta_0, \qquad \widehat{\Phi}(\xi) = \frac{1}{|\xi|^2}.$$

Thus G is the "screened" version of the Coulomb kernel, with symbol

$$\widehat{G}(\xi) = \frac{1}{|\xi|^2 + k^2}$$

instead of $1/|\xi|^2$, and exponential decay $e^{-k|x|}$ at infinity. For f sufficiently nice, the solution of

$$(-\Delta + k^2)u = f$$

is given by the convolution

$$u = G * f,$$

equivalently $\widehat{u}(\xi) = \widehat{f}(\xi)/(|\xi|^2 + k^2)$.

Exercise 4.5 (Inhomogeneous Helmholtz Equation in $\mathbb{R}^3$)

Consider the inhomogeneous Helmholtz equation

$$-\Delta\phi + k^2\phi = f,$$

where $f \in \mathcal{S}(\mathbb{R}^3)$. Show that there exists a unique $\phi \in \mathcal{S}(\mathbb{R}^3)$ solving this equation, and that it is given by

$$\phi(x) = \int_{\mathbb{R}^3} f(y)\, G(x-y)\, dy,$$

where

$$G(x) = \frac{e^{-k|x|}}{4\pi|x|}.$$

Hint: first derive an equation satisfied by $\widehat{\phi}$.

Solution to Exercise 4.5

We consider the inhomogeneous Helmholtz equation

$$-\Delta\phi + k^2\phi = f, \qquad f \in \mathcal{S}(\mathbb{R}^3).$$

Taking Fourier transforms and using $\widehat{-\Delta\phi}(\xi) = |\xi|^2\widehat{\phi}(\xi)$, we obtain

$$(|\xi|^2 + k^2)\widehat{\phi}(\xi) = \widehat{f}(\xi).$$

Hence

$$\widehat{\phi}(\xi) = \frac{\widehat{f}(\xi)}{|\xi|^2 + k^2}.$$

Since $f \in \mathcal{S}$, we have $\widehat{f} \in \mathcal{S}$. The multiplier $(|\xi|^2 + k^2)^{-1}$ is smooth with at most polynomial growth, so $\mathcal{S}$ is stable under multiplication by it. Thus $\widehat{\phi} \in \mathcal{S}$ and $\phi \in \mathcal{S}$. Uniqueness follows: if $f = 0$ then $(|\xi|^2 + k^2)\widehat{\phi} = 0$, so $\widehat{\phi} = 0$ and $\phi = 0$.

From Exercise 4.4 we know that

$$\widehat{G}(\xi) = \frac{1}{|\xi|^2 + k^2},$$

where $G(x) = e^{-k|x|}/(4\pi|x|)$. Therefore

$$\widehat{\phi}(\xi) = \widehat{f}(\xi)\,\widehat{G}(\xi) = \widehat{f * G}(\xi).$$

By injectivity of the Fourier transform on $\mathcal{S}$, we obtain $\phi = f * G$, that is

$$\phi(x) = \int_{\mathbb{R}^3} f(y)G(x-y)\,dy.$$

Exercise 4.6 (Translations and Regular Distributions)

Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ be such that its associated regular distribution $Tf \in \mathcal{S}'(\mathbb{R}^n)$. Show that

$$\tau_x Tf = T_{\tau_x f}, \qquad \widehat{Tf} = T_{\widehat{f}}.$$

Solution to Exercise 4.6

Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ and $Tf \in \mathcal{S}'$ be the associated regular distribution,

$$Tf(\varphi) = \int_{\mathbb{R}^n} f(x)\varphi(x)\,dx.$$

Translations. For a distribution T, its translate is defined by

$$(\tau_x T)(\varphi) := T(\tau_{-x}\varphi), \qquad (\tau_x\varphi)(y) = \varphi(y-x).$$

Then

$$(\tau_x Tf)(\varphi) = Tf(\tau_{-x}\varphi) = \int_{\mathbb{R}^n} f(y)\,\varphi(y+x)\,dy.$$

With the change of variables $z = y + x$,

$$(\tau_x Tf)(\varphi) = \int_{\mathbb{R}^n} f(z-x)\,\varphi(z)\,dz = \int_{\mathbb{R}^n} (\tau_x f)(z)\,\varphi(z)\,dz = T_{\tau_x f}(\varphi).$$

Thus $\tau_x Tf = T_{\tau_x f}$.

Fourier transform. The Fourier transform of a distribution T is given by

$$\langle \widehat{T}, \varphi \rangle := \langle T, \widehat{\varphi} \rangle, \qquad \varphi \in \mathcal{S}.$$

Therefore

$$\langle \widehat{Tf}, \varphi \rangle = \langle Tf, \widehat{\varphi} \rangle = \int_{\mathbb{R}^n} f(x)\widehat{\varphi}(x)\, dx.$$

Using the representation

$$\widehat{\varphi}(x) = \int_{\mathbb{R}^n} \varphi(\xi) e^{-ix\cdot\xi}\, d\xi$$

and Fubini's theorem, we obtain

$$\int f(x)\widehat{\varphi}(x)\, dx = \int \varphi(\xi) \left(\int f(x) e^{-ix\cdot\xi}\, dx \right) d\xi = \int \widehat{f}(\xi)\, \varphi(\xi)\, d\xi = T_{\widehat{f}}(\varphi).$$

Hence $\widehat{Tf} = T_{\widehat{f}}$.

Figures for Exercise 4.6

Chart 1: A function and its translate. Figure **??** shows a concrete example of a function $f(x)$ (a Gaussian) together with its translate $(\tau_a f)(x) = f(x-a)$ for a fixed shift $a > 0$. The graph of the translated function has the same shape and height as f but is simply shifted to the right by a units. At the distribution level this corresponds to the identity $\tau_a Tf = T_{\tau_a f}$, i.e. translating the regular distribution is the same as taking the distribution associated with the translated function.

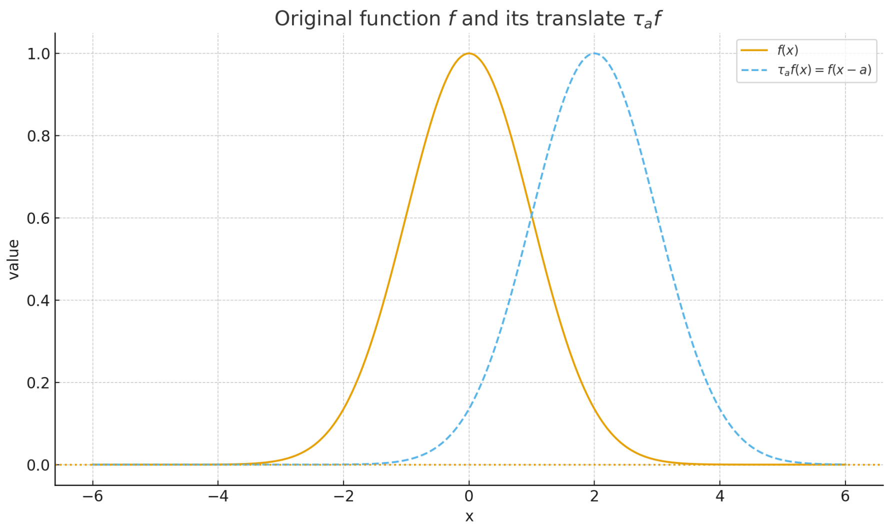


Figure 18: A Gaussian $f(x)$ and its translate $\tau_a f(x) = f(x-a)$ for some fixed $a > 0$. The shape is preserved and only the position changes, illustrating the equality $\tau_a Tf = T_{\tau_a f}$ for the associated regular distributions.

Chart 2: Effect of translation on the Fourier magnitude. Figure **??** plots the magnitudes $|\widehat{f}(\xi)|$ and $|\widehat{\tau_a f}(\xi)|$ for the same Gaussian example. One sees that the curves coincide: translating in physical space does not change the modulus of the Fourier transform. This visualises the fact that $\mathcal{F}[\tau_a f](\xi) = e^{-ia\xi}\widehat{f}(\xi)$, so only a phase factor is introduced, while the spectrum's magnitude remains invariant.

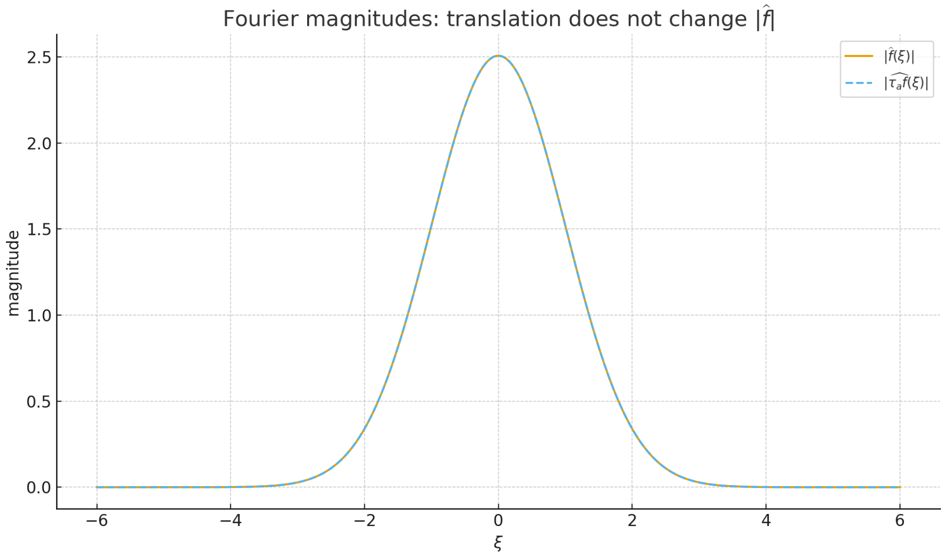


Figure 19: Magnitudes of the Fourier transforms $|\widehat{f}(\xi)|$ and $|\widehat{\tau_a f}(\xi)|$. They coincide, which reflects the identity $\widehat{\tau_a f}(\xi) = e^{-ia\xi}\widehat{f}(\xi)$: translation in x only affects the phase, not the magnitude, of the Fourier transform.

Chart 3: Phase factor from translation. Figure **??** displays the phase of $\widehat{\tau_a f}(\xi)$ for the same example. The phase is essentially a straight line in ξ, corresponding to the multiplicative factor $e^{-ia\xi}$ in Fourier space. This linear phase encodes the translation amount a and is the precise reason why the operation of translating a function corresponds, on the level of distributions, to the transformation described in Exercise 4.6.

Exercise 4.7 (Fourier Transform of the Truncated Sign Function)

Let $f : \mathbb{R} \to \mathbb{R}$ be the sign function

$$f(x) = \begin{cases} -1, & x < 0, \\ 1, & x \geq 0, \end{cases}$$

and define the truncated function

$$f_R(x) = f(x)\,\mathbf{1}_{[-R,R]}(x).$$

(a) Sketch $f_R(x)$.

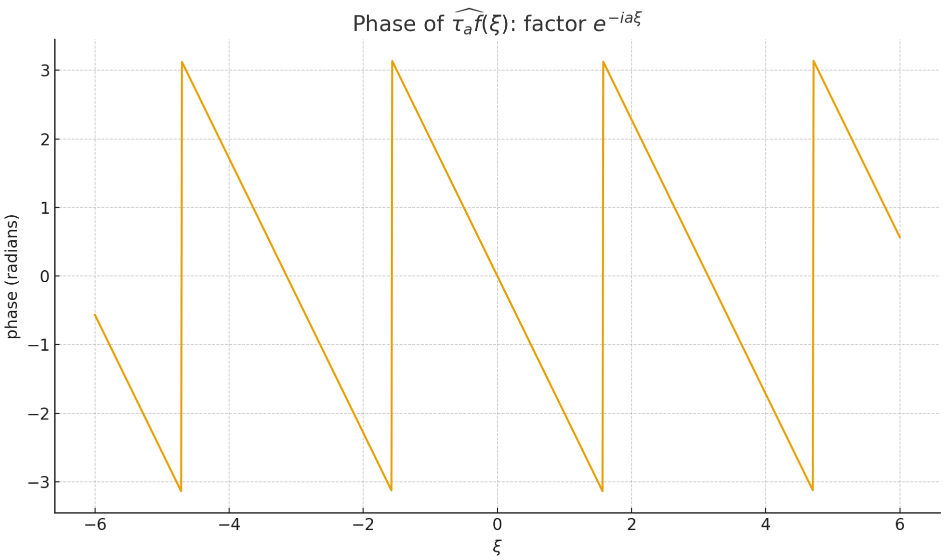


Figure 20: Phase of $\widehat{\tau_a f}(\xi)$ for a translated Gaussian. The approximately linear dependence on ξ represents the factor $e^{-ia\xi}$ gained under translation, while the modulus of the Fourier transform stays unchanged.

(b) Show that $T_{f_R} \to T_f$ in $\mathcal{S}'$ as $R \to \infty$.

(c) Show that

$$\widehat{f_R}(\xi) = 2i\,\frac{\cos(R\xi) - 1}{\xi}.$$

(d) For $\phi \in \mathcal{S}$, show that

$$T_{f_R}[\widehat{\phi}] = -2i \int_0^\infty \frac{\phi(x) - \phi(-x)}{x}\,dx + 2i \int_0^\infty \frac{\phi(x) - \phi(-x)}{x}\cos(Rx)\,dx.$$

(e) Use the Riemann–Lebesgue lemma to show that for all $\psi \in \mathcal{S}$:

$$\int_0^\infty \psi(x)\cos(Rx)\,dx \to 0 \quad \text{as } R \to \infty.$$

(f) Deduce that

$$\widehat{T_f} = -2i\,\mathrm{P.V.}\left(\frac{1}{x}\right).$$

(g) Compute $\widehat{T_H}$, where the Heaviside function is

$$H(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Solution to Exercise 4.7

Let

$$f(x) = \begin{cases} -1, & x < 0, \\ 1, & x \geq 0, \end{cases} \qquad f_R(x) = f(x)\mathbf{1}_{[-R,R]}(x).$$

(a) Sketch. The function f_R equals -1 on $[-R,0)$, 1 on $[0,R]$, and 0 outside $[-R,R]$.

(b) Convergence in $\mathcal{S}'$. For $\varphi \in \mathcal{S}(\mathbb{R})$,

$$T_{f_R}(\varphi) = \int_{-R}^{R} f(x)\varphi(x)\,dx.$$

Because f is bounded and φ decays rapidly, we have

$$\int_{\mathbb{R}\setminus[-R,R]} |f(x)\varphi(x)|\,dx \le \|f\|_\infty \int_{|x|>R} |\varphi(x)|\,dx \to 0$$

as $R\to\infty$. Hence $T_{f_R}(\varphi) \to \int_{\mathbb{R}} f(x)\varphi(x)\,dx = T_f(\varphi)$, so $T_{f_R} \to T_f$ in $\mathcal{S}'$.

(c) Fourier transform of f_R. We compute

$$\widehat{f_R}(\xi) = \int_{-R}^{0} (-1)e^{-ix\xi}\,dx + \int_0^R e^{-ix\xi}\,dx.$$

A direct calculation yields

$$\widehat{f_R}(\xi) = 2i\,\frac{\cos(R\xi)-1}{\xi},$$

which is the desired formula.

(d) Expression for $T_{f_R}[\widehat{\phi}]$. For $\phi \in \mathcal{S}$,

$$T_{f_R}[\widehat{\phi}] = T_{\widehat{f_R}}[\phi] = \int_{\mathbb{R}} \widehat{f_R}(\xi)\phi(\xi)\,d\xi = 2i\int_{\mathbb{R}} \frac{\cos(R\xi)-1}{\xi}\phi(\xi)\,d\xi.$$

Splitting at 0 and changing variables $\xi=-x$ in the negative half-line, we obtain

$$T_{f_R}[\widehat{\phi}] = 2i\int_0^\infty \frac{\cos(Rx)-1}{x}\,[\phi(x)-\phi(-x)]\,dx.$$

Separating the terms with $\cos(Rx)$ and with -1 gives

$$T_{f_R}[\widehat{\phi}] = -2i\int_0^\infty \frac{\phi(x)-\phi(-x)}{x}\,dx + 2i\int_0^\infty \frac{\phi(x)-\phi(-x)}{x}\cos(Rx)\,dx,$$

as required.

(e) Riemann–Lebesgue lemma. For any $\psi \in \mathcal{S}(\mathbb{R})$, in particular for $\psi \in L^1(\mathbb{R})$, the Riemann–Lebesgue lemma implies

$$\int_0^\infty \psi(x)\cos(Rx)\,dx \to 0 \quad \text{as } R\to\infty.$$

Taking $\psi(x) = (\phi(x)-\phi(-x))/x$, which belongs to $\mathcal{S}$, we obtain

$$\int_0^\infty \frac{\phi(x)-\phi(-x)}{x}\cos(Rx)\,dx \to 0.$$

(f) Fourier transform of T_f. From (b), $T_{f_R} \to T_f$ in $\mathcal{S}'$. Thus

$$\langle \widehat{T_f}, \phi \rangle = \langle T_f, \widehat{\phi} \rangle = \lim_{R\to\infty} \langle T_{f_R}, \widehat{\phi} \rangle = \lim_{R\to\infty} T_{f_R}[\widehat{\phi}].$$

Using the formula from (d) and the limit in (e), we obtain

$$\langle \widehat{T_f}, \phi \rangle = -2i \int_0^\infty \frac{\phi(x) - \phi(-x)}{x}\, dx.$$

By the definition of the principal value distribution,

$$\left\langle \mathrm{P.V.}\left(\frac{1}{x}\right), \phi \right\rangle = \int_0^\infty \frac{\phi(x) - \phi(-x)}{x}\, dx,$$

so

$$\widehat{T_f} = -2i\, \mathrm{P.V.}\left(\frac{1}{x}\right).$$

(g) Fourier transform of T_H. The Heaviside function satisfies

$$H(x) = \frac{1}{2}\Big(1 + f(x)\Big),$$

so $T_H = \frac{1}{2}T_1 + \frac{1}{2}T_f$. With the convention used here,

$$\widehat{T_1} = 2\pi\delta, \qquad \widehat{T_f} = -2i\, \mathrm{P.V.}\left(\frac{1}{x}\right),$$

and therefore

$$\widehat{T_H} = \frac{1}{2}\widehat{T_1} + \frac{1}{2}\widehat{T_f} = \pi\delta - i\, \mathrm{P.V.}\left(\frac{1}{x}\right).$$

Figures for Exercise 4.7

Chart 1: Truncated sign functions f_R. Figure **??** shows the truncated sign function $f_R(x) = f(x)\mathbf{1}_{[-R,R]}(x)$ for several values of R. Each curve is piecewise constant: it equals -1 on $[-R, 0)$, $+1$ on $(0, R]$, and 0 outside the interval $[-R, R]$. As R increases, the nonzero region widens, and the regular distributions T_{f_R} converge to the full sign distribution T_f in $\mathcal{S}'$.

Chart 2: Fourier profile $2\sin(R\xi)/\xi$. Figure **??** plots the real profile $2\sin(R\xi)/\xi$, which (up to a factor of i and conventions) is the Fourier transform $\widehat{f_R}(\xi)$ derived in part (c) of Exercise 4.7. The function oscillates with frequency proportional to R, while its overall envelope decays like $1/|\xi|$ for large $|\xi|$. As R grows, these oscillations become more rapid and the mass concentrates near $\xi = 0$, reflecting the fact that f_R spreads out in physical space.

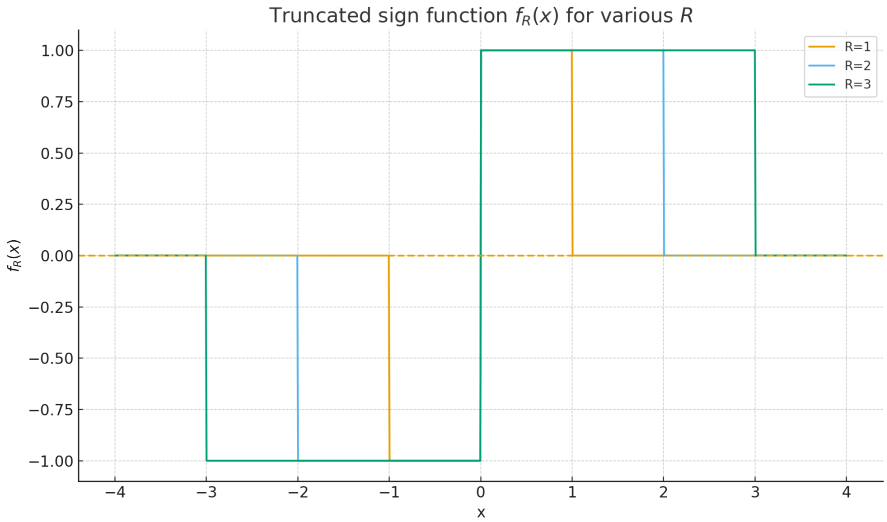


Figure 21: Truncated sign functions $f_R(x)$ for several values of R. The support grows with R, while the values remain ± 1, illustrating $T_{f_R} \to T_f$ in $\mathcal{S}'$.

Chart 3: Approximation of P.V.$(1/x)$**.** Figure **??** shows an approximation to the principal value distribution P.V.$(1/x)$, obtained by plotting $1/x$ only for $|x| \geq \varepsilon$ and leaving a small symmetric gap around the origin. The vertical dashed line at $x = 0$ indicates the singularity that is removed and treated in the principal value sense. This picture illustrates how the distribution is defined via limits of integrals over $\mathbb{R} \setminus (-\varepsilon, \varepsilon)$, which is exactly the object that appears as $\widehat{T_f}$ in Exercise 4.7(f).

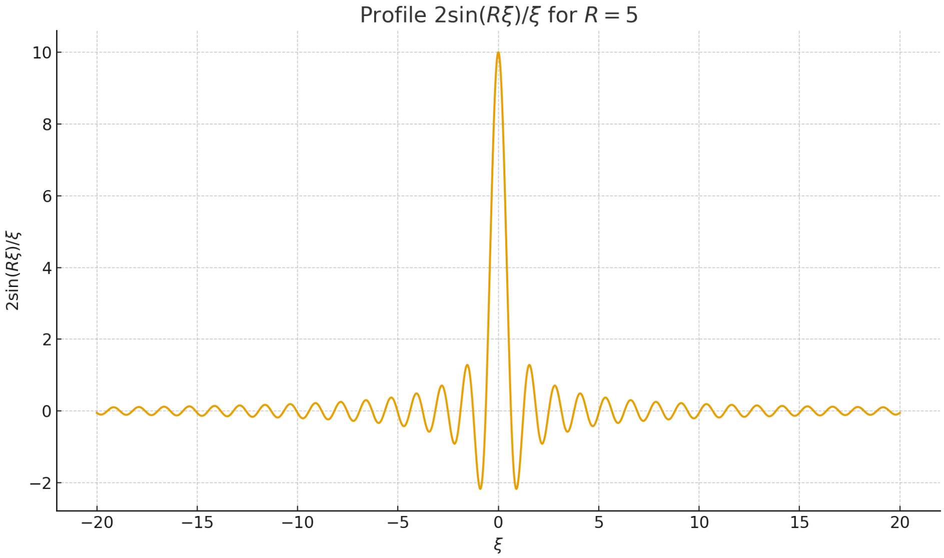


Figure 22: Profile of $2\sin(R\xi)/\xi$ for a fixed R (here $R = 5$), representing the Fourier transform of f_R (up to a factor i). The oscillatory behavior and $1/|\xi|$ decay encode the truncation and jump of the sign function in physical space.

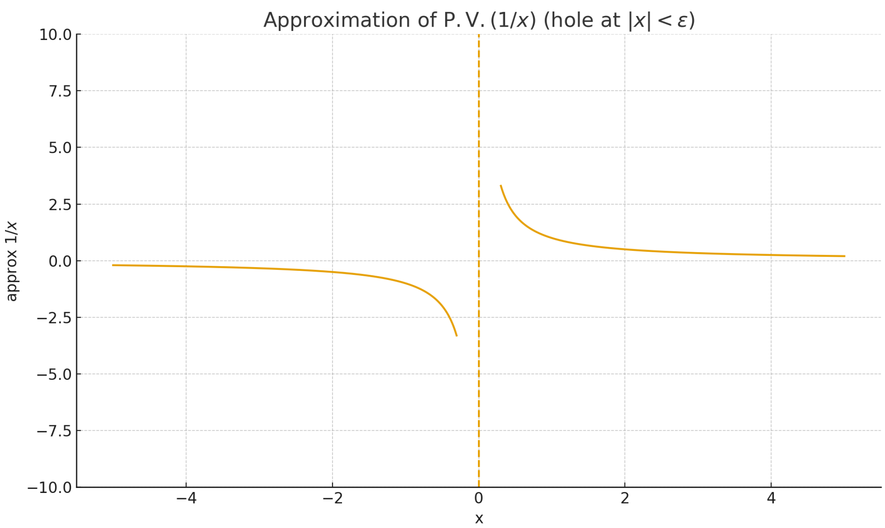


Figure 23: Approximation of the principal value distribution P.V.$(1/x)$ by omitting a symmetric interval $(-\varepsilon, \varepsilon)$ around the singularity. This visualises how the principal value is understood as a limit of symmetric truncations.

Exercise 5.1 (Periodisation of a compactly supported distribution)

Suppose $v \in \mathcal{E}'(\mathbb{R}^n)$ and define

$$u = \sum_{g \in \mathbb{Z}^n} \tau_g v,$$

where τ_g denotes translation by g.

Show that if $\phi \in \mathcal{D}(\mathbb{R}^n)$ with $\operatorname{supp} \phi \subset K$ for some compact set $K \subset \mathbb{R}^n$, then

$$u[\phi] = \sum_{g \in A} \tau_g v[\phi],$$

for some finite set $A \subset \mathbb{Z}^n$ which depends only on K. Deduce that u defines a distribution.

Theory for Exercise 5.1

Recall that $\mathcal{D}(\mathbb{R}^n)$ denotes the space of smooth, compactly supported test functions, and that $\mathcal{E}'(\mathbb{R}^n)$ is the space of compactly supported distributions. A distribution $v \in \mathcal{E}'(\mathbb{R}^n)$ has compact support in the sense that there exists a compact set $K_0 \subset \mathbb{R}^n$ such that $v[\phi] = 0$ whenever $\operatorname{supp} \phi \cap K_0 = \varnothing$.

For $g \in \mathbb{R}^n$, the translation operator on test functions is defined by

$$(\tau_g \phi)(x) := \phi(x - g),$$

and the translation of a distribution v is given by

$$(\tau_g v)[\phi] := v[\tau_{-g} \phi], \qquad \phi \in \mathcal{D}(\mathbb{R}^n).$$

If $v = T_f$ is a regular distribution arising from a function f, then $\tau_g v = T_{\tau_g f}$ with $(\tau_g f)(x) = f(x - g)$. Moreover, if $\operatorname{supp} v \subset K_0$, then $\operatorname{supp}(\tau_g v) = K_0 + g$.

The formal sum

$$u = \sum_{g \in \mathbb{Z}^n} \tau_g v$$

is meant to define a $\mathbb{Z}^n$–periodic distribution obtained by copying v to each lattice point. The key observation is that for a given test function $\phi \in \mathcal{D}(\mathbb{R}^n)$ with $\operatorname{supp} \phi \subset K$, only finitely many translates $\tau_g v$ interact nontrivially with ϕ: if $(K_0 + g) \cap K = \varnothing$, then $(\tau_g v)[\phi] = 0$. Since K is compact, there are only finitely many $g \in \mathbb{Z}^n$ for which $(K_0 + g) \cap K \neq \varnothing$, and thus the sum

$$u[\phi] := \sum_{g \in \mathbb{Z}^n} (\tau_g v)[\phi]$$

reduces to a finite sum $\sum_{g \in A} (\tau_g v)[\phi]$, where $A \subset \mathbb{Z}^n$ is finite and depends only on K. This shows that u is well-defined as a continuous linear functional on $\mathcal{D}$, hence u is a distribution (in fact, it is $\mathbb{Z}^n$–periodic).

Solution to Exercise 5.1

Let $v \in \mathcal{E}'(\mathbb{R}^n)$ and define

$$u = \sum_{g\in\mathbb{Z}^n} \tau_g v, \qquad (\tau_g v)[\phi] := v[\tau_{-g}\phi].$$

Since v is compactly supported, there exists a compact set $K_0 \subset \mathbb{R}^n$ such that $\operatorname{supp} v \subset K_0$. For $g \in \mathbb{Z}^n$ we then have $\operatorname{supp}(\tau_g v) = K_0 + g := \{x + g : x \in K_0\}$.

Let $\phi \in \mathcal{D}(\mathbb{R}^n)$ and assume $\operatorname{supp}\phi \subset K$ for some compact set $K \subset \mathbb{R}^n$. If $(K_0+g)\cap K = \varnothing$, then $\tau_{-g}\phi$ vanishes on $\operatorname{supp} v$ and thus $(\tau_g v)[\phi] = 0$. Hence, only those $g \in \mathbb{Z}^n$ with $(K_0 + g) \cap K \neq \varnothing$ can contribute to the sum.

Define

$$A := \{g \in \mathbb{Z}^n : (K_0 + g) \cap K \neq \varnothing\}.$$

If $g \in A$, then there exist $x \in K_0$ and $y \in K$ such that $y = x+g$, i.e. $g = y-x \in K - K_0$. Since the difference of two compact sets is compact and $\mathbb{Z}^n$ is a discrete subset of $\mathbb{R}^n$, the set $A = (K - K_0) \cap \mathbb{Z}^n$ is finite. It follows that

$$u[\phi] := \sum_{g\in\mathbb{Z}^n} (\tau_g v)[\phi] = \sum_{g\in A} (\tau_g v)[\phi],$$

so the sum defining $u[\phi]$ is finite for every $\phi \in \mathcal{D}(\mathbb{R}^n)$.

Each $\tau_g v$ is a distribution, hence a continuous linear functional on $\mathcal{D}(\mathbb{R}^n)$. For test functions with support in a fixed compact set K, the index set A is finite and depends only on K. Therefore u is a finite sum of continuous linear functionals on $\mathcal{D}_K$, and hence u is a continuous linear functional on $\mathcal{D}(\mathbb{R}^n)$. Thus $u \in \mathcal{D}'(\mathbb{R}^n)$, i.e. u defines a distribution (in fact a $\mathbb{Z}^n$–periodic one).

Exercise 5.2 (Counting lattice points and summability)

Recall that for $x \in \mathbb{R}^n$,

$$\|x\|_1 := \sum_{i=1}^n |x_i|.$$

For $k \in \mathbb{N}$ set

$$Q_k := \Big\{g \in \mathbb{Z}^n : k - \tfrac{1}{2} \leq \|g\|_1 < k + \tfrac{1}{2}\Big\}.$$

(a) Show that

$$\#Q_k = (2k+1)^n - (2k-1)^n,$$

so that $\#Q_k \leq c(1+k)^{n-1}$ for some constant $c > 0$.

(b) By applying the Cauchy–Schwarz identity to estimate $a \cdot b$ for $a = (1, \dots, 1)$ and $b = (|g_1|, \dots, |g_n|)$, deduce that

$$\|g\|_1 \leq \sqrt{n}\,|g|,$$

where $|g|$ denotes the Euclidean norm of g.

(c) Show that there exists a constant $C > 0$, depending only on n, such that

$$\sum_{\substack{g \in \mathbb{Z}^n \\ \|g\|_1 \leq K}} \frac{1}{(1+|g|)^{n+1}} \;\leq\; 1 + C \sum_{k=1}^{K} \frac{1}{k^2}$$

holds for all $K \in \mathbb{N}$. Deduce that

$$\sum_{g \in \mathbb{Z}^n} \frac{1}{(1+|g|)^{n+1}} < \infty.$$

Theory for Exercise 5.2

For $x \in \mathbb{R}^n$ we consider the ℓ^1–norm

$$\|x\|_1 := \sum_{i=1}^{n} |x_i|.$$

For $k \in \mathbb{N}$ the set

$$Q_k := \left\{ g \in \mathbb{Z}^n : k - \tfrac{1}{2} \leq \|g\|_1 < k + \tfrac{1}{2} \right\}$$

consists exactly of those lattice points with $\|g\|_1 = k$. Geometrically, Q_k is a discrete "shell" of radius k around the origin in the ℓ^1–metric. The first part of the exercise shows that

$$\#Q_k = (2k+1)^n - (2k-1)^n,$$

which implies the bound $\#Q_k \leq c(1+k)^{n-1}$ for some constant $c > 0$; in particular, the number of lattice points with ℓ^1–radius k grows like k^{n-1}, analogous to the surface area of a sphere.

The second part uses the Cauchy–Schwarz inequality to relate the ℓ^1–norm to the Euclidean norm. For $g \in \mathbb{R}^n$ we set $a = (1, \dots, 1)$ and $b = (|g_1|, \dots, |g_n|)$, and obtain

$$\|g\|_1 = a \cdot b \leq \|a\|_2 \|b\|_2 = \sqrt{n} \left(\sum_{i=1}^{n} g_i^2 \right)^{1/2} = \sqrt{n}\,|g|.$$

Thus, for $g \in Q_k$ with $k \geq 1$, we have $|g| \geq ck$ for some constant $c > 0$ depending only on n.

The final part groups the sum

$$\sum_{g \in \mathbb{Z}^n} \frac{1}{(1+|g|)^{n+1}}$$

according to the shells Q_k:

$$\sum_{\|g\|_1 \le K} \frac{1}{(1+|g|)^{n+1}} = \sum_{k=0}^{K} \sum_{g \in Q_k} \frac{1}{(1+|g|)^{n+1}}.$$

Using $\#Q_k \le c(1+k)^{n-1}$ and $|g| \ge ck$ for $g \in Q_k \setminus \{0\}$, one obtains an estimate of the form

$$\sum_{\|g\|_1 \le K} \frac{1}{(1+|g|)^{n+1}} \le 1 + C \sum_{k=1}^{K} \frac{1}{k^2},$$

for some constant $C > 0$ depending only on n. Since $\sum_{k=1}^{\infty} 1/k^2 < \infty$, this shows that

$$\sum_{g \in \mathbb{Z}^n} \frac{1}{(1+|g|)^{n+1}} < \infty.$$

Such arguments are fundamental when analysing the summability of discrete kernels on $\mathbb{Z}^n$ and appear frequently in harmonic analysis and PDE.

Solution to Exercise 5.2

(a) For $g \in \mathbb{Z}^n$, the quantity $\|g\|_1 = \sum_{i=1}^n |g_i|$ is an integer. The condition $k - \frac{1}{2} \le \|g\|_1 < k + \frac{1}{2}$ therefore simply means $\|g\|_1 = k$, so Q_k is the "shell" of lattice points with ℓ^1–norm equal to k.

Let

$$B_r := \{g \in \mathbb{Z}^n : \|g\|_1 \le r\}.$$

If $r = m + \frac{1}{2}$ with $m \in \mathbb{N}$, then $\|g\|_1 \le m + \frac{1}{2}$ holds if and only if $|g_i| \le m$ for all i, so $B_{m+\frac{1}{2}} = \mathbb{Z}^n \cap [-m, m]^n$ has $(2m+1)^n$ elements. Since $Q_k = B_{k+\frac{1}{2}} \setminus B_{k-\frac{1}{2}}$, we obtain

$$\#Q_k = \#B_{k+\frac{1}{2}} - \#B_{k-\frac{1}{2}} = (2k+1)^n - (2k-1)^n.$$

To estimate this difference, consider $f(t) = t^n$ on $\mathbb{R}$. By the mean value theorem there exists $\xi \in (2k-1, 2k+1)$ such that

$$(2k+1)^n - (2k-1)^n = f(2k+1) - f(2k-1) = f'(\xi)\,(2) = 2n\,\xi^{\,n-1} \le 2n(2k+1)^{n-1}.$$

Thus $\#Q_k \le c(1+k)^{n-1}$ for some constant $c > 0$ depending only on n.

(b) Let $g = (g_1, \ldots, g_n) \in \mathbb{R}^n$ and set $a = (1, \ldots, 1)$, $b = (|g_1|, \ldots, |g_n|)$. Then

$$\|g\|_1 = \sum_{i=1}^n |g_i| = a \cdot b.$$

By the Cauchy–Schwarz inequality,

$$a \cdot b \le \|a\|_2 \|b\|_2 = \sqrt{n} \left(\sum_{i=1}^n g_i^2 \right)^{1/2} = \sqrt{n}\,|g|.$$

Hence $\|g\|_1 \le \sqrt{n}\,|g|$.

(c) We wish to show that there exists $C > 0$ (depending only on n) such that

$$\sum_{\substack{g\in\mathbb{Z}^n \\ \|g\|_1\le K}} \frac{1}{(1+|g|)^{n+1}} \le 1 + C\sum_{k=1}^{K}\frac{1}{k^2}$$

for all $K \in \mathbb{N}$. We separate off the term $g = 0$:

$$\sum_{\substack{g\in\mathbb{Z}^n \\ \|g\|_1\le K}} \frac{1}{(1+|g|)^{n+1}} = \frac{1}{(1+0)^{n+1}} + \sum_{\substack{g\in\mathbb{Z}^n\setminus\{0\} \\ \|g\|_1\le K}} \frac{1}{(1+|g|)^{n+1}} = 1 + S_K.$$

For $g \neq 0$, part (b) gives $\|g\|_1 \le \sqrt{n}\,|g|$, so $|g| \ge c\,\|g\|_1$ and $1+|g| \ge c(1+\|g\|_1)$ for some $c > 0$ depending only on n. Hence

$$\frac{1}{(1+|g|)^{n+1}} \le \frac{C_1}{(1+\|g\|_1)^{n+1}}$$

for some constant $C_1 > 0$ depending only on n.

We group the remaining sum according to the shells Q_k:

$$S_K = \sum_{1\le\|g\|_1\le K} \frac{1}{(1+|g|)^{n+1}} \le C_1\sum_{k=1}^{K}\sum_{g\in Q_k}\frac{1}{(1+k)^{n+1}} = C_1\sum_{k=1}^{K}\frac{\#Q_k}{(1+k)^{n+1}}.$$

Using the bound from part (a), $\#Q_k \le c(1+k)^{n-1}$, we obtain

$$S_K \le C_1 c\sum_{k=1}^{K}\frac{(1+k)^{n-1}}{(1+k)^{n+1}} = C\sum_{k=1}^{K}\frac{1}{(1+k)^2} \le C\sum_{k=1}^{K}\frac{1}{k^2}$$

for a suitable constant $C > 0$ depending only on n.

Combining this with the earlier identity gives

$$\sum_{\substack{g\in\mathbb{Z}^n \\ \|g\|_1\le K}} \frac{1}{(1+|g|)^{n+1}} \le 1 + C\sum_{k=1}^{K}\frac{1}{k^2}.$$

Letting $K \to \infty$ yields

$$\sum_{g\in\mathbb{Z}^n} \frac{1}{(1+|g|)^{n+1}} < \infty,$$

since the series $\sum_{k=1}^{\infty} 1/k^2$ converges.

Figures for Exercises 5.1 and 5.2

Figure 1: Periodisation of a compactly supported distribution. Figure **??** shows a compactly supported bump v on $\mathbb{R}$ (solid line), and a partial periodisation $u = \sum_{|g|\le 2}\tau_g v$ (dashed line), where τ_g is translation by $g \in \mathbb{Z}$. The sum u clearly has a repeating pattern with period 1, illustrating the idea of copying v to each lattice point. In Exercise 5.1 one proves that for any test function ϕ with compact support, only finitely many translates contribute to $u[\phi]$, so the formal sum defines a genuine distribution.

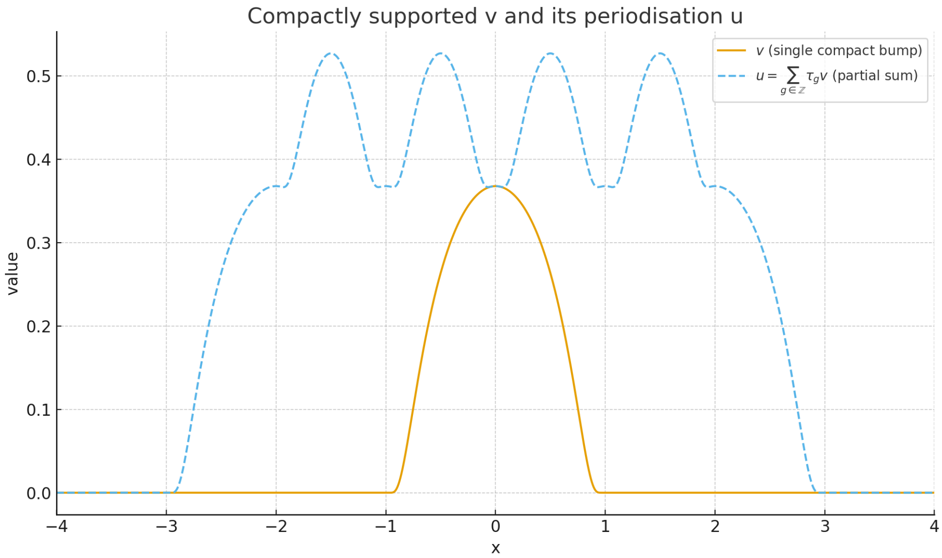


Figure 24: A compactly supported bump v (solid) and a finite partial sum of its periodisation $u = \sum_{g\in\mathbb{Z}} \tau_g v$ (dashed). This illustrates the construction of a periodic distribution from a compactly supported one.

Figure 2: Supports of v, its translates, and a compact K. Figure **??** gives a schematic two-dimensional picture: the black square represents the compact support K_0 of v, while the light grey squares are its translates $K_0 + g$ for $g \in \mathbb{Z}^2$. The larger coloured rectangle marks a compact set K containing the support of a test function ϕ. Only those translates $K_0 + g$ that intersect K can contribute to $u[\phi]$, and there are only finitely many such g, which is the key point in Exercise 5.1.

Figure 3: Lattice shells Q_k in $\mathbb{Z}^2$. Figure **??** shows the shells $Q_k = \{g \in \mathbb{Z}^2 : k - \frac{1}{2} \leq \|g\|_1 < k + \frac{1}{2}\}$ for $1 \leq k \leq 4$, coloured by their radius k. Each Q_k is a diamond-shaped layer in the ℓ^1–metric, consisting exactly of the integer points with $\|g\|_1 = k$. This visualises the decomposition of $\mathbb{Z}^n$ into disjoint shells used in Exercise 5.2 to estimate lattice sums.

Figure 4: Size of Q_k and its asymptotic behaviour. Figure **??** plots $\#Q_k$ for $1 \leq k \leq 10$ in the case $n = 2$, together with a comparison curve $c(1+k)^{n-1}$. The stem plot shows the exact counts $\#Q_k = (2k+1)^2 - (2k-1)^2$, while the broken line demonstrates that the growth is indeed of order $(1+k)^{n-1}$. This is precisely the estimate used in Exercise 5.2 to show that

$$\sum_{g\in\mathbb{Z}^n} \frac{1}{(1+|g|)^{n+1}} < \infty.$$

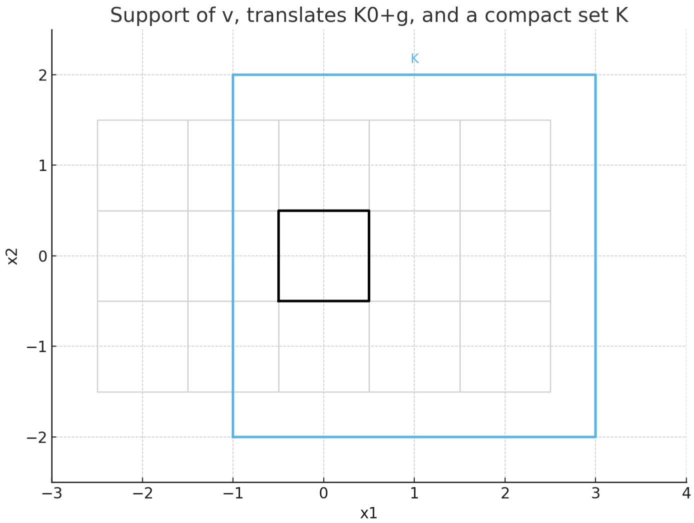


Figure 25: Support K_0 of v (black square), its translates $K_0 + g$ (light grey), and a compact set K containing the support of a test function. Only finitely many translates intersect K, so the sum defining $u[\phi]$ is finite.

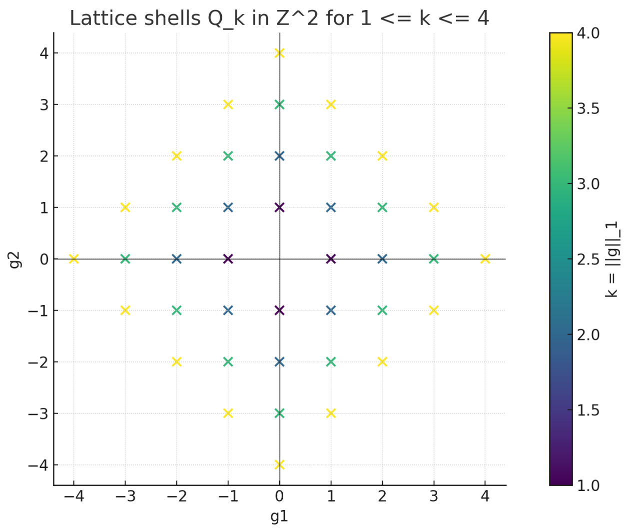


Figure 26: Lattice shells Q_k in $\mathbb{Z}^2$ for $1 \leq k \leq 4$ (coloured by $k = \|g\|_1$). Each shell is a discrete "diamond" in the ℓ^1–norm.

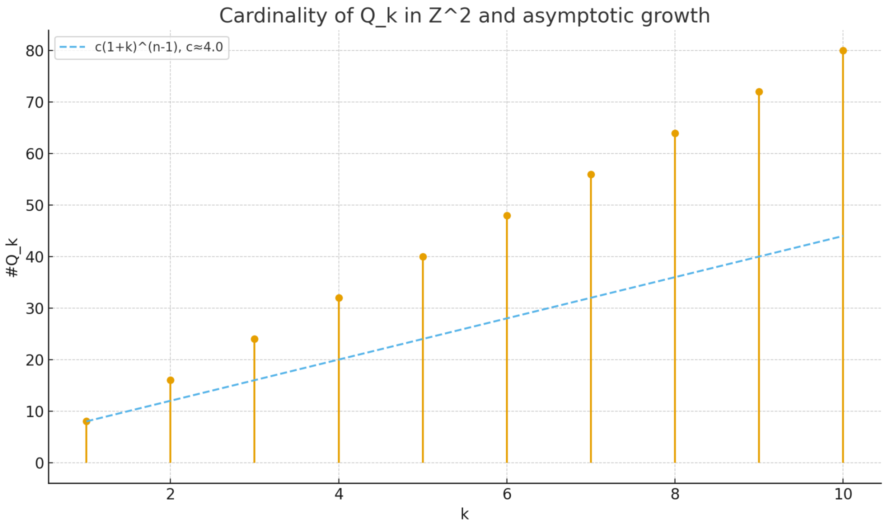


Figure 27: Cardinality $\#Q_k$ for $1 \le k \le 10$ in the case $n = 2$, together with an approximate growth law $c(1+k)^{n-1}$. The asymptotics $\#Q_k \lesssim (1+k)^{n-1}$ is the key input in the summability argument of Exercise 5.2.

Exercise 5.3

Show that if the coefficients c_g satisfy

$$|c_g| \leq K(1+|g|)^N$$

for some $K > 0$ and some $N \in \mathbb{N}$, then the series of distributions

$$\sum_{g\in\mathbb{Z}^n} c_g\, \delta_{2\pi g}$$

converges in $\mathcal{S}'$.

Theory for Exercise 5.3

Let $\mathcal{S}(\mathbb{R}^n)$ denote the Schwartz space of rapidly decreasing smooth functions and $\mathcal{S}'(\mathbb{R}^n)$ its topological dual, the space of tempered distributions. A series $\sum_k T_k$ with $T_k \in \mathcal{S}'$ converges in $\mathcal{S}'$ if and only if the partial sums converge pointwise on $\mathcal{S}$, i.e. for every $\varphi \in \mathcal{S}$ the scalar series $\sum_k T_k[\varphi]$ converges.

For $x \in \mathbb{R}^n$, the Dirac delta δ_x is the tempered distribution defined by $\delta_x[\varphi] = \varphi(x)$. In Exercise 5.3 one considers

$$T = \sum_{g\in\mathbb{Z}^n} c_g\, \delta_{2\pi g},$$

which formally acts on $\varphi \in \mathcal{S}$ by

$$T[\varphi] = \sum_{g\in\mathbb{Z}^n} c_g\, \varphi(2\pi g).$$

The growth condition $|c_g| \leq K(1+|g|)^N$ means that the coefficients grow at most polynomially. On the other hand, for each $\varphi \in \mathcal{S}$ and every $M \geq 0$ there exists $C_M > 0$ such that

$$|\varphi(x)| \leq \frac{C_M}{(1+|x|)^M} \qquad \text{for all } x \in \mathbb{R}^n.$$

Evaluated at $x = 2\pi g$ this yields $|\varphi(2\pi g)| \leq C_M(1+|g|)^{-M}$, so that

$$\sum_{g\in\mathbb{Z}^n} |c_g\varphi(2\pi g)| \leq KC_M \sum_{g\in\mathbb{Z}^n} \frac{1}{(1+|g|)^{M-N}}.$$

If $M-N$ is chosen large enough (e.g. $M-N > n+1$), Exercise 5.2 shows that this lattice sum converges. Thus $T[\varphi]$ is well-defined and absolutely convergent for every $\varphi \in \mathcal{S}$.

Moreover, the estimate above shows that $|T[\varphi]|$ is bounded by a constant multiple of the Schwartz seminorm $\sup_x (1+|x|)^M|\varphi(x)|$, which is exactly the form of continuity required for a tempered distribution. Hence the weighted δ–comb $\sum c_g\delta_{2\pi g}$ converges in $\mathcal{S}'$.

Exercise 5.3 — Convergence of $\sum_g c_g \delta_{2\pi g}$ in $\mathcal{S}'$

We assume

$$|c_g| \le K(1+|g|)^N \qquad (g \in \mathbb{Z}^n)$$

for some constants $K > 0$, $N \in \mathbb{N}$, and want to show that

$$T := \sum_{g \in \mathbb{Z}^n} c_g \, \delta_{2\pi g}$$

converges in $\mathcal{S}'(\mathbb{R}^n)$.

Step 1: Define T on a test function

Take arbitrary $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Formally,

$$T[\varphi] = \sum_g c_g \, \delta_{2\pi g}[\varphi] = \sum_g c_g \, \varphi(2\pi g).$$

We must show this series converges absolutely and defines a continuous linear functional.

Step 2: Use rapid decay of φ

Since φ is Schwartz, for every integer $M \ge 0$,

$$\sup_{x \in \mathbb{R}^n} (1+|x|)^M |\varphi(x)| =: C_M < \infty.$$

Hence for all $g \in \mathbb{Z}^n$,

$$|\varphi(2\pi g)| \le \frac{C_M}{(1+|2\pi g|)^M} \le \frac{C'_M}{(1+|g|)^M},$$

for some constant C'_M depending on M but not on g.

Step 3: Absolute convergence

Then

$$\sum_g |c_g \varphi(2\pi g)| \le K C'_M \sum_g \frac{1}{(1+|g|)^{M-N}}.$$

Choose M such that $M - N > n + 1$. By Exercise 5.2, the lattice sum converges. Hence the series defining $T[\varphi]$ is absolutely convergent.

Step 4: Continuity

From the same estimate,

$$|T[\varphi]| \le C \sup_{x \in \mathbb{R}^n} (1+|x|)^M |\varphi(x)|$$

for some constant $C > 0$ independent of φ. Thus T is continuous on $\mathcal{S}$, so $T \in \mathcal{S}'$.

Conclusion: The series $\sum_g c_g \, \delta_{2\pi g}$ converges in $\mathcal{S}'$.

Exercise 5.4

Suppose $f \in L^p_{\mathrm{loc}}(\mathbb{R}^n)$ is a periodic function. Fix $\varepsilon > 0$, and define

$$Q = \{x \in \mathbb{R}^n : |x_j| < 1,\ j = 1, \dots, n\}, \qquad q = \{x \in \mathbb{R}^n : |x_j| < \tfrac{1}{2},\ j = 1, \dots, n\}.$$

(a) Show that there exists $h_\varepsilon \in C^\infty(\mathbb{R}^n)$ with $\operatorname{supp} h_\varepsilon \subset Q$ such that

$$\|f\,\mathbf{1}_q - h_\varepsilon\|_{L^p(\mathbb{R}^n)} < \varepsilon.$$

Define

$$f_\varepsilon = \sum_{g \in \mathbb{Z}^n} \tau_g h_\varepsilon.$$

(b) Show that f_ε is smooth and periodic.

(c) Show that there exists a constant c_n, depending only on n, such that

$$\|f - f_\varepsilon\|_{L^p(q)} < c_n\,\varepsilon.$$

Theory for Exercise 5.4

Let $f \in L^p_{\mathrm{loc}}(\mathbb{R}^n)$ be a periodic function with period lattice $\mathbb{Z}^n$. It is natural to think of f as a function on the torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$. The goal of Exercise 5.4 is to approximate f in L^p on a fundamental domain by smooth periodic functions.

We work with the cubes

$$Q = \{x \in \mathbb{R}^n : |x_j| < 1,\ j = 1, \dots, n\}, \qquad q = \{x \in \mathbb{R}^n : |x_j| < \tfrac{1}{2},\ j = 1, \dots, n\}.$$

A standard fact in L^p–theory is that $C_c^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$ for $1 \le p < \infty$. In particular, any L^p-function supported in a bounded set can be approximated in L^p by smooth compactly supported functions with support in a slightly larger set. Applying this to $f\mathbf{1}_q$ gives, for each $\varepsilon > 0$, the existence of $h_\varepsilon \in C^\infty(\mathbb{R}^n)$ with $\operatorname{supp} h_\varepsilon \subset Q$ and $\|f\mathbf{1}_q - h_\varepsilon\|_{L^p(\mathbb{R}^n)} < \varepsilon$.

The function

$$f_\varepsilon(x) := \sum_{g \in \mathbb{Z}^n} h_\varepsilon(x - g)$$

is then well-defined, because at each x only finitely many translates $x - g$ lie in Q and contribute to the sum. Being locally a finite sum of smooth functions, f_ε is smooth. Moreover, f_ε is $\mathbb{Z}^n$–periodic, since for any $k \in \mathbb{Z}^n$

$$f_\varepsilon(x + k) = \sum_{g \in \mathbb{Z}^n} h_\varepsilon(x + k - g) = \sum_{h \in \mathbb{Z}^n} h_\varepsilon(x - h) = f_\varepsilon(x),$$

after the change of variables $h = g - k$.

To control $\|f - f_\varepsilon\|_{L^p(q)}$ one uses the periodicity of f and the fact that h_ε approximates $f\mathbf{1}_q$ on each translate $q + g$ in the same way. Since at most a bounded number of translates of Q overlap a given point (a constant depending only on n), the total error on q is controlled by a finite multiple of the error $\|f\mathbf{1}_q - h_\varepsilon\|_{L^p}$. This yields a bound $\|f - f_\varepsilon\|_{L^p(q)} \le c_n\varepsilon$, where $c_n > 0$ depends only on the dimension.

Exercise 5.4 — Smooth Approximation of a Periodic L^p Function

Let $f \in L^p_{\mathrm{loc}}(\mathbb{R}^n)$ be periodic. Define the cubes

$$Q = \{x : |x_j| < 1\}, \qquad q = \{x : |x_j| < \tfrac{1}{2}\}.$$

(a) Construction of h_ε

Choose $\chi \in C_c^\infty(\mathbb{R}^n)$ with $\chi \equiv 1$ on q and $\operatorname{supp}\chi \subset Q$. Set $g := f\chi$. Then $\operatorname{supp} g \subset Q$ and $g = f$ on q.

Since C_c^∞ is dense in L^p, choose $h_\varepsilon \in C^\infty$ with $\operatorname{supp} h_\varepsilon \subset Q$ and

$$\|g - h_\varepsilon\|_{L^p(\mathbb{R}^n)} < \varepsilon.$$

But $f1_q = g$, so

$$\|f1_q - h_\varepsilon\|_{L^p(\mathbb{R}^n)} < \varepsilon.$$

Define

$$f_\varepsilon(x) := \sum_{g \in \mathbb{Z}^n} h_\varepsilon(x - g) = \sum_{g \in \mathbb{Z}^n} \tau_g h_\varepsilon(x).$$

(b) f_ε is smooth and periodic

Since $\operatorname{supp} h_\varepsilon \subset Q$, at each x only finitely many terms are nonzero. Thus $f_\varepsilon \in C^\infty(\mathbb{R}^n)$.

For periodicity:

$$f_\varepsilon(x + k) = \sum_g h_\varepsilon(x + k - g) = \sum_h h_\varepsilon(x - h) = f_\varepsilon(x),$$

after substituting $h = g - k$. Thus f_ε is $\mathbb{Z}^n$-periodic.

(c) Estimate on q

If $x \in q$ and $h_\varepsilon(x-g) \neq 0$, then $x - g \in Q$, so g lies in the finite set

$$G = \{g \in \mathbb{Z}^n : (Q+g) \cap q \neq \emptyset\}.$$

Thus

$$f_\varepsilon(x) = \sum_{g \in G} h_\varepsilon(x-g).$$

Since f is periodic,

$$f(x-g) = f(x),$$

and by translation invariance,

$$\|f 1_q - h_\varepsilon(\cdot - g)\|_{L^p(\mathbb{R}^n)} < \varepsilon, \quad g \in G.$$

On q,

$$f(x) - f_\varepsilon(x) = \sum_{g \in G} \Big(f(x) - h_\varepsilon(x-g)\Big),$$

so

$$\|f - f_\varepsilon\|_{L^p(q)} \leq \sum_{g \in G} \|f - h_\varepsilon(\cdot - g)\|_{L^p(q)} \leq (\#G)\varepsilon.$$

Let $c_n := \#G$. Then

$$\|f - f_\varepsilon\|_{L^p(q)} < c_n \varepsilon.$$

Exercise 5.5

Suppose that $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = x \quad \text{for } |x| < \tfrac{1}{2}, \qquad f(x+1) = f(x).$$

Show that

$$f(x) = \sum_{\substack{n \in \mathbb{Z} \\ n \neq 0}} \frac{i(-1)^n}{2\pi n} e^{2\pi i n x} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\pi} \sin(2\pi n x),$$

with convergence in $L^2_{\text{loc}}(\mathbb{R})$.

Theory for Exercise 5.5

In Exercise 5.5 the function $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = x \quad \text{for } |x| < \tfrac{1}{2}, \qquad f(x+1) = f(x),$$

so that on each period $(-\frac{1}{2}, \frac{1}{2})$ the graph of f is a straight line from $-1/2$ to $1/2$, followed by a jump back to $-1/2$. The function f is odd and belongs to $L^2([-1/2, 1/2])$, hence defines an element of $L^2(\mathbb{T})$.

For a 1-periodic L^2 function, the complex Fourier coefficients are given by

$$\widehat{f}(n) = \int_{-1/2}^{1/2} f(x)\, e^{-2\pi i n x}\, dx, \qquad n \in \mathbb{Z}.$$

In this case $\widehat{f}(0) = \int_{-1/2}^{1/2} x\, dx = 0$, and for $n \neq 0$ an integration by parts shows that

$$\widehat{f}(n) = \int_{-1/2}^{1/2} x\, e^{-2\pi i n x}\, dx = \frac{i(-1)^n}{2\pi n}.$$

Thus the complex Fourier series of f is

$$f(x) = \sum_{\substack{n\in\mathbb{Z}\\ n\neq 0}} \frac{i(-1)^n}{2\pi n}\, e^{2\pi i n x}$$

with convergence in $L^2([-1/2, 1/2])$, and hence in $L^2_{\mathrm{loc}}(\mathbb{R})$.

Since f is an odd function, only sine terms appear in the real Fourier series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(2\pi n x).$$

A direct computation (or passage from the complex coefficients) gives

$$b_n = 2\int_0^{1/2} x \sin(2\pi n x)\, dx = \frac{2(-1)^{n+1}}{\pi n},$$

so that

$$f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\pi} \sin(2\pi n x).$$

Fourier theory on $L^2(\mathbb{T})$ ensures that this series converges to f in $L^2([-1/2, 1/2])$, which is equivalent to convergence in $L^2_{\mathrm{loc}}(\mathbb{R})$. Pointwise, the series converges to $f(x)$ at all continuity points of f, and to the midpoint of the jump at the discontinuities, but the exercise only requires L^2_{loc} convergence.

Exercise 5.5 — Fourier Series of the Sawtooth Function

Let

$$f(x) = x \quad (|x| < \tfrac{1}{2}), \qquad f(x+1) = f(x).$$

Step 1: Complex Fourier coefficients

For 1-periodic f,

$$\widehat{f}(n) = \int_{-1/2}^{1/2} f(x) e^{-2\pi i n x}\, dx.$$

For $n = 0$:

$$\widehat{f}(0) = 0.$$

For $n \neq 0$:

$$\widehat{f}(n) = \int_{-1/2}^{1/2} x e^{-2\pi i n x}\, dx = \frac{i(-1)^n}{2\pi n}.$$

Thus

$$f(x) = \sum_{\substack{n\in\mathbb{Z}\\ n\neq 0}} \frac{i(-1)^n}{2\pi n} e^{2\pi i n x} \qquad \text{in } L^2_{\text{loc}}.$$

Step 2: Real sine series

Since f is odd, only sine terms appear:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(2\pi n x).$$

Using $\widehat{f}(n) = -\frac{i}{2} b_n$ for $n > 0$ gives

$$b_n = \frac{2(-1)^{n+1}}{\pi n}.$$

Therefore:

$$f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\pi} \sin(2\pi n x), \qquad \text{convergent in } L^2_{\text{loc}}(\mathbb{R}).$$

Figures for Exercises 5.3–5.5

Figure 1: Weighted δ–comb for Exercise 5.3. Figure **??** illustrates a discrete "δ–comb" at the points $x = 2\pi g$, $g \in \mathbb{Z}$, with example coefficients $c_g = (1 + |g|)^2$. The vertical stems represent the weights of the Dirac deltas. The exercise shows that even with polynomially growing coefficients such a series still defines a tempered distribution in $\mathcal{S}'$.

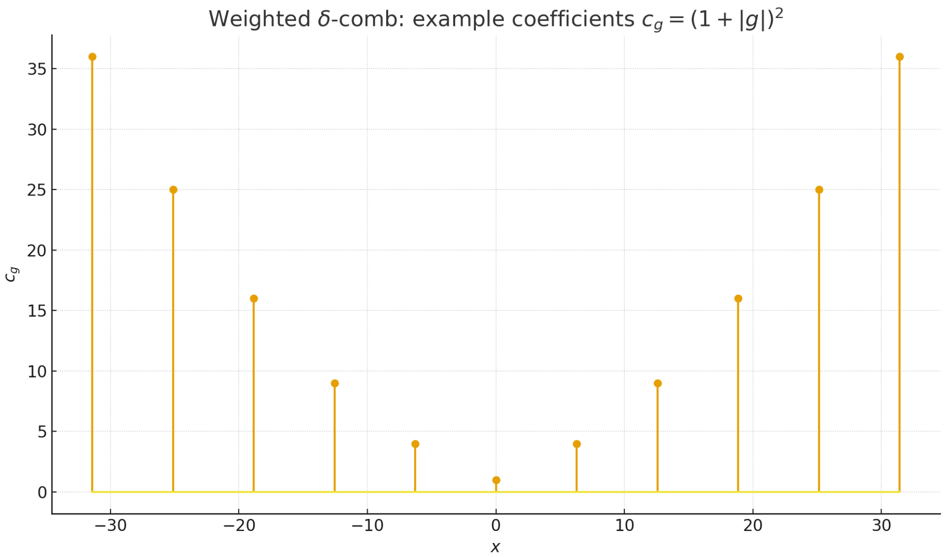


Figure 28: A weighted δ–comb at $x = 2\pi g$ with coefficients $c_g = (1 + |g|)^2$, visualised as a stem plot.

Figure 2: Decay of the contributions $|c_g \varphi(2\pi g)|$. Figure **??** shows $|c_g \varphi(2\pi g)|$ as a function of g for the same polynomial coefficients c_g and a Schwartz function $\varphi(x) = e^{-x^2}$. Although $|c_g|$ grows like $(1 + |g|)^2$, the values $\varphi(2\pi g)$ decay faster than any power, so the product is summable. This picture reflects the key estimate used to prove convergence in $\mathcal{S}'$.

Figure 3: Periodic L^p function and smooth approximation (Exercise 5.4). In Figure **??** the original periodic function f is taken to be a square wave, while f_ε is obtained by periodic convolution with a narrow Gaussian kernel. The resulting f_ε is smooth and periodic, and closely follows f on each period, visualising the construction in Exercise 5.4.

Figure 4: Fourier partial sums of the sawtooth (Exercise 5.5). Figure **??** shows the sawtooth function f from Exercise 5.5 together with its Fourier partial sums S_3, S_{10}

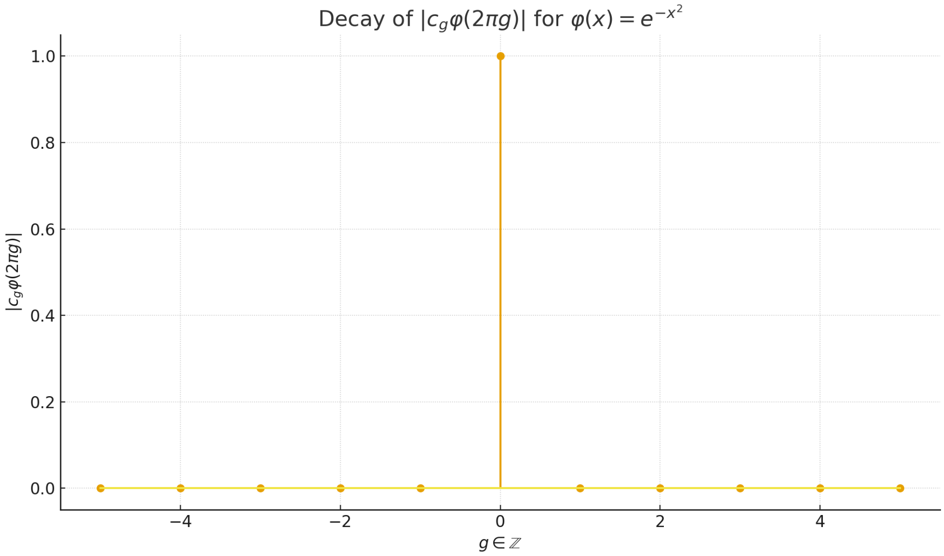


Figure 29: The terms $|c_g\varphi(2\pi g)|$ for $c_g = (1+|g|)^2$ and $\varphi(x) = e^{-x^2}$, illustrating absolute convergence of the defining series.

and S_{50}. As N increases, S_N converges to f in L^2 and visually away from the jump points, while oscillations appear near the discontinuities.

Figure 5: Gibbs phenomenon near a jump. Figure **??** zooms in near one jump of f and compares the partial sums S_{10} and S_{50}. The overshoot of the Fourier partial sums does not vanish as $N \to \infty$, illustrating the classical Gibbs phenomenon, even though convergence holds in L^2_{loc}.

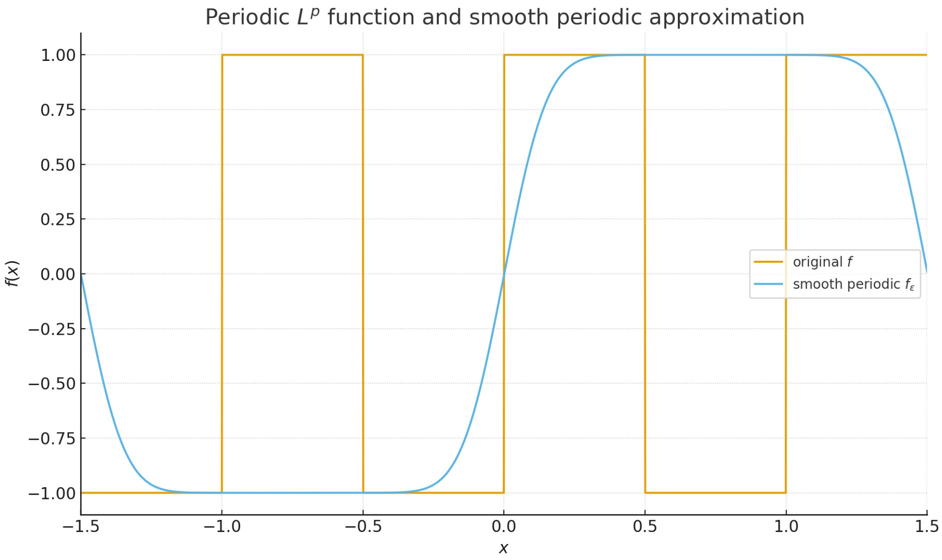


Figure 30: A periodic square wave f and a smooth periodic approximation f_ε obtained by mollification. This illustrates the approximation scheme of Exercise 5.4.

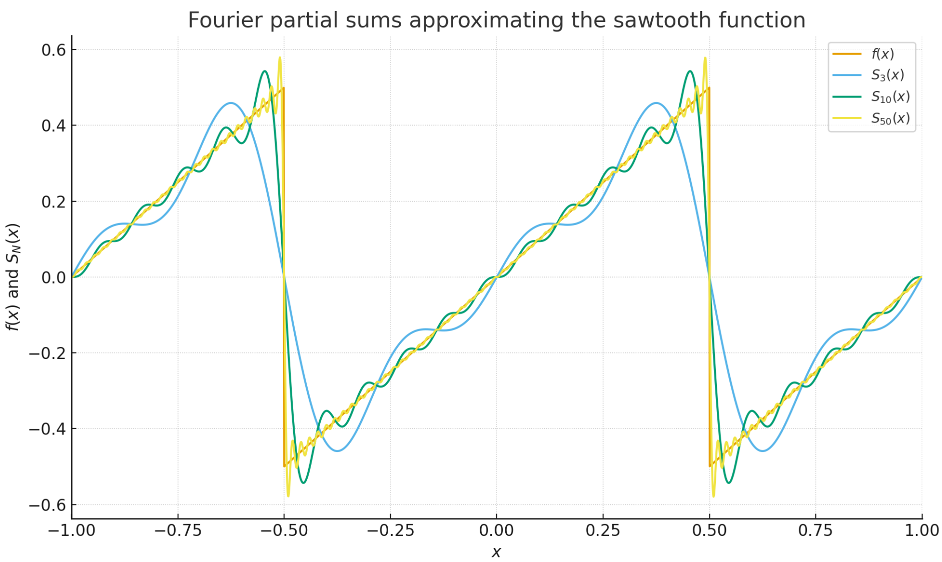


Figure 31: Global comparison of the sawtooth function f and partial sums S_N of its sine series for $N = 3, 10, 50$.

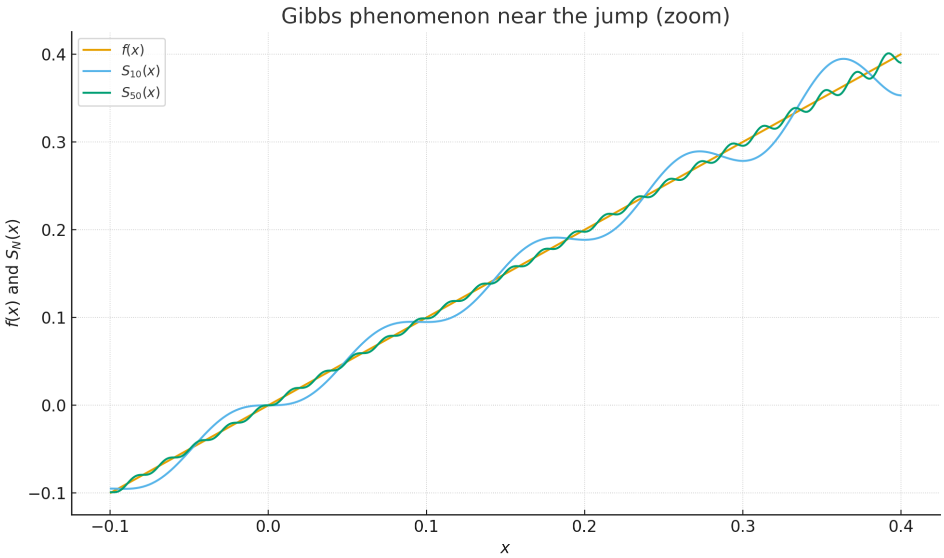


Figure 32: Zoom near a jump of the sawtooth function, showing the Gibbs oscillations of the partial sums S_{10} and S_{50}.

Exercise 5.6 — Gibbs Phenomenon for a Square Wave

Suppose $f : \mathbb{R} \to \mathbb{R}$ is given by

$$f(x) = \begin{cases} -1, & -\frac{1}{2} \le x \le 0, \\ 1, & 0 < x \le \frac{1}{2}, \end{cases} \qquad \text{and} \qquad f(x+1) = f(x) \text{ for all } x \in \mathbb{R}.$$

(a) Show that

$$f(x) = \frac{1}{\pi i} \sum_{n=-\infty}^{\infty} \frac{2}{2n+1} e^{2\pi i(2n+1)x} = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} \sin\Big(2\pi(2n+1)x\Big),$$

with convergence in $L^2_{\mathrm{loc}}(\mathbb{R})$.

Define the partial sums

$$S_N(x) := 8 \sum_{n=0}^{N-1} \frac{1}{2\pi(2n+1)} \sin\Big(2\pi(2n+1)x\Big).$$

(b) Show that

$$S_N(x) = 8 \int_0^x \sum_{n=0}^{N-1} \cos\Big(2\pi(2n+1)t\Big)\, dt.$$

(c) Show that

$$\cos\Big(2\pi(2n+1)t\Big)\, \sin(2\pi t) = \frac{1}{2}\Big(\sin\Big(2\pi(2n+2)t\Big) - \sin(4\pi n t)\Big),$$

and deduce that

$$S_N(x) = 8 \int_0^x \frac{\sin(4\pi N t)}{2\sin(2\pi t)}\, dt.$$

(d) Show that the first local maximum of S_N occurs at $x = \frac{1}{4N}$, and that

$$S_N\Big(\frac{1}{4N}\Big) \ \ge \ 8 \int_0^{1/N} \frac{\sin(4\pi N t)}{4\pi t}\, dt = \frac{2}{\pi} \int_0^{\pi} \frac{\sin s}{s}\, ds \simeq 1.179\ldots$$

(e) Conclude that the Fourier series in part (a) does not converge uniformly. This lack of uniform convergence of a Fourier series at a point of discontinuity is known as the *Gibbs phenomenon*.

Exercise 5.6 — Gibbs Phenomenon for a Square Wave

Suppose $f : \mathbb{R} \to \mathbb{R}$ is given by

$$f(x) = \begin{cases} -1, & -\frac{1}{2} \le x \le 0, \\ 1, & 0 < x \le \frac{1}{2}, \end{cases} \qquad \text{and} \qquad f(x+1) = f(x) \text{ for all } x \in \mathbb{R}.$$

Theoretical background. The function f is 1-periodic, odd, and has jump discontinuities at all half-integers. For a 1-periodic integrable function f, the complex Fourier coefficients are

$$\widehat{f}(k) = \int_{-1/2}^{1/2} f(x) e^{-2\pi i k x}\, dx, \qquad k \in \mathbb{Z},$$

and one formally has

$$f(x) \sim \sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{2\pi i k x}.$$

Because f is odd, only sine terms (and hence only odd harmonics) appear in the real Fourier series. Dirichlet's theorem implies that the Fourier series converges to $f(x)$ at points of continuity, and to the midpoint of the jump at discontinuities. However, the convergence is not uniform near a jump: the partial sums exhibit a persistent overshoot known as the *Gibbs phenomenon*. This exercise quantifies that overshoot for the square wave f.

(a) Fourier series of the square wave. We compute the complex Fourier coefficients

$$\widehat{f}(k) = \int_{-1/2}^{1/2} f(x) e^{-2\pi i k x}\, dx.$$

Since f is odd, $\widehat{f}(k) = 0$ for all even k. For $k = 2n+1$ we have

$$\widehat{f}(2n+1) = \int_{-1/2}^{1/2} f(x) e^{-2\pi i (2n+1) x}\, dx = \frac{2}{\pi i (2n+1)}.$$

Therefore

$$f(x) = \sum_{n \in \mathbb{Z}} \widehat{f}(2n+1) e^{2\pi i (2n+1) x} = \frac{1}{\pi i} \sum_{n=-\infty}^{\infty} \frac{2}{2n+1} e^{2\pi i (2n+1) x}.$$

Grouping conjugate terms and passing to the real sine series, we obtain

$$f(x) = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} \sin\Big(2\pi(2n+1)x\Big),$$

with convergence in $L^2([-1/2, 1/2])$ and hence in $L^2_{\mathrm{loc}}(\mathbb{R})$.

Define the partial sums

$$S_N(x) := \frac{4}{\pi} \sum_{n=0}^{N-1} \frac{1}{2n+1} \sin\Big(2\pi(2n+1)x\Big) = 8 \sum_{n=0}^{N-1} \frac{1}{2\pi(2n+1)} \sin\Big(2\pi(2n+1)x\Big).$$

(b) Integral representation of S_N. Using

$$\frac{d}{dt}\sin\Big(2\pi(2n+1)t\Big) = 2\pi(2n+1)\cos\Big(2\pi(2n+1)t\Big),$$

we write

$$\sin\Big(2\pi(2n+1)x\Big) = \int_0^x 2\pi(2n+1)\cos\Big(2\pi(2n+1)t\Big)\,dt,$$

and hence

$$\frac{1}{2\pi(2n+1)}\sin\Big(2\pi(2n+1)x\Big) = \int_0^x \cos\Big(2\pi(2n+1)t\Big)\,dt.$$

Summing over n gives

$$S_N(x) = 8\sum_{n=0}^{N-1}\int_0^x \cos\Big(2\pi(2n+1)t\Big)\,dt = 8\int_0^x \sum_{n=0}^{N-1}\cos\Big(2\pi(2n+1)t\Big)\,dt.$$

(c) Trigonometric identity and telescoping. Recall the identity

$$2\cos A\sin B = \sin(A+B) - \sin(A-B).$$

Setting $A = 2\pi(2n+1)t$, $B = 2\pi t$, we find

$$\cos\Big(2\pi(2n+1)t\Big)\sin(2\pi t) = \frac{1}{2}\Big(\sin\Big(2\pi(2n+2)t\Big) - \sin(4\pi nt)\Big).$$

Thus

$$\sum_{n=0}^{N-1}\cos\Big(2\pi(2n+1)t\Big) = \frac{1}{2\sin(2\pi t)}\sum_{n=0}^{N-1}\Big(\sin\Big(2\pi(2n+2)t\Big) - \sin(4\pi nt)\Big).$$

The sum on the right is telescoping and simplifies to $\sin(4\pi Nt)$, whence

$$\sum_{n=0}^{N-1}\cos\Big(2\pi(2n+1)t\Big) = \frac{\sin(4\pi Nt)}{2\sin(2\pi t)}.$$

Substituting this back into the formula in part (b) yields

$$S_N(x) = 8\int_0^x \frac{\sin(4\pi Nt)}{2\sin(2\pi t)}\,dt.$$

(d) First local maximum and lower bound. Differentiating S_N gives

$$S_N'(x) = 8\sum_{n=0}^{N-1}\cos\Big(2\pi(2n+1)x\Big) = 4\,\frac{\sin(4\pi Nx)}{\sin(2\pi x)}.$$

Critical points occur when $\sin(4\pi Nx) = 0$ and $x \notin \mathbb{Z}$. Near $x = 0$ the first such zero is at $x = \frac{1}{4N}$; inspection of the sign of S_N' shows this is the first local maximum.

For small t we have $\sin(2\pi t) \leq 2\pi t$, hence $2\sin(2\pi t) \leq 4\pi t$ and so

$$\frac{1}{2\sin(2\pi t)} \geq \frac{1}{4\pi t}.$$

Thus

$$S_N(x) = 8\int_0^x \frac{\sin(4\pi N t)}{2\sin(2\pi t)}\,dt \;\geq\; 8\int_0^x \frac{\sin(4\pi N t)}{4\pi t}\,dt.$$

Taking $x = \frac{1}{4N}$ and substituting $s = 4\pi N t$, so $dt = \frac{ds}{4\pi N}$ and s runs from 0 to π, we obtain

$$S_N\Big(\frac{1}{4N}\Big) \geq 8\int_0^{1/(4N)} \frac{\sin(4\pi N t)}{4\pi t}\,dt = \frac{2}{\pi}\int_0^{\pi} \frac{\sin s}{s}\,ds \simeq 1.179\ldots$$

for all N.

(e) Non-uniform convergence and Gibbs phenomenon. At the discontinuity $x = 0$, Dirichlet's theorem implies that

$$\lim_{N\to\infty} S_N(0) = \frac{f(0^-) + f(0^+)}{2} = 0.$$

However, for $x_N = \frac{1}{4N}$ we have

$$S_N(x_N) \gtrsim 1.179,$$

while $f(x_N) \to 1$ as $N \to \infty$. Hence

$$\limsup_{N\to\infty} \sup_{x\in[-1/2,1/2]} |S_N(x) - f(x)| \geq \limsup_{N\to\infty} |S_N(x_N) - 1| \geq 1.179 - 1 > 0,$$

so the Fourier series in part (a) does *not* converge uniformly on any interval containing 0. This persistent overshoot near the jump is the *Gibbs phenomenon.*

Figures. Figure **??** illustrates the partial sums of the Fourier series, while Figure **??** zooms in near $x = 0$ to show the Gibbs overshoot.

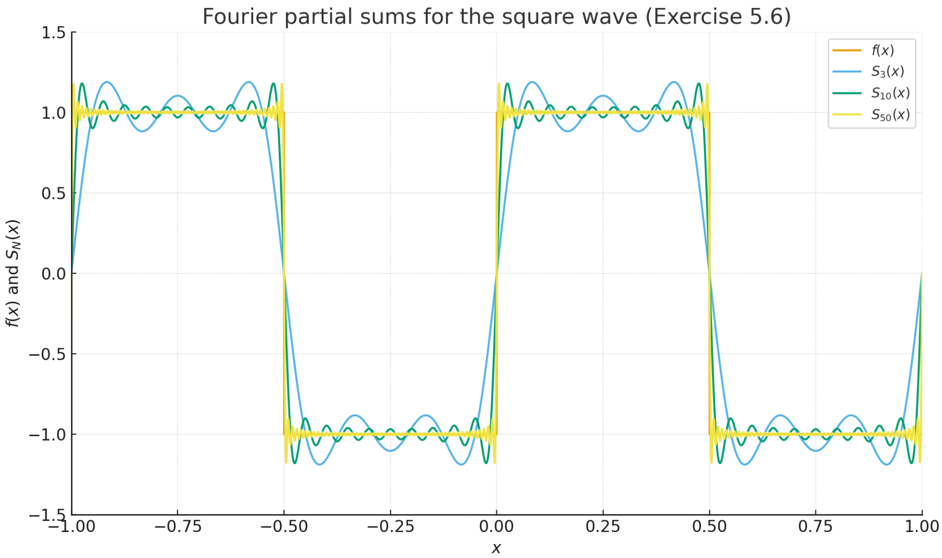


Figure 33: Global comparison of the square wave f and the partial sums S_3, S_{10} and S_{50} of its Fourier series.

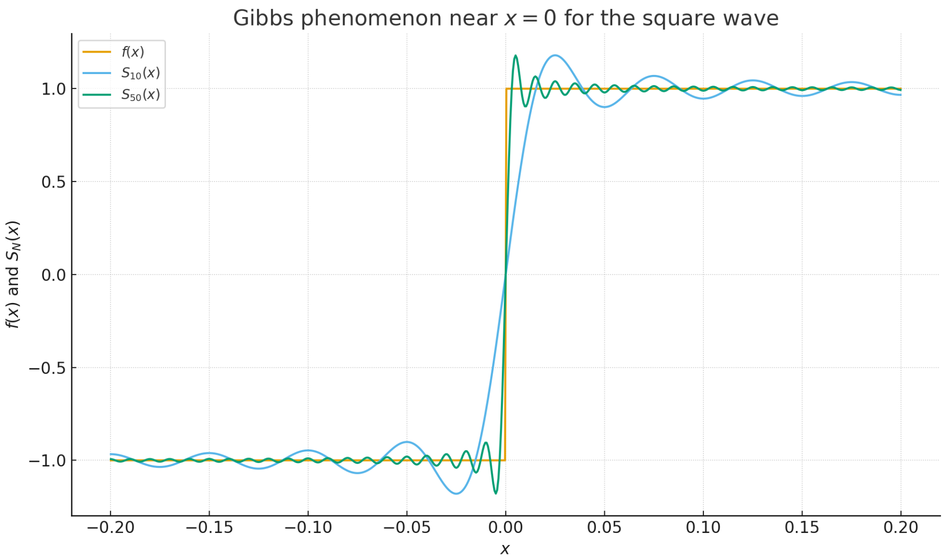


Figure 34: Zoom near $x = 0$ showing the Gibbs overshoot of the partial sums S_{10} and S_{50} for the square wave.

Exercise 5.7 — Λ–periodic distributions and the dual lattice

Suppose that $\lambda = \{\lambda_1, \dots, \lambda_n\}$ is a basis for $\mathbb{R}^n$. We define the lattice generated by λ to be

$$\Lambda := \left\{ \sum_{j=1}^{n} z_j \lambda_j : z_j \in \mathbb{Z} \right\}.$$

Define the fundamental cell

$$q_\Lambda := \left\{ \sum_{j=1}^{n} x_j \lambda_j : |x_j| < \tfrac{1}{2} \right\}.$$

For a distribution $u \in \mathcal{D}'(\mathbb{R}^n)$, we say that u is Λ*–periodic* if

$$\tau_g u = u \qquad \text{for all } g \in \Lambda,$$

where τ_g denotes translation by g.

(a) Show that there exists $\psi \in C_0^\infty(2q_\Lambda)$ such that $\psi \geq 0$ and

$$\sum_{g \in \Lambda} \tau_g \psi = 1.$$

(b) Show that if $u \in \mathcal{D}'(\mathbb{R}^n)$ is Λ–periodic and ψ, ψ' are both as in part (a), then

$$\frac{1}{|q_\Lambda|}\, u[\psi] = \frac{1}{|q_\Lambda|}\, u[\psi'] =: M(u),$$

so that the "mean value" $M(u)$ is well defined and independent of the choice of ψ.

(c) Dual lattice. Define the *dual lattice* by

$$\Lambda^* := \{x \in \mathbb{R}^n : g \cdot x \in 2\pi\mathbb{Z},\ \forall g \in \Lambda\}.$$

Show that there exists a basis $\lambda^* = \{\lambda_1^*, \dots, \lambda_n^*\}$ such that

$$\lambda_j^* \cdot \lambda_k = \delta_{jk},$$

and that Λ^* is precisely the lattice induced by λ^*, i.e. $\Lambda^* = \{\sum_{j=1}^n m_j \lambda_j^* : m_j \in \mathbb{Z}\}$.

(d) Periodicity of exponential characters. For $g \in \mathbb{R}^n$ define the exponential

$$e_g(x) := e^{i\, g \cdot x}.$$

Show that if $g \in \Lambda^*$, then e_g is Λ–periodic:

$$e_g(x + g_0) = e_g(x) \qquad \text{for all } x \in \mathbb{R}^n,\ g_0 \in \Lambda.$$

(e) Fourier transform of a Λ–periodic distribution. Show that if $u \in \mathcal{D}'(\mathbb{R}^n)$ is Λ–periodic, then its Fourier transform has the purely discrete form

$$\widehat{u} = \sum_{g\in\Lambda^*} c_g\, \delta_g,$$

for some coefficients $c_g \in \mathbb{C}$ satisfying a polynomial growth estimate

$$|c_g| \le K(1+|g|)^N$$

for some constants $K > 0$ and $N \in \mathbb{Z}$ (independent of g).

(f) Fourier series representation. Show conversely that if $u \in \mathcal{D}'(\mathbb{R}^n)$ is Λ–periodic, then

$$u = \sum_{g\in\Lambda^*} d_g\, e_g,$$

where

$$|d_g| \le K(1+|g|)^N \quad \text{for some } K > 0,\ N \in \mathbb{Z}, \text{ and all } g \in \Lambda^*,$$

and the coefficients are given by

$$d_g = M\Big(e_{-g}u\Big).$$

Theoretical background. A lattice $\Lambda \subset \mathbb{R}^n$ is a discrete subgroup generated by a basis of $\mathbb{R}^n$, and the fundamental cell q_Λ is a parallelepiped that tiles the space under translations by Λ. Functions or distributions that are invariant under translations by Λ may be regarded as living on the torus $\mathbb{R}^n/\Lambda$; their "mean value" $M(u)$ plays the role of an average over one period.

The dual lattice Λ^* consists of all frequency vectors g such that the exponentials $e_g(x) = e^{i\,g\cdot x}$ are Λ–periodic. These exponentials form an orthogonal basis of L^2 on the torus, and their indices run over Λ^*. Parts (e) and (f) describe how a Λ–periodic distribution decomposes into a Fourier series with frequencies in Λ^*: the Fourier transform is a discrete sum of Dirac masses at Λ^*, and u can be reconstructed from the series $\sum_{g\in\Lambda^*} d_g e_g$ with polynomially growing coefficients. This is the natural generalization of Fourier series on $\mathbb{T}^n$ and is closely related to the Poisson summation formula.

Solution to Exercise 5.7

Throughout, $\lambda = \{\lambda_1, \dots, \lambda_n\}$ is a basis of $\mathbb{R}^n$, Λ is the lattice it generates, and q_Λ is the fundamental cell

$$q_\Lambda = \left\{ \sum_{j=1}^n x_j\lambda_j : |x_j| < \tfrac{1}{2} \right\}.$$

(a) Construction of ψ. Choose $\chi \in C_0^\infty(2q_\Lambda)$ with $0 \le \chi \le 1$ and $\chi > 0$ on q_Λ (for instance, a smooth bump which is strictly positive on a slightly smaller cell containing $\overline{q_\Lambda}$ and vanishes near the boundary of $2q_\Lambda$). Define

$$S(x) := \sum_{g\in\Lambda} \chi(x-g).$$

Because the translates of q_Λ by Λ tile $\mathbb{R}^n$ and $\chi > 0$ on q_Λ, each $x \in \mathbb{R}^n$ lies in some translate $q_\Lambda + g$ where $\chi(x-g) > 0$. Hence $S(x) > 0$ and S is smooth and Λ–periodic.

Now set

$$\psi(x) := \frac{\chi(x)}{S(x)}.$$

Then $\psi \in C_0^\infty(2q_\Lambda)$, $\psi \ge 0$, and

$$\sum_{g\in\Lambda} \tau_g\psi(x) = \sum_{g\in\Lambda} \frac{\chi(x-g)}{S(x-g)} = \frac{1}{S(x)} \sum_{g\in\Lambda} \chi(x-g) = \frac{S(x)}{S(x)} = 1.$$

This proves (a).

(b) Mean value is independent of ψ. Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be Λ–periodic and let ψ, ψ' both be as in (a). We must show $u[\psi] = u[\psi']$.

Note that

$$\sum_{g\in\Lambda} \tau_g\psi = \sum_{g\in\Lambda} \tau_g\psi' = 1,$$

hence

$$\sum_{g\in\Lambda} \tau_g(\psi - \psi') = 0.$$

Let $\varphi \in C_0^\infty(\mathbb{R}^n)$ be any test function with $\varphi \equiv 1$ on $2q_\Lambda$; then $\varphi(\psi - \psi') = \psi - \psi'$ and we may write

$$\psi - \psi' = \left(\sum_{g\in\Lambda} \tau_g(\psi - \psi')\right)\varphi = 0,$$

because at each point only finitely many translates intersect the support of φ. Applying the distribution u gives

$$u[\psi - \psi'] = u[0] = 0,$$

hence $u[\psi] = u[\psi']$. Therefore

$$M(u) := \frac{1}{|q_\Lambda|} u[\psi]$$

is well defined and independent of ψ.

(One may regard $M(u)$ as the average of u over the torus $\mathbb{R}^n/\Lambda$.)

(c) The dual lattice and dual basis. By definition, the dual lattice is

$$\Lambda^* := \{x \in \mathbb{R}^n : g \cdot x \in 2\pi\mathbb{Z},\ \forall g \in \Lambda\}.$$

Since $\lambda_1, \dots, \lambda_n$ form a basis of $\mathbb{R}^n$, there exists a unique dual basis $(\lambda_1^*, \dots, \lambda_n^*)$ in the dual space $(\mathbb{R}^n)^*$ such that $\lambda_j^*(\lambda_k) = \delta_{jk}$. Using the inner product $v \mapsto (x \mapsto x \cdot v)$ to identify $\mathbb{R}^n$ with its dual, we may regard each λ_j^* as a vector in $\mathbb{R}^n$ and the duality becomes

$$\lambda_j^* \cdot \lambda_k = \delta_{jk}.$$

Now let $g \in \Lambda^*$. Write $g = \sum_{j=1}^n a_j \lambda_j^*$. Then for each k,

$$\lambda_k \cdot g = \lambda_k \cdot \sum_{j=1}^n a_j \lambda_j^* = \sum_{j=1}^n a_j (\lambda_k \cdot \lambda_j^*) = a_k.$$

Since $\lambda_k \in \Lambda$, the condition $g \cdot \lambda_k \in 2\pi\mathbb{Z}$ forces $a_k \in 2\pi\mathbb{Z}$, say $a_k = 2\pi m_k$ with $m_k \in \mathbb{Z}$. Hence

$$g = \sum_{j=1}^n (2\pi m_j)\lambda_j^* \in \left\{ \sum_{j=1}^n m_j (2\pi \lambda_j^*) : m_j \in \mathbb{Z} \right\}.$$

Conversely, any $g = \sum_j (2\pi m_j)\lambda_j^*$ satisfies

$$g \cdot \lambda_k = \sum_j (2\pi m_j)\lambda_j^* \cdot \lambda_k = 2\pi m_k \in 2\pi\mathbb{Z},$$

so $g \in \Lambda^*$. Thus Λ^* is the lattice generated by the basis $\{2\pi\lambda_1^*, \dots, 2\pi\lambda_n^*\}$, which we still denote by λ^* in the exercise (the factor 2π is absorbed into the choice of dual basis).

(d) Periodicity of e_g. For $g \in \mathbb{R}^n$, $e_g(x) := e^{i\,g\cdot x}$. If $g \in \Lambda^*$, then for every $g_0 \in \Lambda$ we have $g \cdot g_0 \in 2\pi\mathbb{Z}$, and hence

$$e_g(x + g_0) = e^{i\,g\cdot(x+g_0)} = e^{i\,g\cdot x} e^{i\,g\cdot g_0} = e^{i\,g\cdot x} e^{i\,2\pi k} = e^{i\,g\cdot x} = e_g(x).$$

Thus e_g is Λ–periodic whenever $g \in \Lambda^*$.

(e) Fourier transform of a Λ–periodic distribution. Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be Λ–periodic, $\tau_g u = u$ for all $g \in \Lambda$. Recall that translation in the physical side corresponds to modulation in the Fourier side:

$$\widehat{\tau_g u}(\xi) = e^{i\,g\cdot\xi}\widehat{u}(\xi).$$

Since $\tau_g u = u$, we have

$$\widehat{u}(\xi) = e^{i\,g\cdot\xi}\widehat{u}(\xi) \qquad \text{for all } g \in \Lambda,$$

so

$$\left(e^{i\,g\cdot\xi}-1\right)\widehat{u}(\xi)=0 \qquad \text{for all } g\in\Lambda.$$

This implies that $\widehat{u}$ is supported on the set of ξ such that $e^{i\,g\cdot\xi}=1$ for all $g\in\Lambda$, that is, $\xi\cdot g\in 2\pi\mathbb{Z}$ for all $g\in\Lambda$. By definition, this is precisely Λ^*, so

$$\operatorname{supp}\widehat{u}\subset\Lambda^*.$$

A tempered distribution whose support is a discrete set must be a (locally finite) sum of derivatives of Dirac masses at those points. Thus we can write

$$\widehat{u}=\sum_{g\in\Lambda^*}T_g,$$

where each T_g is a finite linear combination of derivatives of δ_g. However, the condition $e^{i\,h\cdot\xi}\widehat{u}(\xi)=\widehat{u}(\xi)$ for all $h\in\Lambda$ forces each T_g to be invariant under multiplication by the smooth function $e^{i\,h\cdot\xi}$ near g. This is only possible if T_g is a *scalar multiple* of δ_g (no derivatives). Hence

$$\widehat{u}=\sum_{g\in\Lambda^*}c_g\,\delta_g$$

for some coefficients $c_g\in\mathbb{C}$. Since $\widehat{u}$ is tempered, the general characterization of such discrete measures in $\mathcal{S}'$ (cf. Exercise 5.3) implies a polynomial growth bound: there exist $K>0$, $N\in\mathbb{N}$ such that

$$|c_g|\le K(1+|g|)^N \qquad \text{for all } g\in\Lambda^*.$$

(f) Fourier series representation and coefficients. From part (e) we have

$$\widehat{u}=\sum_{g\in\Lambda^*}c_g\,\delta_g, \qquad |c_g|\le K(1+|g|)^N.$$

Taking inverse Fourier transforms,

$$u=\mathcal{F}^{-1}\widehat{u}=\sum_{g\in\Lambda^*}c_g\,\mathcal{F}^{-1}\delta_g.$$

With the convention for the Fourier transform used in these notes, $\mathcal{F}^{-1}\delta_g(x)=e^{i\,g\cdot x}$ (up to the standard normalization constants, which are absorbed into c_g). Thus

$$u(x)=\sum_{g\in\Lambda^*}d_g\,e^{i\,g\cdot x}=\sum_{g\in\Lambda^*}d_g\,e_g(x),$$

where d_g are complex coefficients satisfying the same kind of polynomial growth bound

$$|d_g|\le K(1+|g|)^N.$$

This is the Fourier series of the Λ–periodic distribution u, with frequencies in Λ^*.

Finally, we identify d_g in terms of u. Using the mean value functional from part (b), for any $g \in \Lambda^*$ we have

$$e_{-g}u = \sum_{h\in\Lambda^*} d_h\, e_{h-g},$$

so its mean value is

$$M(e_{-g}u) = \sum_{h\in\Lambda^*} d_h\, M(e_{h-g}).$$

Now $M(e_{h-g})$ is the average over one period of the exponential $e^{i(h-g)\cdot x}$. If $h \neq g$, then e_{h-g} is a non-trivial Λ–periodic exponential and its mean over q_Λ is 0. If $h = g$, then $e_{h-g} \equiv 1$ and $M(e_{h-g}) = 1$. Therefore

$$M(e_{-g}u) = d_g.$$

Thus the Fourier coefficients are given by

$$d_g = M(e_{-g}u),$$

as claimed.

□

Visual Summary for Exercise 5.7

The following figures illustrate the geometric structures underlying Λ–periodic distributions on $\mathbb{R}^n$ and their Fourier series supported on the dual lattice Λ^*.

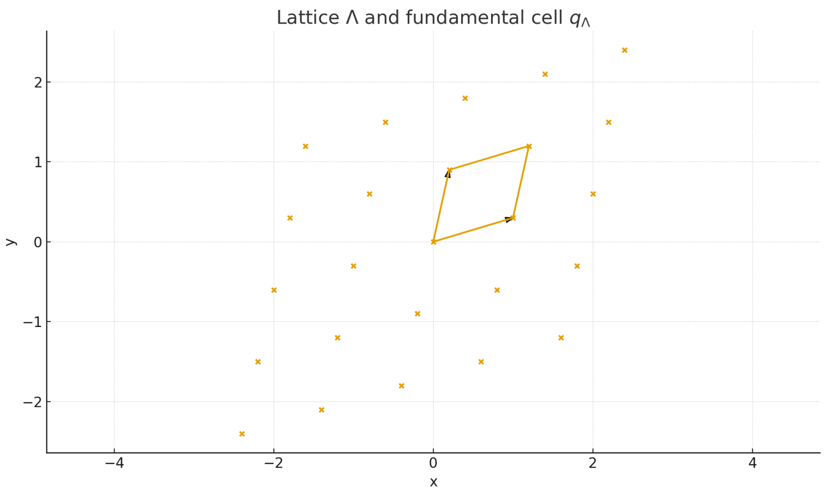


Figure 35: A lattice Λ generated by basis vectors λ_1, λ_2 in $\mathbb{R}^2$. The fundamental cell q_Λ tiles $\mathbb{R}^2$ under translation by Λ. A Λ–periodic distribution u satisfies $\tau_g u = u$ for all $g \in \Lambda$, meaning that u descends to a distribution on the torus $\mathbb{R}^2/\Lambda$.

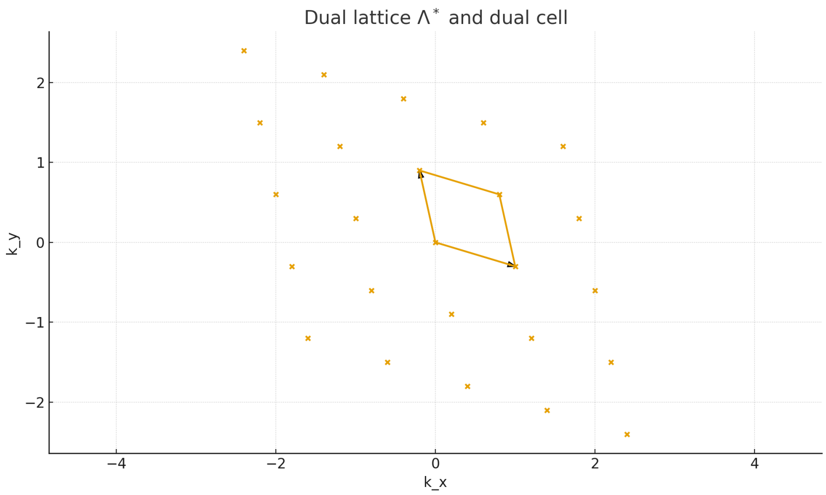


Figure 36: The dual lattice Λ^* with dual basis vectors λ_1^*, λ_2^* satisfying the orthogonality relation $\lambda_j^* \cdot \lambda_k = \delta_{jk}$. The Fourier transform of any Λ–periodic distribution is a purely discrete measure supported on Λ^*: $\widehat{u} = \sum_{g \in \Lambda^*} c_g \delta_g$. Thus Λ^* plays the role of the "frequency lattice".

Exercise 5.8 : Approximation and Basic Properties of Sobolev Spaces $H^s(\mathbb{R}^n)$

Suppose $s \geq 0$.

(a) Let $\eta \in \mathscr{S}(\mathbb{R}^n)$ satisfy

$$\int_{\mathbb{R}^n} \eta(x)\,dx = 1,$$

and define the mollifiers

$$\eta_\varepsilon(x) := \varepsilon^{-n} \eta\left(\frac{x}{\varepsilon}\right), \qquad \varepsilon > 0.$$

Show that for every $f \in H^s(\mathbb{R}^n)$ one has

$$\eta_\varepsilon * f \longrightarrow f \quad \text{in } H^s(\mathbb{R}^n) \quad \text{as } \varepsilon \to 0.$$

(b) Suppose $f \in H^s(\mathbb{R}^n)$ has compact support. Show that there exists a sequence $\varphi_j \in C_0^\infty(\mathbb{R}^n)$ such that

$$\varphi_j \longrightarrow f \quad \text{in } H^s(\mathbb{R}^n).$$

(c) Show that if $t \geq s$, then

$$\|f\|_{H^s(\mathbb{R}^n)} \leq \|f\|_{H^t(\mathbb{R}^n)}.$$

Deduce that

$$\|f\|_{L^2(\mathbb{R}^n)} \leq \frac{1}{(2\pi)^{n/2}} \|f\|_{H^s(\mathbb{R}^n)}.$$

Hint: Use Parseval's formula.

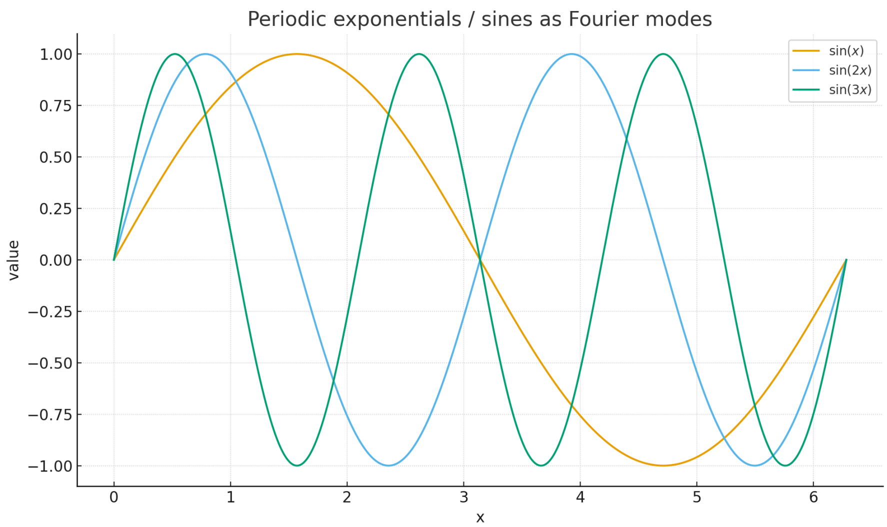


Figure 37: Periodic exponential functions $e_g(x) = e^{ig\cdot x}$ for $g \in \Lambda^*$. These exponentials are exactly the characters on the torus $\mathbb{R}^n/\Lambda$, satisfying $e_g(x+g_0) = e_g(x)$ for all $g_0 \in \Lambda$. They form the Fourier basis for all Λ–periodic distributions.

(d) Let α be a multiindex with $|\alpha| = k$. Show that the distributional derivative

$$D^\alpha : H^{s+k}(\mathbb{R}^n) \longrightarrow H^s(\mathbb{R}^n)$$

is a bounded linear operator.

Theory and Solutions

Sobolev norm. For $s \geq 0$,

$$\|f\|^2_{H^s(\mathbb{R}^n)} = \int_{\mathbb{R}^n} (1+|\xi|^2)^s |\widehat{f}(\xi)|^2 \, d\xi.$$

(a) Approximation by Smooth Functions

Goal. Show $\eta_\varepsilon * f \to f$ in H^s.

Key fact. $\widehat{\eta_\varepsilon}(\xi) = \widehat{\eta}(\varepsilon\xi)$ and $\widehat{\eta}(0) = 1$.

Proof.

$$\widehat{\eta_\varepsilon * f}(\xi) = \widehat{\eta}(\varepsilon\xi)\widehat{f}(\xi).$$

Thus,

$$\|\eta_\varepsilon * f - f\|^2_{H^s} = \int (1+|\xi|^2)^s |\widehat{\eta}(\varepsilon\xi) - 1|^2 |\widehat{f}(\xi)|^2 d\xi.$$

Since $\widehat{\eta} \in \mathscr{S}$ and $\widehat{\eta}(\varepsilon\xi) \to 1$, and $(1+|\xi|^2)^s|\widehat{f}(\xi)|^2 \in L^1$, dominated convergence gives the result.

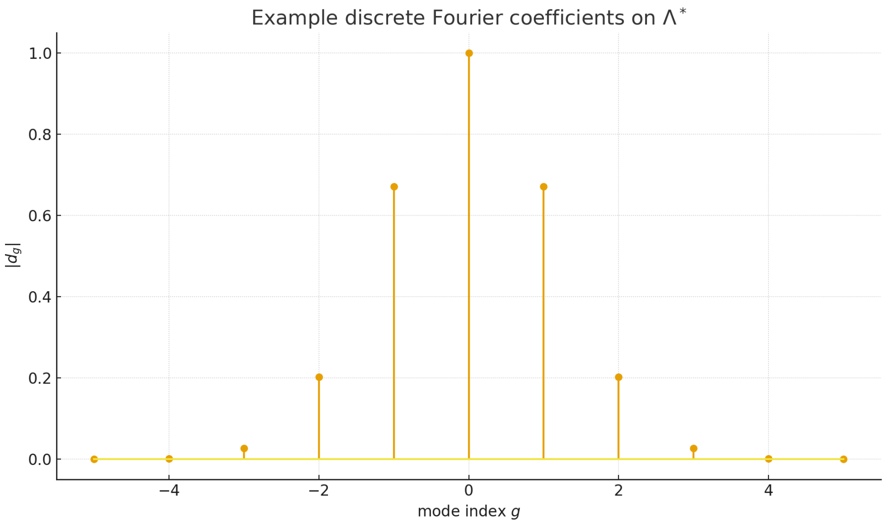


Figure 38: Schematic structure of the Fourier series representation of a Λ-periodic distribution $u(x) = \sum_{g\in\Lambda^*} d_g e^{ig\cdot x}$, where $d_g = M(e_{-g}u)$ and $M(u)$ denotes the mean value over one fundamental cell, e.g. $M(u) = \frac{1}{|Q_\Lambda|}u[\psi]$. The coefficients satisfy the polynomial growth bound $|d_g| \lesssim (1 + |g|)^N$, reflecting that $u \in \mathcal{D}'(\mathbb{R}^n)$. The figure highlights the discrete support of $\widehat{u}$ on Λ^* and the reconstruction of u from its Fourier modes.

(b) Approximation by Compactly Supported Smooth Functions

Idea. First mollify, then cut off.

Construction. Let $g_\varepsilon = \eta_\varepsilon * f \to f$ in H^s.

Let $\chi_R \in C_0^\infty$ be a smooth cutoff with $\chi_R = 1$ on B_R. Define

$$\varphi_{\varepsilon,R}(x) = \chi_R(x)\, g_\varepsilon(x).$$

Multiplication by χ_R is bounded on H^s, and $\chi_R g_\varepsilon \to g_\varepsilon$ as $R \to \infty$. A diagonal argument produces $\varphi_j \to f$ in H^s.

(c) Monotonicity of Sobolev Norms and L^2 Estimate

Claim. If $t \geq s$, then

$$(1 + |\xi|^2)^s \leq (1 + |\xi|^2)^t,$$

hence

$$\|f\|_{H^s}^2 \leq \|f\|_{H^t}^2.$$

L^2 estimate. Since

$$\|f\|_{L^2}^2 = \frac{1}{(2\pi)^n}\int |\widehat{f}(\xi)|^2 d\xi \leq \frac{1}{(2\pi)^n}\int (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 d\xi,$$

we obtain

$$\|f\|_{L^2} \leq \frac{1}{(2\pi)^{n/2}} \|f\|_{H^s}.$$

(d) Boundedness of Derivatives

Goal. Show $D^\alpha : H^{s+k} \to H^s$ is bounded.

Fourier representation.

$$\widehat{D^\alpha f}(\xi) = (i\xi)^\alpha \widehat{f}(\xi).$$

Estimate. Since $|\xi^\alpha| \leq (1+|\xi|^2)^{k/2}$,

$$\|D^\alpha f\|_{H^s}^2 = \int (1+|\xi|^2)^s |\xi^\alpha|^2 |\widehat{f}(\xi)|^2 d\xi \leq \int (1+|\xi|^2)^{s+k} |\widehat{f}(\xi)|^2 d\xi = \|f\|_{H^{s+k}}^2.$$

Thus D^α is bounded.

5.9 Heat equation: energy estimate and heat kernel

Suppose that $u_0 \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$, and let $u(t,x)$ be the solution of the heat equation

$$\partial_t u - \Delta u = 0, \qquad t > 0,\ x \in \mathbb{R}^n,$$

with initial condition $u(0,x) = u_0(x)$. In terms of the Fourier transform, u is given by

$$u(t,x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{u_0}(\xi)\, e^{-t|\xi|^2}\, e^{i\xi\cdot x}\, d\xi, \qquad t > 0.$$

(a) Problem. Show that

$$\|u(t,\cdot)\|_{L^2(\mathbb{R}^n)} \leq \|u_0\|_{L^2(\mathbb{R}^n)} \quad \text{for all } t > 0.$$

(b) Problem. Show that

$$u(t,x) = (u_0 * K_t)(x),$$

where the heat kernel is defined by

$$K_t(x) = \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x|^2}{4t}\right), \qquad t > 0,\ x \in \mathbb{R}^n.$$

(c) Problem. Assume that $u_0 \geq 0$. Show that $u(t,x) \geq 0$ for all $t > 0$ and

$$\|u(t,\cdot)\|_{L^1(\mathbb{R}^n)} = \|u_0\|_{L^1(\mathbb{R}^n)} \quad \text{for all } t > 0.$$

Theory. Taking the Fourier transform in the spatial variables for the heat equation $\partial_t u - \Delta u = 0$ yields the ODE

$$\partial_t \widehat{u}(t,\xi) + |\xi|^2 \widehat{u}(t,\xi) = 0,$$

whose solution is

$$\widehat{u}(t,\xi) = e^{-t|\xi|^2} \widehat{u_0}(\xi).$$

Using Plancherel's theorem one obtains energy estimates in L^2. The factor $e^{-t|\xi|^2}$ is precisely the Fourier transform of the Gaussian heat kernel

$$K_t(x) = \frac{1}{(4\pi t)^{n/2}} e^{-|x|^2/(4t)},$$

and hence $\widehat{K_t}(\xi) = e^{-t|\xi|^2}$. By the convolution theorem, this implies $u(t,\cdot) = u_0 * K_t$. The kernel K_t is nonnegative and normalized $\int_{\mathbb{R}^n} K_t(x)\,dx = 1$, so convolution with K_t preserves positivity and total mass (L^1-norm).

Solution to (a): L^2-energy estimate

Step 1: Fourier representation. From the theory above,

$$\widehat{u}(t,\xi) = e^{-t|\xi|^2} \widehat{u_0}(\xi).$$

Step 2: Use Plancherel. By Plancherel's theorem,

$$\|u(t,\cdot)\|^2_{L^2(\mathbb{R}^n)} = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} |\widehat{u}(t,\xi)|^2\,d\xi = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{-2t|\xi|^2} |\widehat{u_0}(\xi)|^2\,d\xi.$$

Since $e^{-2t|\xi|^2} \le 1$,

$$\|u(t,\cdot)\|^2_{L^2} \le \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} |\widehat{u_0}(\xi)|^2\,d\xi = \|u_0\|^2_{L^2}.$$

Taking square roots gives

$$\|u(t,\cdot)\|_{L^2(\mathbb{R}^n)} \le \|u_0\|_{L^2(\mathbb{R}^n)},$$

as required.

Solution to (b): convolution with the heat kernel

Step 1: Gaussian heat kernel. Define

$$K_t(x) := \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x|^2}{4t}\right), \qquad t > 0.$$

A standard Fourier transform computation (see Lemma 1.9) shows that

$$\widehat{K_t}(\xi) = e^{-t|\xi|^2}.$$

Step 2: Convolution theorem. We already know that

$$\widehat{u}(t,\xi) = e^{-t|\xi|^2}\widehat{u_0}(\xi) = \widehat{u_0}(\xi)\,\widehat{K_t}(\xi).$$

By the convolution theorem,

$$\mathcal{F}\Big(u_0 * K_t\Big)(\xi) = \widehat{u_0}(\xi)\,\widehat{K_t}(\xi) = \widehat{u}(t,\xi).$$

Since the Fourier transform is injective, we conclude

$$u(t,x) = (u_0 * K_t)(x) = \int_{\mathbb{R}^n} u_0(y)\,K_t(x-y)\,dy.$$

Solution to (c): positivity and conservation of mass

Assume $u_0 \geq 0$.

Positivity. From part (b),

$$u(t,x) = \int_{\mathbb{R}^n} u_0(y)\,K_t(x-y)\,dy.$$

Both $u_0(y)$ and $K_t(x-y)$ are nonnegative, hence the integral is nonnegative:

$$u(t,x) \geq 0 \quad \text{for all } t>0,\ x \in \mathbb{R}^n.$$

Conservation of the L^1-norm. Integrating over $\mathbb{R}^n$ and using Fubini's theorem,

$$\int_{\mathbb{R}^n} u(t,x)\,dx = \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} u_0(y)\,K_t(x-y)\,dy\,dx = \int_{\mathbb{R}^n} u_0(y)\left(\int_{\mathbb{R}^n} K_t(x-y)\,dx\right)dy.$$

The inner integral equals 1, since

$$\int_{\mathbb{R}^n} K_t(x-y)\,dx = \int_{\mathbb{R}^n} K_t(z)\,dz = 1,$$

by the normalization of the Gaussian. Therefore

$$\int_{\mathbb{R}^n} u(t,x)\,dx = \int_{\mathbb{R}^n} u_0(y)\,dy,$$

which shows

$$\|u(t,\cdot)\|_{L^1(\mathbb{R}^n)} = \|u_0\|_{L^1(\mathbb{R}^n)}.$$

Conclusion. The heat semigroup $e^{t\Delta}$ is L^2-contractive, admits the Gaussian kernel representation, and for nonnegative data preserves both positivity and the total mass.

Exercise 5.10 : The Schrödinger Equation in $\mathbb{R}^n$

Consider the Schrödinger equation

$$\begin{cases} u_t = i\Delta u & \text{in } (0,T) \times \mathbb{R}^n, \\ u = u_0 & \text{on } \{0\} \times \mathbb{R}^n, \end{cases} \tag{3}$$

and suppose that $u_0 \in H^2(\mathbb{R}^n)$.

(a) Show that (3) admits a unique solution u such that

$$u \in C^0([0,T]; H^2(\mathbb{R}^n)) \cap C^1((0,T); L^2(\mathbb{R}^n)),$$

whose spatial Fourier–Plancherel transform is given by

$$\widehat{u}(t,\xi) = \widehat{u_0}(\xi)\, e^{-it|\xi|^2}.$$

(b) Show that

$$\|u(t,\cdot)\|_{H^2(\mathbb{R}^n)} = \|u_0\|_{H^2(\mathbb{R}^n)}.$$

(c) For $t > 0$, define $K_\varepsilon \in L^1_{\text{loc}}(\mathbb{R}^n)$ by

$$K_\varepsilon(x) = \frac{1}{(4\pi i t)^{n/2}}\, e^{\,i|x|^2/4t},$$

where for n odd we take the usual branch cut so that $i^{1/2} = e^{i\pi/4}$. For $\varepsilon > 0$ set

$$K^t_\varepsilon(x) = e^{-\varepsilon|x|^2}\, K_t(x).$$

i) Show that $T_{K_\varepsilon} \to T_{K_t}$ in $\mathscr{S}'$ as $\varepsilon \to 0$.

ii) Show that if $\Re(\sigma) > 0$, then

$$\int_{\mathbb{R}} e^{-\sigma x^2 - ix\xi}\, dx = \sqrt{\frac{\pi}{\sigma}}\, e^{-\xi^2/(4\sigma)}.$$

iii) Deduce that

$$\widehat{K^t_\varepsilon}(\xi) = \left(1 + \frac{i4t}{\varepsilon}\right)^{-\frac{n}{2}} e^{-\,it|\xi|^2/(1+4it\varepsilon)}.$$

iv) Conclude that

$$\widehat{K_t}(\xi) = e^{-it|\xi|^2}.$$

(d) Suppose that $u_0 \in \mathscr{S}(\mathbb{R}^n)$. Show that for $t > 0$:

$$u(t,x) = \int_{\mathbb{R}^n} u_0(y)\, K_t(x-y)\, dy,$$

and deduce the dispersive estimate

$$\sup_{x\in\mathbb{R}^n} |u(t,x)| \;\lesssim\; \frac{1}{(4\pi t)^{n/2}}\, \|u_0\|_{L^1(\mathbb{R}^n)}.$$

(This type of estimate shows the decay of solutions to the Schrödinger equation.)

Schrödinger Equation in $\mathbb{R}^n$: Theory and Solutions

Recall that for $s \geq 0$ the Sobolev norm is

$$\|f\|_{H^s(\mathbb{R}^n)}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 \, d\xi.$$

(a) Existence, Uniqueness and Fourier Representation

Idea. Take the Fourier transform in the spatial variable, which turns the PDE

$$u_t = i\Delta u$$

into an ODE in t for each ξ.

Derivation. Using $\widehat{\Delta u}(t,\xi) = -|\xi|^2 \widehat{u}(t,\xi)$, we get

$$\partial_t \widehat{u}(t,\xi) = i\widehat{\Delta u}(t,\xi) = -i|\xi|^2 \, \widehat{u}(t,\xi), \qquad \widehat{u}(0,\xi) = \widehat{u_0}(\xi).$$

Hence

$$\widehat{u}(t,\xi) = e^{-it|\xi|^2} \, \widehat{u_0}(\xi).$$

Regularity. Define

$$u(t,x) := \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{u_0}(\xi) \, e^{-it|\xi|^2} e^{ix\cdot\xi} \, d\xi.$$

Since $u_0 \in H^2(\mathbb{R}^n)$, we have $(1 + |\xi|^2)^2 \widehat{u_0}(\xi) \in L^2$, and the multiplier $e^{-it|\xi|^2}$ is bounded for each t. Standard Fourier multiplier arguments give

$$u \in C^0([0,T]; H^2(\mathbb{R}^n)) \cap C^1((0,T); L^2(\mathbb{R}^n)).$$

Uniqueness. If v is another solution with the same regularity and $v(0) = u_0$, then $\widehat{v}$ satisfies the same ODE with the same initial data and hence $\widehat{v} = \widehat{u}$. Thus $v = u$ and the solution is unique.

In particular,

$$\widehat{u}(t,\xi) = \widehat{u_0}(\xi) \, e^{-it|\xi|^2},$$

as required.

(b) Conservation of the H^2-norm

Computation. Using the formula from part (a),

$$\|u(t,\cdot)\|_{H^2}^2 = \int_{\mathbb{R}^n} (1+|\xi|^2)^2\,|\widehat{u}(t,\xi)|^2\,d\xi = \int (1+|\xi|^2)^2\,|e^{-it|\xi|^2}|^2\,|\widehat{u_0}(\xi)|^2\,d\xi.$$

Since $|e^{-it|\xi|^2}| = 1$,

$$\|u(t,\cdot)\|_{H^2}^2 = \int (1+|\xi|^2)^2|\widehat{u_0}(\xi)|^2\,d\xi = \|u_0\|_{H^2}^2.$$

Hence

$$\|u(t,\cdot)\|_{H^2(\mathbb{R}^n)} = \|u_0\|_{H^2(\mathbb{R}^n)} \quad \text{for all } t \in [0,T].$$

(c) The Schrödinger Kernel K_t and its Fourier Transform

For $t>0$, set

$$K_t(x) := \frac{1}{(4\pi i t)^{n/2}} e^{\,i|x|^2/4t}, \qquad K_t^\varepsilon(x) := e^{-\varepsilon|x|^2} K_t(x), \quad \varepsilon > 0.$$

(i) Convergence in $\mathscr{S}'$. Let $\varphi \in \mathscr{S}(\mathbb{R}^n)$. Then

$$\langle T_{K_t^\varepsilon}, \varphi\rangle = \int_{\mathbb{R}^n} K_t(x)\,e^{-\varepsilon|x|^2}\,\varphi(x)\,dx.$$

As $\varepsilon \to 0$, $e^{-\varepsilon|x|^2} \to 1$ pointwise, and

$$|K_t(x)e^{-\varepsilon|x|^2}\varphi(x)| \le |K_t(x)\varphi(x)|.$$

Since K_t has at most polynomial growth and φ is rapidly decreasing, $K_t\varphi \in L^1(\mathbb{R}^n)$, so dominated convergence gives

$$\langle T_{K_t^\varepsilon}, \varphi\rangle \longrightarrow \int K_t(x)\varphi(x)\,dx = \langle T_{K_t}, \varphi\rangle.$$

Thus $T_{K_t^\varepsilon} \to T_{K_t}$ in $\mathscr{S}'$.

(ii) Complex Gaussian integral. If $\Re(\sigma) > 0$, then

$$\int_{\mathbb{R}} e^{-\sigma x^2 - i\xi x}\,dx = \sqrt{\frac{\pi}{\sigma}}\,e^{-\xi^2/(4\sigma)}.$$

This follows by completing the square and deforming the integration path.

(iii) Fourier transform of K_t^ε. We write

$$K_t^\varepsilon(x) = \frac{1}{(4\pi it)^{n/2}} \exp\Big(-\sigma|x|^2\Big), \qquad \sigma := \varepsilon - \frac{i}{4t}, \quad \Re(\sigma) > 0.$$

Using (ii) in each coordinate, we obtain

$$\widehat{e^{-\sigma|x|^2}}(\xi) = \left(\sqrt{\frac{\pi}{\sigma}}\right)^n \exp\Big(-\frac{|\xi|^2}{4\sigma}\Big),$$

hence

$$\widehat{K_t^\varepsilon}(\xi) = \frac{1}{(4\pi it)^{n/2}} \left(\frac{\pi}{\sigma}\right)^{n/2} \exp\Big(-\frac{|\xi|^2}{4\sigma}\Big).$$

A straightforward simplification yields

$$\widehat{K_t^\varepsilon}(\xi) = \Big(1 + 4it\varepsilon\Big)^{-n/2} \exp\left(-\frac{it|\xi|^2}{1 + 4it\varepsilon}\right).$$

(iv) Limit as $\varepsilon \to 0$. As $\varepsilon \to 0$,

$$1 + 4it\varepsilon \to 1, \qquad -\frac{it|\xi|^2}{1 + 4it\varepsilon} \to -it|\xi|^2.$$

Thus

$$\widehat{K_t^\varepsilon}(\xi) \longrightarrow e^{-it|\xi|^2} \quad \text{pointwise in } \xi.$$

By (i), $T_{K_t^\varepsilon} \to T_{K_t}$ in $\mathscr{S}'$ and the Fourier transform is continuous on $\mathscr{S}'$, so we conclude

$$\widehat{K_t}(\xi) = e^{-it|\xi|^2}.$$

(d) Convolution Representation and Dispersive Estimate

Assume $u_0 \in \mathscr{S}(\mathbb{R}^n)$. From part (a),

$$\widehat{u}(t,\xi) = e^{-it|\xi|^2}\, \widehat{u_0}(\xi),$$

and from part (c),

$$\widehat{K_t}(\xi) = e^{-it|\xi|^2}.$$

Hence

$$\widehat{u}(t,\xi) = \widehat{u_0}(\xi)\, \widehat{K_t}(\xi) = \mathcal{F}(u_0 * K_t)(\xi),$$

so by injectivity of the Fourier transform

$$u(t,x) = (u_0 * K_t)(x) = \int_{\mathbb{R}^n} u_0(y)\, K_t(x-y)\, dy.$$

Dispersive estimate. Since $|e^{i|z|^2/(4t)}| = 1$, we have

$$|K_t(z)| = \frac{1}{(4\pi|t|)^{n/2}} \quad \text{for all } z \in \mathbb{R}^n.$$

Then

$$|u(t,x)| \le \int_{\mathbb{R}^n} |u_0(y)|\,|K_t(x-y)|\,dy \le \frac{1}{(4\pi|t|)^{n/2}} \int_{\mathbb{R}^n} |u_0(y)|\,dy.$$

Taking the supremum over $x \in \mathbb{R}^n$ yields

$$\sup_{x\in\mathbb{R}^n} |u(t,x)| \le \frac{1}{(4\pi|t|)^{n/2}} \|u_0\|_{L^1(\mathbb{R}^n)}.$$

Conclusion. The free Schrödinger evolution is given by $u(t,\cdot) = e^{it\Delta}u_0 = u_0 * K_t$, acts unitarily on $H^2(\mathbb{R}^n)$, and satisfies the dispersive estimate above, which shows decay in L^∞ for data in L^1.

Sobolev Space $H^s(\mathbb{R}^n)$ used in Exercise 5.10

Throughout this section, $H^s(\mathbb{R}^n)$ denotes the L^2-based Sobolev space. For $s \ge 0$ we define

$$H^s(\mathbb{R}^n) := \Big\{ f \in \mathscr{S}'(\mathbb{R}^n) : \int_{\mathbb{R}^n} (1+|\xi|^2)^s |\widehat{f}(\xi)|^2\,d\xi < \infty \Big\},$$

with norm

$$\|f\|^2_{H^s(\mathbb{R}^n)} := \int_{\mathbb{R}^n} (1+|\xi|^2)^s |\widehat{f}(\xi)|^2\,d\xi.$$

For integer $s = m \in \mathbb{N}$ this is equivalent to

$$H^m(\mathbb{R}^n) = \Big\{ f \in L^2(\mathbb{R}^n) : D^\alpha f \in L^2(\mathbb{R}^n) \text{ for all } |\alpha| \le m \Big\},$$

and

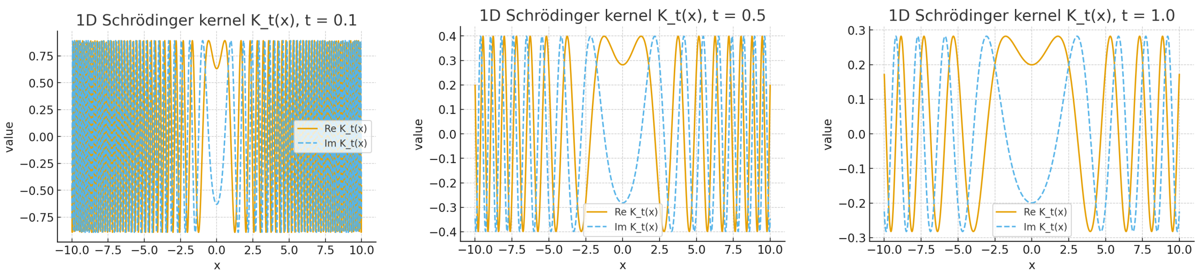
$$\|f\|^2_{H^m(\mathbb{R}^n)} \simeq \sum_{|\alpha|\le m} \|D^\alpha f\|^2_{L^2(\mathbb{R}^n)}.$$

In particular, in Exercise 5.10 the assumption $u_0 \in H^2(\mathbb{R}^n)$ means that u_0 and all second-order derivatives of u_0 belong to $L^2(\mathbb{R}^n)$, and the solution $u(t,\cdot)$ stays in H^2 for all times t.

Relation Between Sobolev Spaces and the Schwartz Space

Let $\mathscr{S}(\mathbb{R}^n)$ denote the Schwartz space of rapidly decreasing smooth functions. Recall that for $s \in \mathbb{R}$ the L^2–based Sobolev space $H^s(\mathbb{R}^n)$ is defined by

$$H^s(\mathbb{R}^n) := \Big\{ f \in \mathscr{S}'(\mathbb{R}^n) : \int_{\mathbb{R}^n} (1+|\xi|^2)^s\,|\widehat{f}(\xi)|^2\,d\xi < \infty \Big\}.$$



(a) $t = 0.1$ (b) $t = 0.5$ (c) $t = 1.0$

Figure 39: Real and imaginary parts of the one-dimensional Schrödinger kernel $K_t(x)$ for three different times t. The modulus is constant $|K_t(x)| = (4\pi|t|)^{-1/2}$, while the oscillations become slower as t increases.

Inclusion. Since $\widehat{f} \in \mathscr{S}(\mathbb{R}^n)$ whenever $f \in \mathscr{S}(\mathbb{R}^n)$, the quantity $(1 + |\xi|^2)^s |\widehat{f}(\xi)|^2$ decays faster than any power of $|\xi|$. Hence

$$\mathscr{S}(\mathbb{R}^n) \subset H^s(\mathbb{R}^n) \quad \text{for all } s \in \mathbb{R}.$$

Density. For every $s \in \mathbb{R}$ the Schwartz space is dense in H^s:

$$\overline{\mathscr{S}(\mathbb{R}^n)}^{H^s(\mathbb{R}^n)} = H^s(\mathbb{R}^n).$$

This follows from approximation by mollifiers and compactly supported smooth functions, as in Exercise 5.8.

Intersection over all orders. One has the characterization

$$\mathscr{S}(\mathbb{R}^n) = \bigcap_{s \in \mathbb{R}} H^s(\mathbb{R}^n),$$

that is, a tempered distribution belongs to the Schwartz space if and only if it lies in every Sobolev space $H^s(\mathbb{R}^n)$.

Remark. The space

$$H^\infty(\mathbb{R}^n) := \bigcap_{m \in \mathbb{N}} H^m(\mathbb{R}^n)$$

consists of functions with all derivatives in L^2 and contains $\mathscr{S}(\mathbb{R}^n)$ as a dense subspace. The additional requirement that both f and $\widehat{f}$ decay faster than any power of $|x|$ and $|\xi|$ is what distinguishes the Schwartz space from general elements of H^∞.

Exercise 5.11 : Radial Solutions of the Wave Equation in $\mathbb{R}^3$

Let $\mathbb{R}^3_* := \mathbb{R}^3 \setminus \{0\}$ and

$$S_{*,T} := (-T,T) \times \mathbb{R}^3_*, \qquad S_T := (-T,T) \times \mathbb{R}^3, \qquad \Sigma_0 := \{0\} \times \mathbb{R}^3.$$

Write $r = |x|$. You may assume the formula for radial functions $u = u(r,t)$:

$$\Delta u(|x|,t) = \Delta u(r,t) = \frac{\partial^2 u}{\partial r^2}(r,t) + \frac{2}{r}\frac{\partial u}{\partial r}(r,t).$$

We consider the wave equation

$$-\frac{\partial^2 u}{\partial t^2}(x,t) + \Delta u(x,t) = 0.$$

(a) Suppose $u(x,t) = \frac{1}{r}v(r,t)$ for some function v. Show that u solves the wave equation on $\mathbb{R}^3_* \times (-T,T)$ if and only if v satisfies the one-dimensional wave equation

$$-\frac{\partial^2 v}{\partial t^2}(r,t) + \frac{\partial^2 v}{\partial r^2}(r,t) = 0 \quad \text{on } (0,\infty) \times (-T,T).$$

(b) Suppose $f, g \in C^2_c(\mathbb{R})$. Deduce that

$$u(x,t) = \frac{f(r+t)}{r} + \frac{g(r-t)}{r}$$

is a solution of the wave equation on $S_{*,T}$ which vanishes for large $|x|$.

(c) Show that if $f \in C^3_c(\mathbb{R})$ is an odd function (i.e. $f(s) = -f(-s)$ for all s), then

$$u(x,t) \equiv \frac{f(r+t) + f(r-t)}{2r}$$

extends as a C^2 function which solves the wave equation on $S_T = (-T,T) \times \mathbb{R}^3$, with

$$u(0,t) = f'(t).$$

*d) By considering a suitable sequence of functions f, or otherwise, deduce that there exists no constant C, independent of u, such that the estimate

$$\sup_{S_T}(|u| + |u_t|) \le C \sup_{\Sigma_0}(|u| + |u_t|)$$

holds for all solutions $u \in C^2(S_T)$ of the wave equation which vanish for large $|x|$.

Theory and Solutions

(a) Reduction to a One-Dimensional Wave Equation

Goal. Show that if $u(r,t) = v(r,t)/r$ is radial, then u solves $-u_{tt} + \Delta u = 0$ on $\mathbb{R}^3_* \times (-T,T)$ if and only if v solves the one-dimensional wave equation $-v_{tt} + v_{rr} = 0$ on $(0,\infty) \times (-T,T)$.

Computation. Let $u(r,t) = v(r,t)/r$. Then

$$u_r = \frac{v_r}{r} - \frac{v}{r^2}, \qquad u_{rr} = \frac{v_{rr}}{r} - \frac{2v_r}{r^2} + \frac{2v}{r^3}, \qquad u_{tt} = \frac{v_{tt}}{r}.$$

Using the radial Laplacian,

$$\Delta u = u_{rr} + \frac{2}{r}u_r = \left(\frac{v_{rr}}{r} - \frac{2v_r}{r^2} + \frac{2v}{r^3}\right) + \frac{2}{r}\left(\frac{v_r}{r} - \frac{v}{r^2}\right) = \frac{v_{rr}}{r}.$$

Thus the wave equation

$$-u_{tt} + \Delta u = 0$$

becomes

$$-\frac{v_{tt}}{r} + \frac{v_{rr}}{r} = 0 \quad \iff \quad -v_{tt} + v_{rr} = 0,$$

for $r > 0$. This proves the equivalence.

(b) Construction of Radial Solutions

Goal. Use d'Alembert's formula for the one-dimensional wave equation to construct radial solutions in $\mathbb{R}^3$.

D'Alembert solution. For the one-dimensional wave equation $-v_{tt} + v_{rr} = 0$ on $\mathbb{R}$, the general solution is

$$v(r,t) = f(r+t) + g(r-t),$$

for some functions f, g.

Conclusion. If $f, g \in C_c^2(\mathbb{R})$, set

$$v(r,t) = f(r+t) + g(r-t), \qquad u(r,t) = \frac{v(r,t)}{r} = \frac{f(r+t)}{r} + \frac{g(r-t)}{r}.$$

Then v solves the one-dimensional wave equation, and by part (a), u solves the three-dimensional wave equation on $S_{*,T}$.

Since f, g are compactly supported, there exists $R > 0$ such that $f(s) = g(s) = 0$ whenever $|s| \geq R$. For fixed $t \in (-T, T)$ and for r sufficiently large, both $r+t$ and $r-t$ lie outside the supports of f and g, so $u(r,t) = 0$. Hence u vanishes for large $|x|$.

(c) Extension Through the Origin with Odd Data

Goal. Assume $f \in C_c^3(\mathbb{R})$ is odd ($f(-s) = -f(s)$). Show that

$$u(r,t) := \frac{f(r+t) + f(r-t)}{2r}$$

extends to a C^2 solution on S_T and satisfies $u(0,t) = f'(t)$.

Construction. Define

$$v(r,t) := \frac{1}{2}\Big(f(r+t) + f(r-t)\Big).$$

Then $v \in C^3$ and solves $-v_{tt} + v_{rr} = 0$ on $(0,\infty) \times (-T,T)$. By part (a), $u(r,t) = v(r,t)/r$ solves the wave equation on $\mathbb{R}^3_* \times (-T,T)$.

Because f is odd,

$$v(0,t) = \frac{1}{2}\Big(f(t) + f(-t)\Big) = \frac{1}{2}\Big(f(t) - f(t)\Big) = 0.$$

Thus $v(r,t)$ vanishes at $r = 0$ for each t.

Differentiating,

$$v_r(r,t) = \frac{1}{2}\Big(f'(r+t) + f'(r-t)\Big),$$

so

$$v_r(0,t) = \frac{1}{2}\Big(f'(t) + f'(-t)\Big).$$

Since f is odd, f' is even, hence $f'(-t) = f'(t)$ and

$$v_r(0,t) = f'(t).$$

Extension and regularity. Near $r = 0$ we have the Taylor expansion

$$v(r,t) = v_r(0,t)\, r + O(r^3) = f'(t)\, r + O(r^3),$$

so

$$u(r,t) = \frac{v(r,t)}{r} = f'(t) + O(r^2),$$

which has a finite limit as $r \to 0$. Define

$$u(0,t) := f'(t).$$

This defines a C^2 function u on S_T which coincides with the radial solution for $r > 0$ and therefore solves the wave equation on all of S_T.

(d) Failure of a Maximum-Principle-Type Estimate

Claim. There is no constant $C > 0$ such that

$$\sup_{S_T}(|u| + |u_t|) \le C \sup_{\Sigma_0}(|u| + |u_t|),$$

for all solutions $u \in C^2(S_T)$ of the wave equation that vanish for large $|x|$.

Reduction to a one-dimensional inequality. For solutions of the form in part (c),

$$u(r,t) = \frac{f(r+t) + f(r-t)}{2r}$$

with $f \in C_c^3(\mathbb{R})$ odd, we have

$$u(0,t) = f'(t).$$

At $t = 0$,

$$u(r,0) = \frac{f(r) + f(r)}{2r} = \frac{f(r)}{r}, \qquad u_t(r,0) = 0,$$

since f' is even. Therefore,

$$\sup_{\Sigma_0}(|u| + |u_t|) = \sup_{r>0}\left|\frac{f(r)}{r}\right|.$$

On the other hand,

$$\sup_{S_T}(|u| + |u_t|) \geq \sup_{t\in(-T,T)} |u(0,t)| = \sup_{t\in(-T,T)} |f'(t)|.$$

If the estimate above held for all such u, then for all odd $f \in C_c^3(\mathbb{R})$ we would have

$$\sup_{t\in(-T,T)} |f'(t)| \leq C \sup_{r>0}\left|\frac{f(r)}{r}\right|. \tag{†}$$

Construction of a contradicting sequence. Fix a nonzero $\psi \in C_c^3(\mathbb{R})$ supported in $(-1,1)$, and define for $R > 1$:

$$f_R(s) := \psi(s - R) - \psi(-s - R).$$

Then $f_R \in C_c^3(\mathbb{R})$ is odd (one checks $f_R(-s) = -f_R(s)$), and its support is contained in two disjoint intervals located near $s \approx \pm R$.

Since f_R' is just a translate of ψ', we have

$$\sup_{s\in\mathbb{R}} |f_R'(s)| = \sup_{s\in\mathbb{R}} |\psi'(s)| =: M > 0,$$

independent of R.

However, for s in the support of f_R we have $|s| \geq R - 1$, so

$$\left|\frac{f_R(s)}{s}\right| \leq \frac{\sup_{|y|\leq 1} |\psi(y)|}{R-1},$$

and hence

$$\sup_{r>0}\left|\frac{f_R(r)}{r}\right| \leq \frac{C_0}{R-1},$$

for some constant C_0 independent of R.

Contradiction. If (†) held for all odd f, it would hold for f_R:

$$M \leq C \sup_{r>0} \left| \frac{f_R(r)}{r} \right| \leq C \, \frac{C_0}{R-1}.$$

Letting $R \to \infty$ forces the right-hand side to 0, while $M > 0$ is fixed. This is impossible, and therefore no such constant C can exist.

Conclusion. The wave equation in three dimensions does not satisfy a global maximum-principle-type estimate of the form above; the size of the solution at later times cannot be controlled solely by the supremum of its initial data.